

ModelDesk

Scenario Creation

For ModelDesk 2023-A

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About This Document

Contents

This document introduces you to the Scenario Editor that is used to create maneuvers and traffic scenarios.

Symbols

dSPACE user documentation uses the following symbols:

Symbol	Description
 DANGER	Indicates a hazardous situation that, if not avoided, will result in death or serious injury.
 WARNING	Indicates a hazardous situation that, if not avoided, could result in death or serious injury.
 CAUTION	Indicates a hazardous situation that, if not avoided, could result in minor or moderate injury.
 NOTICE	Indicates a hazard that, if not avoided, could result in property damage.
 Note	Indicates important information that you should take into account to avoid malfunctions.
 Tip	Indicates tips that can make your work easier.
	Indicates a link that refers to a definition in the glossary, which you can find at the end of the document unless stated otherwise.
	Follows the document title in a link that refers to another document.

Naming conventions

dSPACE user documentation uses the following naming conventions:

%name% Names enclosed in percent signs refer to environment variables for file and path names.

< > Angle brackets contain wildcard characters or placeholders for variable file and path names, etc.

Special Windows folders

Windows-based software products use the following special folders:

Common Program Data folder A standard folder for application-specific program data that is used by all users.

%PROGRAMDATA%\dSPACE\<InstallationGUID>\<ProductName>

or

%PROGRAMDATA%\dSPACE\<ProductName>\<VersionNumber>

Documents folder A standard folder for application-specific files that are used by the current user.

%USERPROFILE%\Documents\dSPACE\<ProductName>\<VersionNumber>

Local Program Data folder A standard folder for application-specific program data that is used by the current, non-roaming user.

%USERPROFILE%\AppData\Local\dSPACE\<InstallationGUID>\<ProductName>

Accessing dSPACE Help and PDF Files

After you install and decrypt Windows-based dSPACE software, the documentation for the installed products is available in dSPACE Help and as PDF files.

dSPACE Help (local) You can open your local installation of dSPACE Help:

- On its home page via Windows Start Menu
- On specific content using context-sensitive help via **F1**

PDF files You can access PDF files via the  icon in dSPACE Help. The PDF opens on the first page.

dSPACE Help (Web) Independently of the software installation, you can access the Web version of dSPACE Help at <https://www.dspace.com/go/help>.

To access the Web version, you must have a *mydSPACE* account.

For more information on the mydSPACE registration process, refer to www.dspace.com/faq?097.

Basics and Instructions

Where to go from here

Information in this section

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The introduction gives you basic information on the Scenario Editor.	
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Provides basic information on the specification of a scenario.	
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A maneuver specifies the movement of the ASM vehicle.	
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You can use ModelDesk to start, stop, or reset maneuvers for vehicle dynamics test or driving cycles for engine tests.	

Introduction to the Scenario Editor

Introduction The introduction gives you basic information on the Scenario Editor.

Where to go from here

Information in this section

[Basics of the Scenario Editor..... 20](#)

With the Scenario Editor, you can create maneuvers and traffic scenarios for controlling the movement of the ASM vehicle and simulating traffic situations such as vehicles overtaking and changing lanes.

[Graphical User Interface of the Scenario Editor..... 22](#)

Describes all the elements of the Scenario Editor.

[Filtering, Collapsing, or Expanding Blocks in the Scenario Editor..... 24](#)

To focus on the blocks you are working on, you can reduce the number of visible blocks and collapse other blocks.

Basics of the Scenario Editor

Introduction With the Scenario Editor, you can create maneuvers and traffic scenarios for controlling the movement of the ASM vehicle and simulating traffic situations such as vehicles overtaking and changing lanes.

Features

When you create a scenario with the Scenario Editor, you can:

- Specify the movement of the ASM vehicle.
- Specify the movements of the traffic participants, such as fellow vehicles or pedestrians
- Specify the road the scenario is linked to
- Specify the transition from one activity to the next
- Specify user signals for the maneuver, fellows, or globally to communicate with the simulation model
- Download the scenario parameters to Simulink or generate a parameter file to be loaded to the simulation platform via an experiment software

Definitions

Scenario A scenario contains a maneuver and an optional traffic scenario. Depending on your test scenario, you can define only the movement of the ASM vehicle (maneuver) or additionally the movement of traffic objects around the ASM vehicle (traffic scenario).

Maneuver A maneuver defines the movement of the ASM vehicle. A maneuver can consist of several sequences that are executed one after the other.

You can specify specific start properties, for each sequence. Each sequence has a start segment to specify the maneuver start. After a start segment, several segments can be specified. Each segment consists of an activity and a transition. In an activity, one kind of movement is specified. A transition contains one or more conditions that stops the activity.

Traffic A traffic defines the movements of objects (fellows) around the ASM vehicle. Traffics consist of several segments.

ASM vehicle The ASM vehicle is the initial vehicle in a scenario. Its movement is specified by the maneuver.

Fellow A fellow vehicle (fellow) is a vehicle defined in the traffic. Its movement is defined by specifying one or more sequences for it. To visualize the fellow, a suitable traffic object can be selected.

User signal User signals can be used to communicate with the maneuver scheduler in the simulation model by triggering user-defined actions. You can specify user signals globally in the scenario (global user signals) or as part of the maneuver or a fellow definition (maneuver user signal or fellow user signal). Global user signals are specified in the same way as fellow definitions.

Activity An activity describes the behavior of the maneuver, a fellow, or global user signal. The behavior of the maneuver and a fellow can be described in longitudinal and lateral direction. This means that you can specify the acceleration of a fellow or a lane change. For global user signals, only a value can be specified.

The activity can be defined with absolute values or relative to the ASM vehicle or other traffic participants.

Transition A transition contains one or more conditions from that one must be met to end the current activity of the maneuver, a fellow, or global user signal and start the next one.

Segment A segment is the combination of an activity and a transition. You should use the first segment for initializing the maneuver, fellow, or global user signal.

Sequence A sequence is a chain of segments. You can specify several segments, each containing an activity and a transition. The segments are connected in a chain so that the ASM vehicle and fellows can make different kinds of movements or the global user signals can have different values in the scenario. The first segment of a sequence is used for initializing the maneuver or fellow.

Start segment In the start segment of the maneuver or a fellow sequence you can specify the route and direction of travel.

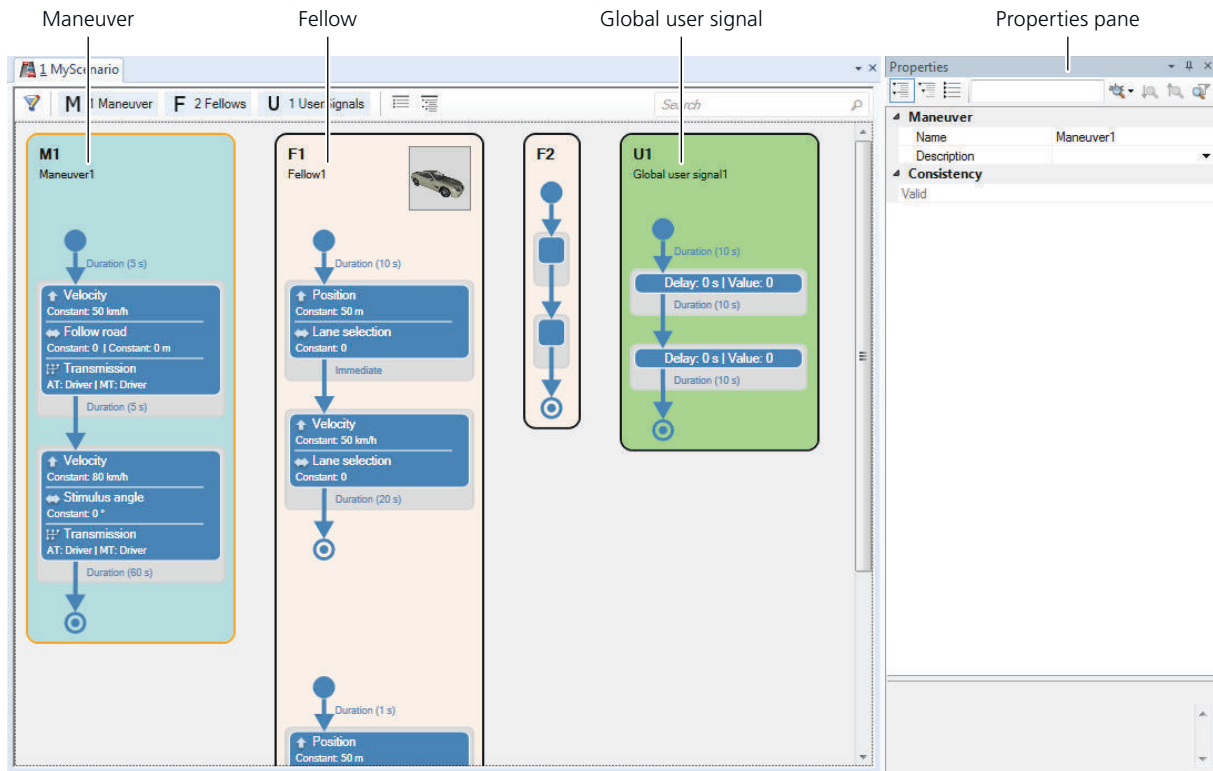
Example

For an example of traffic scenario and its modeling, refer to [Example of a Scenario \(ASM Traffic Reference !\[\]\(0fb13ad0bfa3d86868cdd3883e5665b3_img.jpg\)](#)).

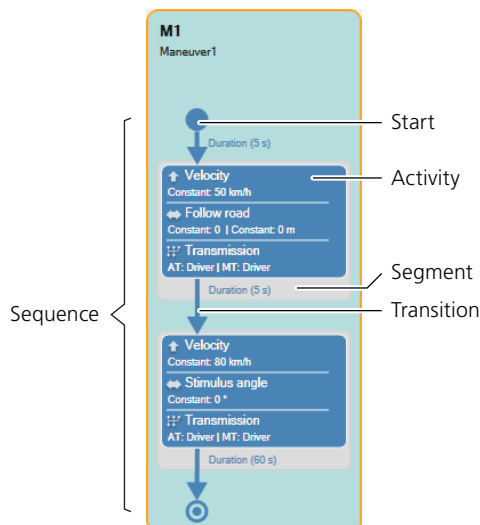
Graphical User Interface of the Scenario Editor

Overview of the graphical user interface

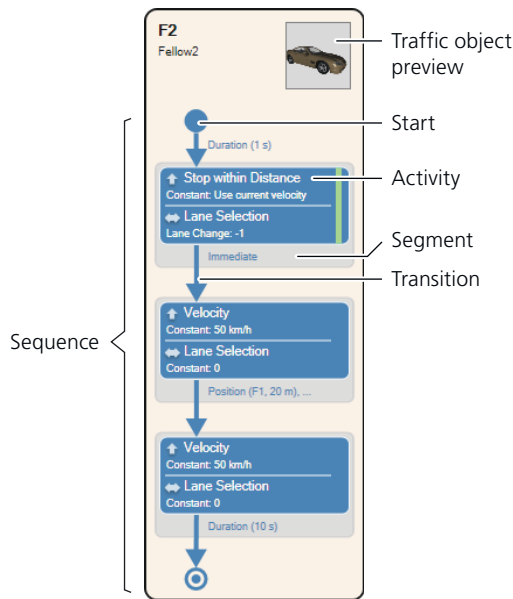
The Scenario Editor is integrated in ModelDesk's main window.



Maneuver block The Maneuver block is the graphical representation of sequences of the maneuver of the ASM vehicle. You can click the contained elements, such as activity or transition, and specify their properties in the Properties pane.



Fellow block The fellow block is the graphical representation of sequences of a fellow. You can click the contained elements, such as activity or transition, and specify their properties in the Properties pane.



Global user signal block The global user signal block is the graphical representation of a global user signal. You can click the contained elements, such as activity or transition, and specify their properties in the Properties pane. You can also click an element's name to modify it.

Context menu The Scenario Editor has a context menu on the workspace. The context menu contains various commands for editing the maneuver, fellow definition, and global user signal. The commands that are available depend on the selected element.

Properties pane Displays and lets you specify properties for different elements. The properties are grouped in categories. The contents of the pane depend on the pane and element that is currently active. The Properties pane can be hidden or shown by using the Switch Controlbars command in the View - Controlbar ribbon group.

Related topics

Basics

[Filtering, Collapsing, or Expanding Blocks in the Scenario Editor](#)..... 24

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[Layout Diagram](#)..... 107

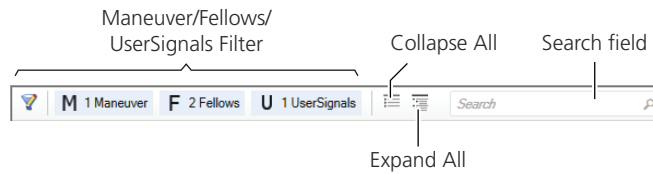
Filtering, Collapsing, or Expanding Blocks in the Scenario Editor

Introduction

To focus on the blocks you are working on, you can reduce the number of visible blocks and collapse other blocks.

Filter and collapse/expand

The Scenario Editor has some commands that you can use to filter or collapse/expand the maneuver, fellow and global user signal blocks.



Maneuver/Fellows/UserSignals Filter Use the Maneuver, Fellows, and Global User Signals Filter to hide or display all the blocks of this type. When a filter is active, the Scenario Editor displays the number of visible blocks and the total number of blocks.

Collapse All/Expand All buttons Use the Collapse All button to collapse all the blocks.

Use the Expand All button to expand all the blocks.

Tip

You can also double-click a block to collapse/expand this block individually.

Search field Use the search field to filter all the Maneuver, Fellow, and Global User Signal blocks. When you enter a string, the Scenario Editor displays only the blocks whose names or values contain the search string.

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Basics of Scenarios

Introduction

Scenarios define the movement of the ASM vehicle and the movements of fellows.

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Scenarios define the movement of the ASM vehicle and the movements of fellows relative to the ASM vehicle or to the road.	
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A fellow is a vehicle or a pedestrian whose movements are defined in a scenario.	
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You can specify global user signals independently of the maneuver and fellow definitions.	
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An activity specifies the movement of the maneuver or a fellow or the value of a global user signal.	
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Transitions specify the conditions that end one activity and start the following activity.	
Example of Scenarios with Roads.....	36
A scenario can be combined with a road. Only if it combined with the correct road, the simulation results make sense.	

Scenarios

Introduction

Scenarios define the movement of the ASM vehicle and the movements of fellows relative to the ASM vehicle or to the road. You can simulate traffic situations such as vehicles overtaking and changing lanes.

Connection to road

A scenario can be connected to a road network. So it can use all the specification of the road, for example, the route that can be driven by the ASM vehicle and the fellows or the lanes that can be selected.

If you want to specify a maneuver only, i.e., the stimulation and movements of the ASM vehicle, connecting to a road network is optional. If you want to

specify the movements of fellows, connecting to a road network is mandatory. In addition, you must assign routes to the fellows. Therefore, routes must be specified for the road network.

Use the Road Generator to specify a road network for the scenario, refer to [Introduction to the Road Generator \(ModelDesk Road Creation !\[\]\(3dfb8d66e81160ad61421a3452093d1b_img.jpg\)](#)).

It is also possible to specify a maneuver without connection to a road.

Traffic scenario components

A scenario consists of a maneuver, fellows and global user signals:

- A maneuver defines the movement of the ASM vehicle. You can specify only one maneuver in a scenario. For details, refer to [Basics of Maneuvers](#) on page 26.
- A fellow defines the movement of a traffic participant. Fellows are optional. You can specify any number of fellows in a scenario. For details, refer to [Basics of Fellows](#) on page 28.
- A global user signal defines a signal that can be read in the simulation model and is independent of the maneuver and fellow definition. Global user signals are optional. You can specify any number of global user signals in a scenario. For details, refer to [Basics of Global User Signals](#) on page 29.

A segment describes the maneuver, a fellow or the value of a global user signal. A segment has an activity and a transition. The activity specifies the kind of movement or value. The transition specifies when the current activity ends and the next activity starts. Refer to:

- [Basics of Activities](#) on page 30
- [Basics of Transitions](#) on page 34

Related topics

Basics

[Basics of the Scenario Editor](#)..... 20

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Basics of Maneuvers

Introduction

In scenarios, you can simulate the movement of the ASM vehicle. This is defined in the maneuver.

Maneuver

Maneuvers are predefined sequences of driving instructions for simulating a variety of driving situations with the ASM vehicle. Typical maneuvers are braking on a road with different friction on two lanes (μ -split maneuver), performing a

double lane change, and step steering. The maneuver subsystem of the ASM Vehicle Dynamics Simulation Package controls the ASM vehicle, road, and driver models to perform maneuvers.

Maneuver types

There are stimulus maneuvers and controlled maneuvers. The two maneuver types are usually combined in the maneuver segments.

Stimulus maneuvers In stimulus maneuvers, the vehicle model is controlled by stimulus signals for accelerator pedal, brake pedal, clutch pedal, gear, and steering wheel. The Scenario Editor makes it easy to enter individual stimulus signals. In addition, you can import stimulus signals from measured data (MAT file).

Controlled maneuvers In controlled maneuvers, the driver model controls the vehicle model. Accelerator, brake and clutch pedal positions, and gear are controlled by the driver model's longitudinal controller. The steering wheel is controlled by the lateral controller. For more information on the driver model, refer to [Driver \(ASM Environment Reference\)](#).

Specifying maneuver segments

The movement of the ASM vehicle is described in sequences that are subdivided into segments arranged in a chain. A segment contains an activity and a transition. An activity specifies the movement of the ASM vehicle in longitudinal and lateral direction and the clutch and gear parameters. A transition specifies when an activity ends and the following activity starts. A transition can have one or more conditions.

For details, refer to [Basics of Activities](#) on page 30 and [Basics of Transitions](#) on page 34.

Specifying initial values

The first segment of a maneuver is used to specify the initial values for the ASM vehicle. You can specify the initial position, initial velocity, and initial vehicle height in this segment. In the corresponding transition, specify a small duration for transient effects, for example, 2 seconds.

When the ASM vehicle is to follow a route on the associated road, so you must specify the route on the start segment of each sequence. In addition, you can specify the direction of travel.

Maneuver user signals

To communicate with other elements of the simulation model, user signals are used. Some user signals can be selected in the activity of a maneuver. These are the maneuver user signals.

You can select maneuver user signal and assign a source type to it. The Scenario Editor provides four kind of source types: Constant, ramp, pulse, and table. The selected source type will be set when the corresponding activity is active. It is possible to specify a delay value so that the source type will be set after a specified time.

If a maneuver user signal is specified, the maneuver block on the Scenario pane is marked with a green bar.

Related topics

Basics

[Workflow for Maneuver Creation](#)..... 38

Basics of Fellows

Introduction

In scenarios, you can simulate the movements of fellows (for example, fellow vehicles or pedestrians) in the environment of the ASM vehicle. The movement of the ASM vehicle is defined in the maneuver sequences. The movements of the fellows are defined in fellow sequences.

Fellow

A definition is the specification of a traffic participant in the environment of an ASM vehicle. Traffic participants can be any kind of object. They can be other vehicles or pedestrians, depending on what traffic scenario is to be simulated.

You can specify the movements of the fellows in absolute values or relative to the position of the ASM vehicle or other fellows.

Geometry

The geometry for the visualization of a fellow is specified by traffic objects. The traffic objects also define the settings for object detection sensors.

Traffic objects can have various geometries. They can be based on vehicles so you can use them for simulating driving situations such as overtaking or parking. Traffic objects can also be based on humans or animals so you can simulated traffic scenarios such as pedestrian crossing.

Note

Do not use grouped objects for fellows. The scene generation fails if fellows have a geometry of grouped objects.

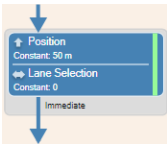
Refer to [Basics of Traffic Objects \(ModelDesk Traffic Object Management !\[\]\(248b91fcdac4810ffd15cf33fb6aec6f_img.jpg\)](#)).

Specifying movements

You can specify the movements of each fellow individually. The movements of a fellow are described in sequences that are subdivided into segments arranged in a chain. A segment contains an activity and a transition. An activity specifies the movement of a fellow in longitudinal and lateral direction. A transition specifies when an activity ends and the following activity starts. A transition can have one or more conditions.

Refer to [Basics of Activities](#) on page 30 and [Basics of Transitions](#) on page 34.

Specifying initial values	The first segment of a fellow definition is used to specify initial position for the fellow. You can specify the initial longitudinal and lateral position in this segment. In the corresponding transition, you can specify the Immediate condition so that the segment sets only the initial value and has no duration. A fellow is to follow a route on the associated road, so you must specify the route on the start segment of each sequence. In addition, you can specify the direction of travel.
Fellow user signals	To communicate with other elements of the simulation model, user signals are used. Some user signals can be selected in the activity of a fellow. These are the fellow user signals. You can select fellow user signal and assign a source type to it. The Scenario Editor provides four kind of source types: Constant, ramp, pulse, and table. The selected source type will be set when the corresponding activity is active. It is possible to specify a delay value so that the source type will be set after a specified time. If a fellow user signal is specified, the fellow block on the Traffic Scenario pane is marked with a green bar, see the following illustration.



Related topics	Basics
	Workflow for Traffic Creation..... 56

Basics of Global User Signals

Introduction	You can specify global user signals independently of the maneuver and fellow definitions.
Global user signals	Global user signals are signals that you can specify independently of the maneuver and fellow definitions. The value of global user signals can change during the execution of the traffic scenario. You can specify different values and the conditions that must be fulfilled to switch from one value to another.
Specifying global user signals	To specify the global user signals, the same mechanism as for maneuver and fellow definitions is used. The global user signals are described in sequences,

which are subdivided in segments arranged in a chain. A segment contains an activity and a transition. An activity specifies the source type of the global user signal. The Scenario Editor provides four kind of source types: Constant, ramp, pulse, and table. The selected source type will be set when the corresponding activity is active. A transition specifies when an activity ends and the following activity, which can have another source type for the global user signal, starts. A transition can have one or more conditions.

Refer to [Basics of Activities](#) on page 30 and [Basics of Transitions](#) on page 34.

Related topics

HowTos

[How to Specify a Global User Signal.....](#) 64

Basics of Activities

Introduction

An activity specifies the movement of the maneuver or a fellow or the value of a global user signal.

Activity for a maneuver

The activity of the maneuver is used to specify its movement. You can specify a longitudinal and lateral profile. The longitudinal profile describes the movement in the direction of travel, such as acceleration or constant velocity. The lateral profile describes the movement at an angle to the direction of travel, such as lane changing. In addition, you can specify the clutch and gear settings of an ASM vehicle with manual transmission or the selector lever settings of an ASM vehicle with automatic transmission.

The settings are used until the transition of the corresponding segment ends the activity and starts the next activity.

Longitudinal profiles You can define different types of stimulation of the ASM vehicle in the longitudinal direction, see the following table.

Longitudinal Type	Description
Velocity	The velocity of the ASM vehicle is specified in absolute values or relative to a fellow. The velocity is constant (as specified by the value in the segment or the value of the previous segment) or changes (as specified by a ramp, a final value, a sine function, a table, or an external value).
Stop	The ASM vehicle brakes until standstill within a specified distance or brakes until it stops. If it brakes until standstill, you can specify a velocity at the beginning of the segment or keep the velocity from the previous segment.

Longitudinal Type	Description
Pedal stimulus	The position of the acceleration and brake pedals are specified. You can specify the position of the pedals using several source types: - Specified values (constant, ramp, table) - Manually by the experiment software - Externally by the simulation model
Standstill	The ASM vehicle stands still in this segment.
Lateral acceleration	All the longitudinal definitions of the current segment are ignored. The longitudinal velocity depends on the lateral acceleration. This type can be used only if the lateral type is Basic road and the steering type is Circle .

The parameters to be specified depend on the selected longitudinal type.

Lateral profiles You can define different types of stimulation of the ASM vehicle in the lateral direction, see the following table.

Lateral Type	Description
Not used	The lateral profile is not defined, for example, if the ASM vehicle stands.
Stimulus angle	The lateral profile is specified by an angle of the steering. ModelDesk provides several source types for setting the angle: - Specified values (constant, ramp, sine, fixed, step, pulse, sine with dwell, table) - Manually by the experiment software - Externally by the simulation model
Stimulus torque	The lateral profile is specified by a torque at the steering. ModelDesk provides several source types for setting the torque: - Specified values (constant, ramp, sine, fixed, step, pulse, sine with dwell, table) - Manually by the experiment software - Externally by the simulation model
Basic road	The ASM vehicle is driven in a stationary driving situations such as driving straight forward or in a circle at a given lateral acceleration. This type can be used only in scenarios if no road is referenced. For this lateral profile, you should use the <code>Lateral_Control2</code> subsystem in the simulation model. Refer to Scenario Properties Dialog on page 93 and Lateral Controller 2 (ASM Environment Reference) .
Follow road	The ASM vehicle follows the road on a selected route. For this lateral profile, you should use the <code>Lateral_Control1</code> subsystem in the simulation model. Refer to Scenario Properties Dialog on page 93 and Lateral Controller 1 (ASM Environment Reference) .

The parameters to be specified depend on the selected lateral type.

Selector lever You can define different types of stimulation of the selector lever of the ASM vehicle with automatic transmission, see the following table.

Selector Lever Type	Description
Driver	The selector lever is specified by the driver model.
Stimulus	The selector lever is specified using source types. ModelDesk provides several source types for setting the selector lever: - Specified values (constant or table) - Manually by the experiment software - Externally by the simulation model

Clutch and gear You can define different types of stimulation of the clutch and gear of the ASM vehicle with manual transmission, see the following table.

Clutch and Gear Type	Description
Open clutch	The clutch is open.
Driver	The clutch and the gear is specified by the driver model.
Stimulus	The clutch and the gear is specified using different source types. ModelDesk provides several source types for setting the clutch and gear: - Specified values (constant, ramp, or table) - Manually by the experiment software - Externally by the simulation model
Reference gear	The gear is specified using different source types. ModelDesk provides several source types for setting the gear: - Specified values (constant or table) - Manually by the experiment software - Externally by the simulation model

Activity for a fellow definition

The activity of a fellow is used to specify their movement. You can specify a longitudinal and lateral profile. The longitudinal profile describes the movement in the direction of travel, such as acceleration or constant velocity. The lateral profile describes the movement at an angle to the direction of travel, such as lane changing.

The profiles are used until the transition of the corresponding segment ends the activity and starts the next activity.

Longitudinal profiles You can define several types of movement of a fellow in the longitudinal direction, see the following table.

Longitudinal Type	Description
Not used	The fellow is not used in this segment.
Continue	The fellow continues the last valid segment definition.

Longitudinal Type	Description
Acceleration	The fellow accelerates in longitudinal direction. The acceleration is constant or changes (as specified by a ramp or a table).
Distance [m]	The longitudinal position is specified by the distance of the fellow to the ASM vehicle or another fellow. You can specify where the reference points of the objects are (main point, front, rear, left, right). The distance is constant or changes (as specified by a ramp, a final value, an external value, or a table).
Distance [s]	The longitudinal position is specified by the time gap of the fellow to the ASM vehicle or another fellow. The time gap is constant or changes (as specified by a ramp, a final value, an external value, or a table).
Position	The longitudinal position is specified by the distance of the fellow to the road start position. The distance is constant (a specified value in the segment or the value of the previous segment) or changes (as specified by a ramp, an external value, or a table).
Velocity	The velocity of the fellow is specified in absolute values or relative to the ASM vehicle or another fellow. The velocity is constant (as specified by the value in the segment or the value of the previous segment) or changes (as specified by a ramp, a final value, a sine function, an external value, or a table).
Stop	The fellow brakes until standstill within a specified distance. You can specify a velocity at the beginning of the segment or keep the velocity from the previous segment.

The parameters to be specified depend on the selected longitudinal type.

Lateral profiles You can define several types of movement of a fellow in the lateral direction, see the following table.

Lateral Type	Description
Not used	The fellow is not used in this segment.
Continue	The fellow continues the last valid segment definition.
Lateral deviation	The fellow's lateral position is defined as a fixed distance to the road's reference line by absolute values or relative to the ASM vehicle or another fellow. The lateral position is constant (a specified value in the segment or the value of the previous segment) or changes (as specified by a ramp, a smooth lane change, a final value, an external value, or a table).
Lane selection	The fellow's lateral position is defined relative to the lanes of the road or relative to the ASM vehicle or another fellow. The lateral position is constant (a specified lane index or the value of the previous segment) or changes (a lane change, a smooth lane change, a final value, an external value, or

Lateral Type	Description
	a table). You can enable a force-to-road option that ensures that the fellow drives on the road even if the specified lane is outside the road. If the lane index is 0, the preferred lane of the direction of travel is used.

The parameters to be specified depend on the selected lateral type.

Activity for a global user signal

The activity of a global user signal is used to specify a value. You can select one of four source types that calculates the value. So the value is constant or changes during the activity.

This source type is valid until the transition of the corresponding segment ends the activity and starts the next activity.

Source type You can use four different source types, see the following table.

Source Type	Description
Constant	The value of the signal is constant.
Ramp	The value of the signal changes with a ramp function.
Pulse	The value of the signal changes with a pulse function.
Table	The value of the signal is read from a table. The values of the table can be imported from a MAT file.

Related topics

References

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Basics of Transitions

Introduction

Transitions specify the conditions that end one activity and start the following activity.

Basics

A segment contains a transition and an activity. You can specify one or more conditions for the transition that must be met to end the segment. The first condition to be fulfilled ends the segment.

Conditions

A transition can have various kinds of conditions. The transition to the following segment can be delayed. The following conditions for segment transitions are available.

Duration The segment ends after a specified time.

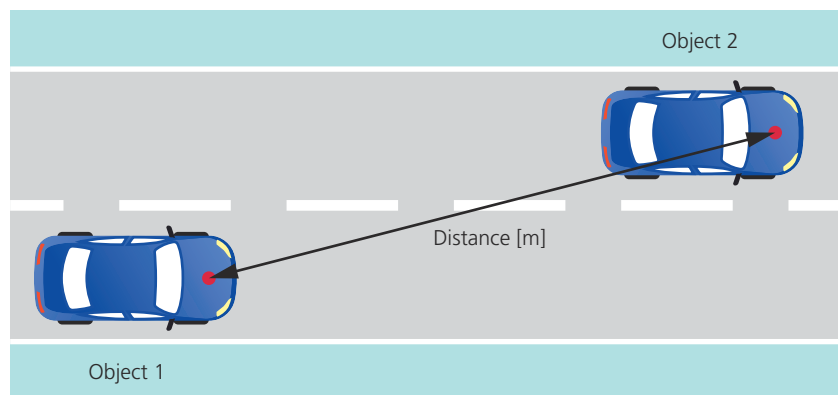
Endless The segment never ends.

Immediate The segment ends immediately. You can use this condition to use a segment for initialization.

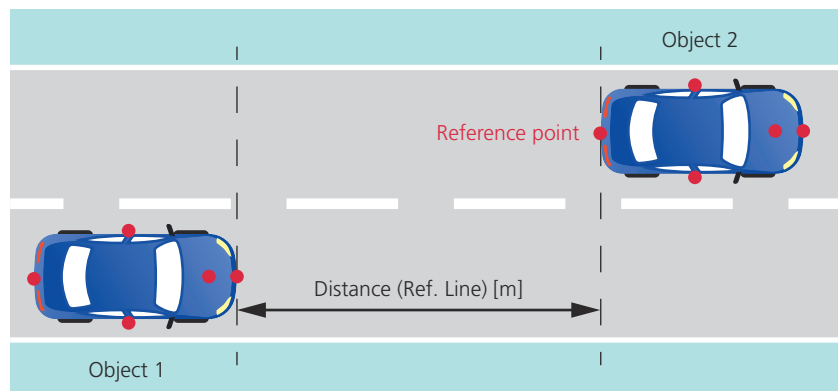
Position The segment ends when the ASM vehicle or fellow reaches a specified position.

Driven Distance The segment ends when the ASM vehicle or fellow traveled the specified distance.

Distance (between main points) The segment ends when the distance between the main points of two objects (ASM vehicle, fellow, or position marker) is greater or smaller than a specified value.



Distance (along road reference line) The segment ends when the distance between two objects (ASM vehicle, fellow, or position marker) is greater or smaller than a specified value. The distance is measured between selectable reference points of the objects along the reference line.



Lateral acceleration (Only for maneuvers) The segment ends when the lateral acceleration of the ASM vehicle is greater or smaller than a specified value.

Time Gap (Only for fellows and global user signals) The segment ends when the time gap between two objects (ASM vehicle, fellow, or position marker) is greater or smaller than a specified value.

Velocity The segment ends when the velocity of the ASM vehicle or fellow is greater or smaller than a specified value.

Trigger The segment ends when it is externally triggered.

Related topics

HowTos

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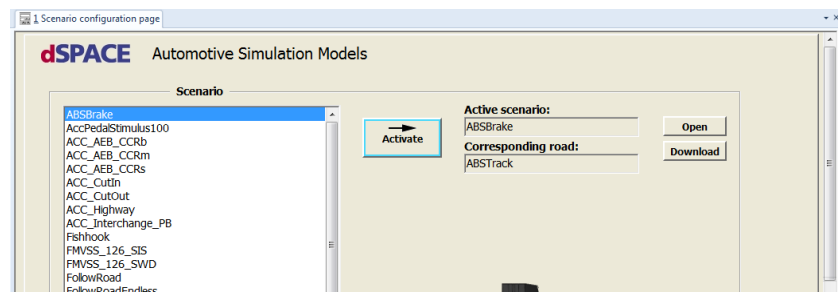
Example of Scenarios with Roads

Introduction

A scenario can be combined with a road. Only if it combined with the correct road, the simulation results make sense.

Combination examples

A scenario usually is related to a road. When you activate a scenario, the related road is displayed on the Scenario configuration page. The following illustration shows the example of the **ABSBrake** scenario that is related to the **ABSTrack** road.



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Creating Maneuvers

Introduction

A maneuver specifies the movement of the ASM vehicle.

Where to go from here

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A maneuver specifies the stimulation of the ASM vehicle. The maneuver can consist of one or more sequences. A sequence consist of one or more segments. A segment consists of one activity and one transition.	
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You can control the vehicle's accelerating and braking behavior by defining stimulus signals for accelerator and brake pedal or by defining a reference velocity profile.	
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You can control the ASM vehicle's gear shifting for manual transmission by specifying clutch and gear, and automatic transmission by specifying the selector lever.	
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For a maneuver on a road network, you can specify road-following in your maneuver. So it is not necessary to define the steering.	
How to Specify a Lane Change Maneuver.....	48
You can specify a lane change maneuver when the ASM vehicle follows a road with multiple lanes.	
How to Steer in a Maneuver.....	49
You can specify one of the stimulus signals provided, choose a predefined steering profile, or fix the steering wheel.	
How to Specify a Transition for a Maneuver.....	51
A transition specifies the condition when an activity ends and the succeeding activity begins.	
How to Specify the Properties of the Driver Model.....	52
The steering wheel, the accelerator and brake pedals, and the clutch and gear are controlled by a driver model, which is specified by a driver type. In ModelDesk, you can specify the parameters for the driver model.	
How to Select and Specify Maneuver User Signals.....	53
You can define signals at any point or time in the maneuver segment for use as user signals in the Simulink model.	

Workflow for Maneuver Creation

Introduction

This overview shows the workflow for creating a maneuver.

Workflow

A maneuver is necessary if you want to specify the movement of the ASM vehicle. To create a maneuver, you must perform certain steps.

1. You use the Scenario Editor to create a new scenario. Refer to [How to Start Scenario Creation](#) on page 39.
2. A maneuver is structured in sequences and segments. Refer to [How to Add Sequences and Segments for the Maneuver](#) on page 41.
3. In an activity of a segment, you specify the signals for stimulating the ASM vehicle.
 - To accelerate or brake, you can stimulate the acceleration pedal and brake pedal. Refer to [How to Accelerate and Brake in a Maneuver](#) on page 43.
 - To shift the gears, you can stimulate signals for automatic and manual transmission systems. Refer to [How to Shift Gears in a Maneuver](#) on page 44.
 - To steer the ASM vehicle, you can stimulate the steering wheel by specifying an angle or torque of the steering wheel. Refer to [How to Steer in a Maneuver](#) on page 49.

When the ASM vehicle is to follow the road, stimulating the steering wheel is not necessary. Refer to [How to Follow a Road Automatically](#) on page 46.
4. In the transition of a segment, you define when an activity ends and the next one starts. Refer to [How to Specify a Transition for a Maneuver](#) on page 51.
5. You can specify the parameters of the driver model that controls the ASM vehicle. Refer to [How to Specify the Properties of the Driver Model](#) on page 52.
6. To communicate with the simulation model, you can use maneuver user signals. Refer to [How to Select and Specify Maneuver User Signals](#) on page 53.
7. If you want to simulate traffic situation using ASM Traffic, you can specify the movements of the traffic participants (fellows). Refer to [Creating Traffic of Scenarios](#) on page 56.
8. If the scenario is complete, you can download the parameters for simulation. Refer to [Using Scenarios](#) on page 68.

Related topics

Basics

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How to Start Scenario Creation

Objective

A maneuver specifies the movement of the ASM vehicle. It is specified with the Scenario Editor.

This instruction shows the following steps:

1. Creating a new scenario.
2. Specifying the global properties of a scenario, for example, selecting a road.
3. Specifying the start properties of a maneuver, for example, route selection, initial vehicle position and velocity.
4. Specifying a name for the scenario.

Preconditions

A project must be open in ModelDesk.

Method

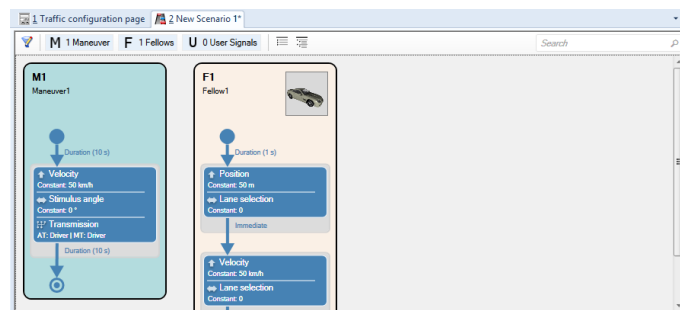
To start scenario creation

- 1 On the Environment ribbon, click Scenario – New.

Or

In the Project Navigator, open the context menu of Scenario and select New.

ModelDesk creates a new scenario and opens the Scenario pane.

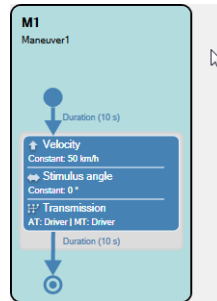


If the ModelDesk experiment is connected to an ASM Traffic Model, the scenario has one Maneuver block and one Fellow block by default. If the ModelDesk experiment is connected to an ASM Vehicle Dynamics Model, the scenario has only the Maneuver block.

You can add more Fellow blocks and Global User Signal blocks. If you want to specify a maneuver only, you can delete the Fellow block.

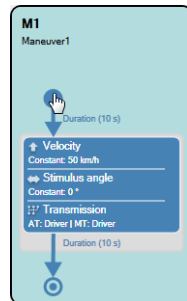
- 2 To delete the Fellow block, select the block, open its context menu and select Delete Fellow.

- 3 To specify the properties of the scenario, click on a free area of the Scenario pane.



The Properties pane displays the scenario properties.

- 4 Specify the properties of the scenario. For details, refer to [Traffic Scenario Properties](#) on page 156.
- 5 To specify the start properties, click the start in the Maneuver block.



The Properties pane displays the start properties.

- 6 Specify the start properties. For details, refer to [Start Properties \(Maneuver\)](#) on page 155.
- 7 To specify the scenario properties, for example, driver properties and controller type, go to the Environment ribbon and click **Scenario – Properties**.
The Scenario Properties dialog opens.
- 8 Open the Maneuver page, and specify the properties . For details, refer to [Scenario Properties Dialog](#) on page 93.
- 9 The scenario name is specified when you save it:
In the Project Navigator, open the context menu of Scenario and select **Save As**.
The Specify File Name opens.
- 10 Specify a file name and click **Save**.

Result

A new scenario is created and the name is specified.

The scenario can be edited, but it is not linked to the current experiment. If you want to link it, use the **Activate** command.

Next Step

You can specify the maneuver:

- [How to Accelerate and Brake in a Maneuver](#) on page 43
- [How to Shift Gears in a Maneuver](#) on page 44
- [How to Follow a Road Automatically](#) on page 46
- [How to Steer in a Maneuver](#) on page 49
- [How to Select and Specify Maneuver User Signals](#) on page 53

You can specify the traffic of the scenario, refer to [Workflow for Traffic Creation](#) on page 56.

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How to Add Sequences and Segments for the Maneuver

Objective

A maneuver specifies the stimulation of the ASM vehicle. The maneuver can consist of one or more sequences. A sequence consist of one or more segments. A segment consists of one activity and one transition.

Preconditions

A scenario with a maneuver block must be created and open in the Scenario pane, refer to [How to Start Scenario Creation](#) on page 39.

Segment and sequence

Segment To specify several activities of a maneuver that are executed one after the other, you use multiple segments.

You can append a after the last segment or insert a new segment between existing segments. Refer to [Part 1](#) on page 42.

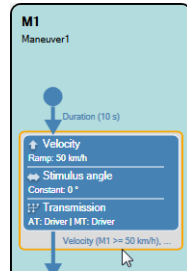
Sequence Using additional sequences, you can specify different initial values. So you can specify different vehicle positions and initial velocities, for example. You can also assign each sequence to a specific route. However, you can only select one road network for the active scenario.

You can only append a sequence after the last existing sequence. Refer to [Part 2](#) on page 42.

Part 1

To add a segment to the maneuver

- 1 Open the context menu of an existing sequence.

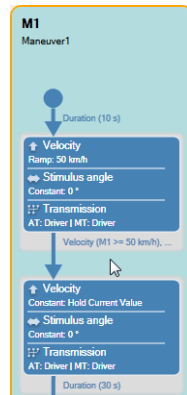


- 2 Select one of the following commands:
 - Append Segment
 - Insert Segment – Before
 - Insert Segment – After
 A segment is inserted.

Part 2

To add a sequence to the maneuver

- 1 Open the context menu of the maneuver block.



- 2 Select Append Sequence.

Result

A sequence and/or segment is appended.

You can specify the behavior in the activities of the maneuver, refer to

- [How to Accelerate and Brake in a Maneuver](#) on page 43
- [How to Shift Gears in a Maneuver](#) on page 44
- [How to Follow a Road Automatically](#) on page 46
- [How to Steer in a Maneuver](#) on page 49
- [How to Select and Specify Maneuver User Signals](#) on page 53

You can specify the transition, refer to [How to Specify a Transition for a Maneuver](#) on page 51

Related topics**Basics**

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How to Accelerate and Brake in a Maneuver

Objective

You can control the vehicle's accelerating and braking behavior by defining stimulus signals for accelerator and brake pedal or by defining a reference velocity profile.

Basics of acceleration and braking

Acceleration and braking Acceleration and braking is specified using the longitudinal properties of a maneuver's activity.

- You can control the accelerator and brake pedal by individual stimulus signals. ModelDesk provides simplified profiles (constant, ramp, table). Profiles specified in a MAT file can be imported via a table. It is also possible to stimulate the pedals using the Simulink model or via an experiment software.
- You can specify a reference velocity. ModelDesk provides several profiles (constant, ramp, sine, table). Profiles specified in a MAT file can be imported via a table. It is also possible to set the reference velocity using the Simulink model or via an experiment software.
- You can specify a simplified brake profile.
- You can specify a standstill.

Hold current value You can specify whether the position of the acceleration pedal or brake pedal reached in the previous maneuver segment is held for the current maneuver segment or not by enabling the **Hold current value** option (available only for Constant pedal source). If enabled, you cannot specify values for the current maneuver segment.

Start with current value You can specify whether the position of the acceleration pedal or brake pedal reached in the previous maneuver segment is used as the start value or not by enabling the **Start with current value** option (available only for Ramp pedal source). If enabled, you cannot specify a start value for the current maneuver segment.

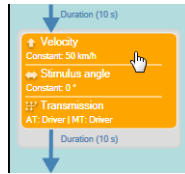
Preconditions

A scenario with a Maneuver block must be created and the Scenario pane must be open, refer to [How to Start Scenario Creation](#) on page 39.

Method

To accelerate and brake in a maneuver

- 1 Select the activity of the maneuver which longitudinal properties you want to specify.



The Properties pane displays the properties of the selected activity. For specifying the acceleration and braking of the ASM vehicle, you must specify the properties of the Longitudinal category. For details, refer to [Longitudinal Type Properties \(Maneuver\)](#) on page 143.

- 2 In the Longitudinal category, select a longitudinal type.
Now, the properties are displayed in the category that are required by the selected type.
- 3 Specify the properties of the selected type.

Result

You specified the acceleration and brake of the ASM vehicle.

Related topics

HowTos

[How to Shift Gears in a Maneuver](#)..... 44

References

[Longitudinal Type Properties \(Maneuver\)](#)..... 143
[Maneuver Scheduler \(ASM Environment Reference !\[\]\(626ce8ac21792b9405bfddfea8e0c96a_img.jpg\)\)](#)

How to Shift Gears in a Maneuver

Objective

You can control the ASM vehicle's gear shifting for manual transmission by specifying clutch and gear, and automatic transmission by specifying the selector lever.

Basics of shifting gears

For manual transmission, the clutch and gear are specified in the Clutch and gear category of a maneuver's activity. For automatic transmission, the selector lever is specified in the Selector lever category of a maneuver's activity.

- You can specify the clutch, gear, and selector lever individually by stimulus signals. ModelDesk provides simplified profiles (constant, table, ramp (only for the clutch)). Profiles specified in a MAT file can be imported via a table. It is

also possible to stimulate the clutch and gear using the Simulink model or via an experiment software.

- You can also let the driver model of ASM control the gear shifting by selecting the driver mode in the pedal stimulus or desired velocity profile. If you select the final velocity or brake profile (with open clutch disabled) for the longitudinal type, the driver model controls the gear shifting automatically.
- You can specify the gear individually by stimulus signals. ModelDesk provides simplified profiles (constant and table). Profiles specified in a MAT file can be imported via a table. It is also possible to stimulate the gear using the Simulink model or via an experiment software.
- You can open the clutch.

Hold current value You can specify whether the clutch position or gear reached in the previous maneuver segment is held for the current maneuver segment or not by enabling the Hold current value option (only available for Constant clutch pedal or gear source). If enabled, you cannot specify values for the current maneuver segment.

Start with current value You can specify whether the clutch position reached in the previous maneuver segment is used as the start value or not by enabling the Start with current value option (available only for Ramp clutch pedal source). If enabled, you cannot specify a start value for the current maneuver segment.

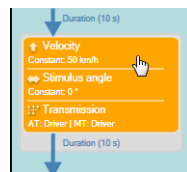
Preconditions

A scenario with a Maneuver block must be created and the Scenario pane must be open, refer to [How to Start Scenario Creation](#) on page 39.

Method

To shift gears in a maneuver

- 1 Select the activity of the maneuver which gear shifting properties you want to specify.



The Properties pane displays the properties of the selected activity. For specifying the acceleration and braking of the ASM vehicle, you must specify the properties of the Clutch and gear category (manual transmission) or Selector lever category (automatic transmission). For details, refer to [Clutch and Gear \(MT\) Properties](#) on page 117 or [Selector Lever \(AT\) Properties](#) on page 151.

- 2 In the Clutch and gear or Selector lever category, select the Clutch and gear type or Selector lever type.
The properties that are displayed in the category depends on the selected type.
- 3 Specify the properties of the selected type.

Result	You specified the gear shifting.
---------------	----------------------------------

Related topics**References**

Clutch and Gear (MT) Properties.....	117
Lateral Controller 1 (ASM Environment Reference )	
Lateral Controller 2 (ASM Environment Reference )	
Selector Lever (AT) Properties.....	151

How to Follow a Road Automatically

Objective	For a maneuver on a road network, you can specify road-following in your maneuver. So it is not necessary to define the steering.
------------------	---

Basics of following the road automatically

If the maneuver follows the road automatically, you do not have to define any steering behavior for the vehicle model, because the driver model controls the steering wheel. You can also adapt the velocity to the road by selecting Adapt velocity to road as driver type in the Lateral category. Then the driver model will drive the vehicle using preview information for road-following and speed adjustment. For information on specifying the global driver settings, refer to [How to Specify the Properties of the Driver Model](#) on page 52.

Lane change maneuver

A road can have several lanes that the ASM vehicle can use for driving. If the ASM vehicle drives on lane 0, it uses the preferred lane in the specified direction of travel. You can change the lane within an activity when the maneuver follows a route on a road network.

When you select the Force to road property, you can specify that the ASM vehicle stays on the road even if the specified final lane index is outside the road.

Selecting a route

If a road network consists of several road elements connected by junction elements, a route defines the sequence of road elements that a vehicle can use. You can specify the routes that can be used on a road network in the Road Generator. For details, refer to [How to Specify Routes on Road Networks \(ModelDesk Road Creation !\[\]\(104fbf564e2e5a8fbd84f31656d114c7_img.jpg\)\)](#).

If your maneuver is to follow a route, you must select the route in the Start segment of the maneuver. Refer to [How to Start Scenario Creation](#) on page 39.

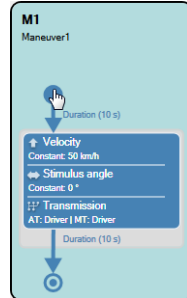
Preconditions

A scenario with a Maneuver block must be created and the Scenario pane must be open, refer to [How to Start Scenario Creation](#) on page 39.

Method

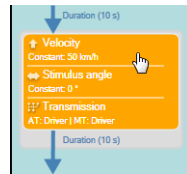
To follow a road automatically

- 1 Select the Start of the maneuver.



The Properties pane displays the properties of the Start activity.

- 2 In the Start category, select the route for the ASM vehicle.
- 3 Select the activity of the maneuver.



The Properties pane displays the properties of the selected activity.

- 4 In the Lateral type of the Lateral category, select **Follow road**.
- 5 Specify the other properties of the Lateral category. For details, refer to [Lateral Type Properties \(Maneuver\)](#) on page 127.

Result

The ASM vehicle drives along the related route.

Next steps

To specify a lane change maneuver, refer to [How to Specify a Lane Change Maneuver](#) on page 48.

Related topics

HowTos

[How to Steer in a Maneuver](#)..... 49

References

[Lateral Type Properties \(Maneuver\)](#)..... 127
[Start Properties \(Maneuver\)](#)..... 155

How to Specify a Lane Change Maneuver

Introduction	You can specify a lane change maneuver when the ASM vehicle follows a road with multiple lanes.
Basics of lane change	When an activity of a maneuver has the Follow road lateral type, the ASM vehicle drives on the road while being controlled by a controller model. It is therefore not necessary to specify the steering via the properties. However, the ASM vehicle can change lane, for example, to simulate overtaking.
Possible methods	In the Scenario Editor, you can specify the index of the lane to which the ASM vehicle changes in three ways: ▪ Lane index in relation to the center lane A lane section of a road element has always one center lane with the lane index 0. The lanes right to the center lane in the direction of travel have negative indices (-1, -2, ...). The lanes left to the center lane have positive indices (1, 2, ...). The lane index can be one of these lane indices. Refer to Method 1 on page 48. ▪ Lane index in relation to preferred lane In the lane section of a road element, one lane is marked as the preferred lane for the direct direction. You can specify a lane index in relation to this lane. Specifying positive numbers steers the ASM vehicle to a lane to the left in the direction of travel. Refer to Method 2 on page 49. ▪ Lane index in relation to the ASM vehicle You can specify a lane relative to the lane that is used by the ASM vehicle. Specifying positive numbers steers the ASM vehicle to a lane to the left in the direction of travel. Refer to Method 3 on page 49.
Preconditions	The maneuver has an activity with a Follow road lateral type. Refer to How to Follow a Road Automatically on page 46.
Method 1	To specify a lane change maneuver with a lane in relation to the center lane 1 In the Environment ribbon, click Scenario – Properties to open the Scenario Properties dialog. 2 On the Global Properties page of the dialog, select Relative to center lane (absolute lane index). 3 Click the activity block of the maneuver. The Properties pane displays the properties of the maneuver's activity. 4 In the Lateral category, set Reference type to Relative to lane. 5 In the Parameters category, specify the lane index.

Method 2**To specify a lane change maneuver with a lane in relation to the preferred lane**

- 1** In the Environment ribbon, click Scenario – Properties to open the Scenario Properties dialog.
- 2** On the Global Properties page of the dialog, select Relative to preferred lane and driving direction.
- 3** Click the activity block of the maneuver.
The Properties pane displays the properties of the maneuver's activity.
- 4** In the Lateral category, set Reference type to Relative to lane.
- 5** In the Parameters category, specify the lane index.

Method 3**To specify a lane change maneuver relative to the lane used by the ASM vehicle**

- 1** Click the activity block of the maneuver.
The Properties pane displays the properties of the maneuver's activity.
- 2** In the Lateral category, set Reference type to Relative to object.
- 3** In the Parameters category, specify the lane index (number of lanes to be changed).

Result

The lane to which the ASM vehicle is changes is specified.

Related topics**References**

Lateral Type Properties (Maneuver)	127
Scenario Properties Dialog	93

How to Steer in a Maneuver

Objective

You can specify one of the stimulus signals provided, choose a predefined steering profile, or fix the steering wheel.

Basics of steering

Steering of the ASM vehicle is specified using the lateral properties of a maneuver's activity.

- You can specify circular or straight activities to make the ASM vehicle drive on them automatically.
- You can control the steering angle by individual stimulus signals. ModelDesk provides simplified profiles (constant, ramp, sine, sine with dwell, step, pulse, table). Profiles specified in a MAT file can be imported via a table. It is also

possible to stimulate the steering angle using the Simulink model or via an experiment software.

- You can control the steering torque by individual stimulus signals. ModelDesk provides simplified profiles (constant, ramp, table). Profiles specified in a MAT file can be imported via a table. It is also possible to stimulate the steering angle using the Simulink model or via an experiment software.
- You can hold the current steering wheel angle in the maneuver segment.
- You can also follow the road. Then, the driver model specifies the steering. Refer to [How to Follow a Road Automatically](#) on page 46.

Hold current value You can specify whether the steering wheel position reached in the previous maneuver segment is held for the current maneuver segment or not by enabling the Hold current value option (only available for Constant steering source). If enabled, you cannot specify values for the current maneuver segment.

Start with current value You can specify whether the steering wheel reached in the previous maneuver segment is used as the start value or not by enabling the Start with current value option (available only for Ramp steering source). If enabled, you cannot specify a start value for the current maneuver segment.

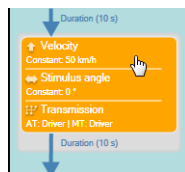
Preconditions

A scenario with a Maneuver block must be created and the Scenario pane must be open, refer to [How to Start Scenario Creation](#) on page 39.

Method

To steer in a maneuver

- 1 Select the activity of the maneuver which longitudinal properties you want to specify.



The Properties pane displays the properties of the selected activity. For specifying the steering of the ASM vehicle, you must specify the properties of the Lateral category. For details, refer to [Lateral Type Properties \(Maneuver\)](#) on page 127.

- 2 In the Lateral category, select a lateral type.
Now, the properties are displayed in the category that are required by the selected type.
- 3 Specify the properties of the selected type.

Result

You have specified the steering of the ASM vehicle.

Related topics

HowTos

[How to Follow a Road Automatically](#)..... 46

References

[Lateral Type Properties \(Maneuver\)](#)..... 127

How to Specify a Transition for a Maneuver

Objective

A transition specifies the condition when an activity ends and the succeeding activity begins.

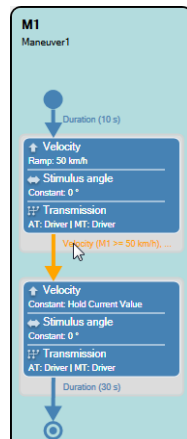
Preconditions

A scenario with a sequence for a maneuver must be created, refer to [How to Add Sequences and Segments for the Maneuver](#) on page 41.

Method

To specify a transition for a maneuver

- 1 In the Scenario pane, click the transition.



The Properties pane displays the properties of the selected transition.

By default, a transition contains one Duration condition. You can remove the default condition and/or add further conditions.

- 2 Add further conditions and specify their properties. Note that some condition types can only be used multiple times if it makes sense. For example, the Duration type can only be used once because multiple different durations are inconsistent.

Result

You specified a transition.

Related topics

Basics

[Basics of Maneuvers](#)..... 26

References

[Transition Properties](#)..... 157

How to Specify the Properties of the Driver Model

Objective

The steering wheel, the accelerator and brake pedals, and the clutch and gear are controlled by a driver model, which is specified by a driver type. In ModelDesk, you can specify the parameters for the driver model.

Basics

ASM contains a driver model that you can select for controlling the ASM vehicle. The driver model support three different driver types and two controller types. For details, refer to [Lateral Controller 1 \(ASM Environment Reference !\[\]\(dd161862f9164df98f62b726e9846241_img.jpg\)](#)) and [Lateral Controller 2 \(ASM Environment Reference !\[\]\(370afeb5bfccb68f3befb985d1441328_img.jpg\)](#)).

In ModelDesk, you can specify the parameters for the driver model.

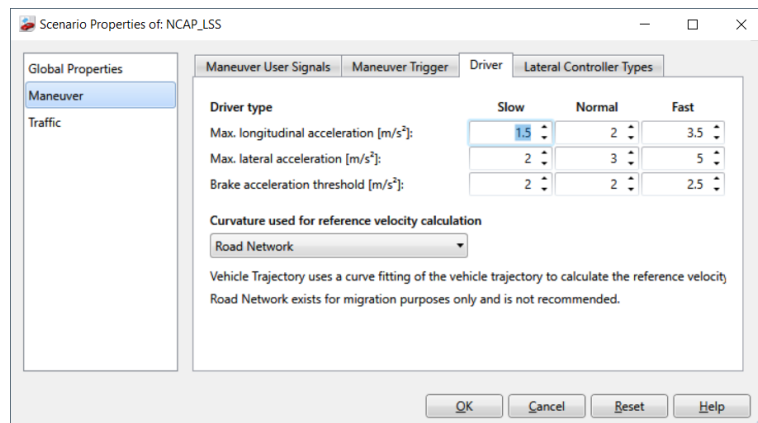
Preconditions

A scenario must be open in ModelDesk.

Method

To specify global driver properties

- 1 On the Environment ribbon, click Scenario – Properties.
The Scenario Properties dialog opens.
- 2 In the dialog, click Maneuver and then Driver.



- 3 Specify the properties.

Result The properties of the global driver are specified.

Related topics

References

Lateral Controller 1 (ASM Environment Reference )
 Lateral Controller 2 (ASM Environment Reference )
 Scenario Properties Dialog..... 93

How to Select and Specify Maneuver User Signals

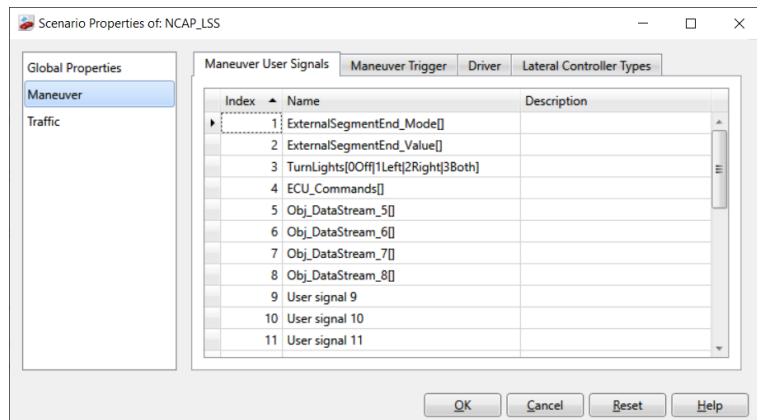
Objective You can define signals at any point or time in the maneuver segment for use as user signals in the Simulink model.

Basics of user signals

When a maneuver segment with user signals runs, signals are generated which can be used in the Simulink model, for example, as triggers. You can access these signals from the ASMSignalBus or from the user signal output of the MANEUVER_SCHEDULER block of the ASM. Refer to [Maneuver Scheduler \(ASM Environment Reference !\[\]\(0b5e7e25e8775f7e7e80906ada4f0021_img.jpg\)\)](#).

Maneuver user signals

You can create maneuver user signals including a description on the **Maneuver User Signals** page of the Scenario Properties dialog. The dialog displays the maneuver user signals of the model configuration.



The settings are valid for all maneuver segments that specifies user signals in the current experiment. User signals can be selected in the activity of a maneuver, see below.

Selecting and specifying user signals in the activity

You can select and specify maneuver user signals for each maneuver activity. ModelDesk provides four signal types: Constant, ramp, pulse, and table. You can

edit user signal parameters in the Properties pane. Whenever you select a signal type in a user signal row, the appropriate parameters are displayed.

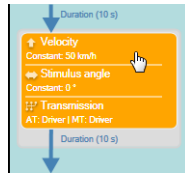
Preconditions

A scenario with a Maneuver block must be created and the Scenario pane must be open, refer to [How to Start Scenario Creation](#) on page 39.

Method

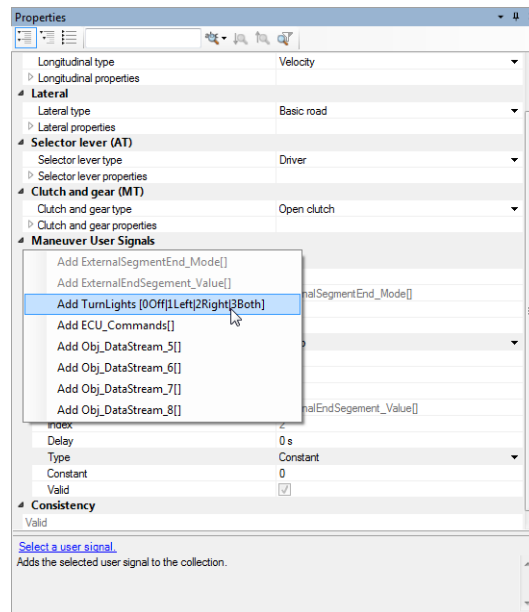
To select and specify maneuver user signals

- 1 Select the activity of the maneuver where you want to add a maneuver user signal.



The Properties pane displays the properties of the selected activity. For selecting and specifying a maneuver user signal, you must specify the properties of the Maneuver User Signals category.

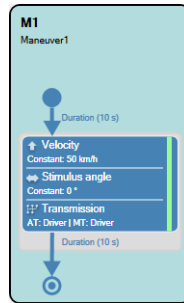
- 2 In the Maneuver User Signals category, click **Select a maneuver user signal** and select a signal.



- 3 Specify the properties of the maneuver user signal.
- 4 To specify further maneuver user signals, repeat the previous steps.

Result

The maneuver user signals are specified. When you have specified a maneuver user signal in an activity, its block is marked with a green bar.



Related topics

Basics

[ASMSignalBus \(ASM User Guide !\[\]\(8d0f0e0fe25b320c33272c52aec1fbca_img.jpg\)](#))

References

[Environment Configuration Dialog.....](#) 88

Creating Traffic of Scenarios

Introduction

To create a traffic of a scenario, you must specify all the fellows and global user signals.

Where to go from here

Information in this section

Workflow for Traffic Creation.....	56
This overview shows the workflow for creating a traffic of a scenario.	
How to Create a Fellow User Signal.....	57
Fellow user signals are used to set values of the associated signals in the simulation model.	
How to Create and Specify a Fellow.....	59
Describes how to create a fellow and to select a route and traffic object.	
How to Initialize and Specify the Movements of a Fellow.....	60
The movements of fellows are specified by segments, which contain an activity and a transition.	
How to Specify a Global User Signal.....	64
Describes how to create and specify global user signals to communicate with the simulation model independently of fellows.	
How to Specify the Properties of a Traffic Driver.....	66
You can specify the properties of a traffic driver in ModelDesk.	

Information in other sections

Example of a Scenario (ASM Traffic Reference )
The example illustrates the implementation of a scenario in the Scenario Editor and the corresponding signal data from simulation.

Workflow for Traffic Creation

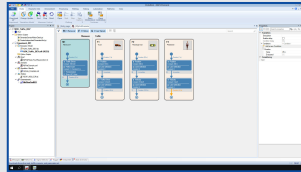
Introduction

This overview shows the workflow for creating a traffic of a scenario.

Tutorial video

Creating and Customizing a Scenario

This video show you how to specify scenarios. Scenarios define the movement of the ego-vehicle as well as fellows.



To watch this video, click the following link or scan the QR code:
<https://www.dspace.com/dspace-help/rg4NR>

Workflow for creating a traffic of a scenario

To create a traffic of a scenario, you must specify the fellows and global user signals.

1. Create the scenario. Refer to [How to Start Scenario Creation](#) on page 39.
2. When you create a fellow, you can select a traffic object from the Pool. The traffic object specifies the geometry of the fellow and the 3-D object used in the visualization. Refer to [How to Create and Specify a Fellow](#) on page 59.
3. A new fellow is created with one sequence and two segments by default. The first segment is used to initialize the position of the fellow. The second segment is used to specify its first movement. You can add more segments to the fellow definition to specify different movements. Refer to [How to Initialize and Specify the Movements of a Fellow](#) on page 60.
4. Global user signals are user signals which can be set independently from the fellows. You can create them in the same way as fellow definitions. Refer to [How to Specify a Global User Signal](#) on page 64.

Related topics

Basics

[Scenarios](#)..... 25

How to Create a Fellow User Signal

Objective

Fellow user signals are used to set values of the associated signals in the simulation model.

Fellow user signals

Fellow user signals are used to set values of signals for fellows in the simulation model (for details on the Simulink block, refer to [Traffic Scheduler \(ASM Traffic Reference\)](#)). The signals can be used, for example, to set states of fellows.

When fellows are visualized in MotionDesk and state objects are used, the turn signals or brake lights can be switched for the fellow. For details, refer to [Basics of Using State Objects in the Scene \(MotionDesk Scene Animation !\[\]\(529949c2c3dadbaa4e538e8c643454bc_img.jpg\)](#)).

Fellow user signals are created in the Scenario dialog. When you specify the activities of fellows, you can select one or more of them and assign values for the states.

Fellow user signals and model configuration

The fellow user signals that you specify in ModelDesk must correspond to the signals in the model configuration. The number and names of fellow user signals in the Scenario dialog must be identical to the fellow user signals specified in the Model Configuration dialog. Otherwise the fellow user signals are invalid.

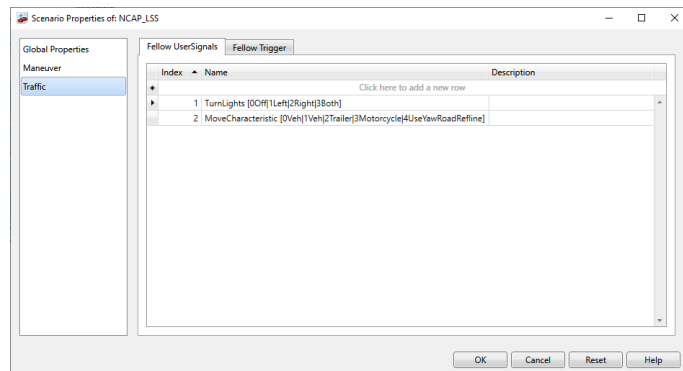
Preconditions

A scenario must be created and open in the Scenario Editor.

Method

To specify a fellow user signal

- 1 To open the Scenario dialog, go to the Environment ribbon and click Scenario – Properties.
- 2 Select the Traffic page.



- 3 Click on the first row to add a new row.
- 4 Specify the name according to the name specified in the Model Configuration dialog.
You can also reset the settings of the Scenario Properties dialog. When you do this, all the settings of this dialog are set to the settings of the model configuration in the Model Configuration dialog. This also applies to the fellow user signals.
- 5 Click Close.

Result

You specified fellow user signals.

Related topics	References
	Environment Configuration Dialog.....88 Settings.....112

How to Create and Specify a Fellow

Objective You must select a traffic object and a route for a fellow.

Specifying a fellow When you want to create a fellow, you have to specify several parts.

Route As a fellow is to follow a route, you must select one that is defined for the selected road network.

Traffic object Select a traffic object to specify the geometry of the fellow. The traffic object defines the parameters for the object detection sensors for the simulation and the 3-D object that will be used to visualize the fellow in MotionDesk.

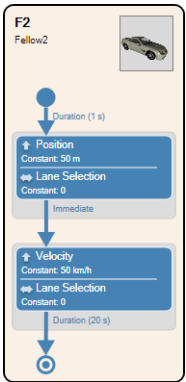
Preconditions

- A scenario must be created and opened in the Scenario Editor.
- The Properties pane must be open.
- The traffic object to be used must be created. Refer to [How to Create a Traffic Object \(ModelDesk Traffic Object Management !\[\]\(a03a7eb2f4046e1d3c76772003e549ea_img.jpg\)](#)).

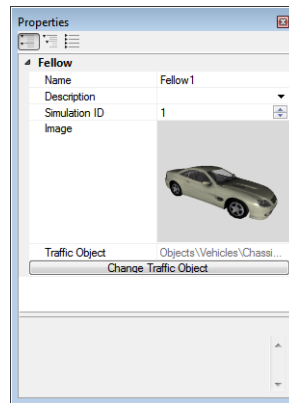
Method

To create and specify a fellow

- 1 On the Scenario pane, open the context menu and select New – Fellow. ModelDesk creates a new fellow with one sequence and two segments.



- 2 Select the fellow block.
The Properties pane displays the properties of the fellow.



- 3 Click **Change Traffic Object** to select a traffic object.
ModelDesk opens the **Traffic Object Browser**, which displays all the traffic objects of the Pool.
- 4 Select a suitable traffic object and click **OK**.
- 5 Specify the other properties of the fellow. For a description of the properties, refer to [Fellow Properties](#) on page 120.

Result You specified a fellow.

Related topics

Basics

[Basics of Fellows](#)..... 28

References

[New – Fellow](#)..... 108
[Segment Properties](#)..... 151

How to Initialize and Specify the Movements of a Fellow

Objective

The movements of fellows are specified by segments, which contain an activity and a transition. The specification method is independent of the fellow type. You can specify fellow vehicles in the same way as other traffic participants, such as pedestrians.

Specifying the movement

To specify the segments for a fellow, you have to specify several parts:

Initialization The first segment is used to initialize the fellow. In the activity, you can specify information such as the position of the fellow. In the transition, you can set different types, such as the immediate type. This type ends the segment immediately.

If the first segment is not used for specifying the position, the simulation model automatically sets the fellow's starting position to the origin ($s = 0$ and $d = 0$).

Movement To define the movement of the fellow, you can specify a chain of segments. Each segment contains an activity and a transition.

An activity specifies the movement in the longitudinal and lateral directions. For detailed information on the possible settings, refer to [Basics of Activities](#) on page 30.

A transition specifies the conditions that end an activity. A transition can have one or more conditions of different types. The condition that is fulfilled first ends the segment. For detailed information on the conditions, refer to [Basics of Transitions](#) on page 34.

You can add further segments to specify other activities of the fellow. The segments are executed consecutively.

Selected route When routes are defined on the road network, you must select a route for the fellow to use. In each sequence, the fellow can follow a different route. If you want the fellow to follow other routes, you must create additional sequences.

Fellow user signals Fellow user signals are used to set the values of the associated signals in the simulation model. You can select fellow user signals in the activities. For each fellow user signal, you can specify the value that will be set when the activity of the fellow is executed. For details, refer to [Basics on Movement Characteristics \(ASM Traffic Model Description\)](#).

Reusing settings

You can copy settings of an element of a fellow (sequence, segment, activity, or transition) to another element of the same type. You can do this via the Copy and Paste commands of the context menu or via drag & drop.

Preconditions

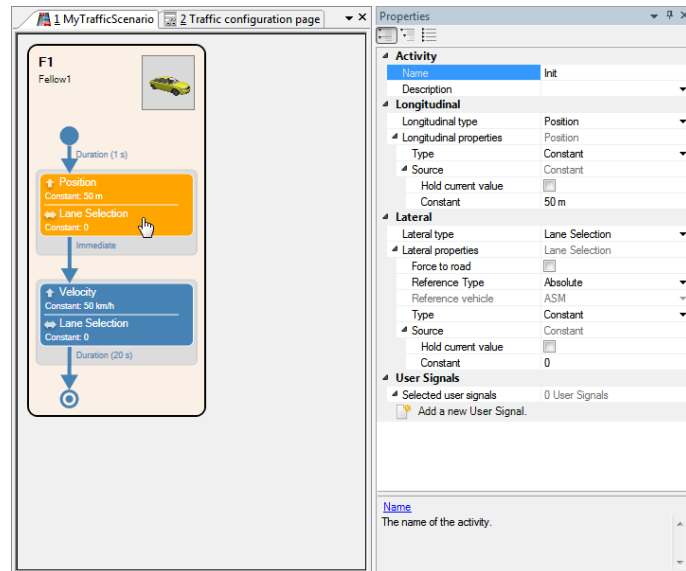
- A scenario must be created and open in the Scenario Editor.
- A road network must be selected for the scenario.
- A fellow must exist (see [How to Create and Specify a Fellow](#) on page 59).
- Fellow user signals must exist (see [How to Create a Fellow User Signal](#) on page 57).
- The Properties pane must be open.

Method

To initialize and specify the movement of a fellow

- 1 On the Scenario pane, select the activity of the first segment.

The Properties pane displays the properties of the activity.

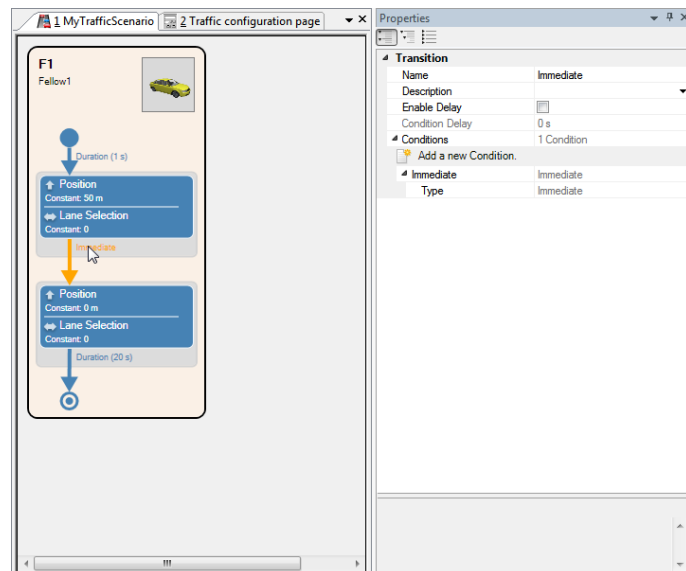


The first segment is used to initialize the position of the fellow.

- 2 In the Longitudinal category, specify a value in longitudinal direction (s coordinate) for the fellow. For details of the properties, refer to [Longitudinal Type Properties \(Fellow\)](#) on page 135.
- 3 In the Lateral category, select a lane or specify a value (d coordinate) for the fellow. For details of the properties, refer to [Lateral Type Properties \(Fellow\)](#) on page 123.

When the position is specified, you can specify the transition.

- 4 On the Scenario pane, select the transition of the first segment. The Properties pane displays the properties of the transition.



Because the first segment is used to initialize the fellow, the transition has the Immediate condition type by default.

- 5 In the Conditions category, specify the conditions that end the activity. You can specify several conditions. The first condition that is fulfilled ends the activity. You should therefore delete the Immediate condition if you want to use another condition type. For details of the properties, refer to [Transition Properties](#) on page 157.

Tip

Observe the Evaluated property. It shows you whether the appropriate condition will be evaluated in the simulation.

- 6 Click the activity of the second segment. You can modify its properties as desired. For details of the properties, refer to [Longitudinal Type Properties \(Fellow\)](#) on page 135 and [Lateral Type Properties \(Fellow\)](#) on page 123.
- 7 Click the transition of the second segment. You can modify its properties as desired. For details of the properties, refer to [Transition Properties](#) on page 157.
- 8 To add a segment to the sequence, open the context menu of a segment and select
 - Insert Segment - After to add the segment after the selected segment
 - Insert Segment - Before to add the segment before the current segment
 - Append Segment to add the segment after the last segment of the sequence
- 9 Specify the segment, activity, and transition of the new segment as described above.
- 10 To add a new sequence for the fellow, open the context menu of the fellow and select [Append Sequence](#) on page 100. A further sequence is necessary, for example, when you want to select another route for the same fellow.
- 11 To select a route, select the Start element. On the Properties pane, you can select the route and direction of travel.
- 12 To select a fellow user signal, select the activity. On the Properties pane in the User Signals category, select a fellow user signal.

Result

You specified the segments of the fellow.

Related topics

Basics

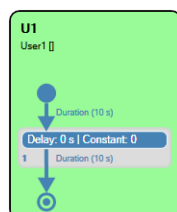
[Basics of Fellows](#)..... 28

References

[Append Segment](#)..... 100
[Insert Segment - After](#)..... 106
[Insert Segment - Before](#)..... 107
[New – Fellow](#)..... 108

How to Specify a Global User Signal

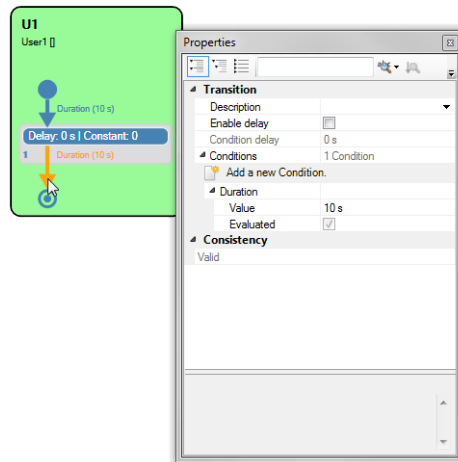
Objective	You can create and specify global user signals to communicate with the simulation model independently of fellows.
Global user signals	You can use global user signals for communicating with the maneuver scheduler to trigger user-defined actions in the simulation model. Global user signals are independent from the fellows definitions. For details on the maneuver scheduler, refer to Maneuver Scheduler (ASM Environment Reference ).
Specifying global user signals	For specifying global user signals, the same method is used as for specifying fellows. You can specify a chain of segments. Each segment contains an activity and a transition. An activity specifies a constant value for the global user signal. A transition specifies the conditions that end an activity. A transition can have one or more conditions of the same or different types. The condition that is fulfilled first ends the segment. For detailed information on the conditions, refer to Basics of Transitions on page 34.
Reusing settings	You can copy settings of an element of a global user signal (segment, activity, or transition) to another element of the same type. You can do this via the Copy and Paste commands of the context menu or via drag & drop. You can also copy the settings of a fellow's transition and paste them to the transition of a global user signal and vice versa.
Preconditions	▪ A scenario must be created and opened in the Scenario Editor. ▪ A road network must be selected for the scenario. ▪ The Properties pane must be open.
Method	To specify a global user signal 1 On the Scenario pane, open the context menu and select New – User Signal. ModelDesk creates a new global user signal with one segment containing an activity and a transition.



- 2 Select the segment.
The Properties pane displays the properties of the global user signal.
- 3 Specify the properties of the segment. Refer to [Global User Signal Properties](#) on page 122.
- 4 Select the activity of the segment.



- 5 Specify the properties of the activity. Refer to [Activity Properties \(Global User Signal\)](#) on page 115.
- 6 Select the transition of the segment.



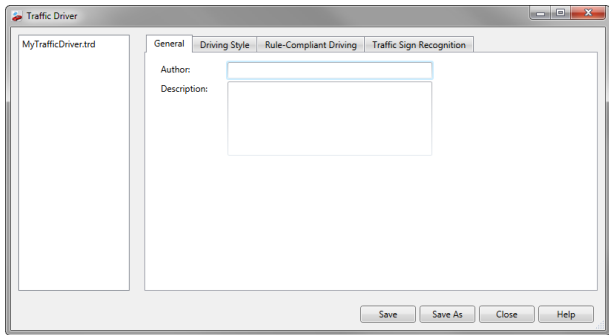
- 7 Specify the properties of the transition. Refer to [Transition Properties](#) on page 157.
- 8 To add a segment to the sequence, open the context menu of a segment and select:
 - Insert Segment - After to add the segment after the selected segment
 - Insert Segment - Before to add the segment before the current segment
 - Append Segment to add the segment after the last segment of the sequence
- 9 Specify the segment, activity, and transition of the new segment as described above.

Result	You specified a global user signal.
Related topics	References Append Segment..... 100 Insert Segment - After..... 106 Insert Segment - Before..... 107 Segment Properties..... 151

How to Specify the Properties of a Traffic Driver

Objective	You can specify the properties of a traffic driver in ModelDesk.
Basics of traffic driver	The ASM traffic scheduler has the traffic driver feature that provides the following functionalities: ▪ Fellow vehicles and the ASM vehicle detect other vehicles on the same lane and implement a plausible follow behavior. ▪ Fellow vehicles and the ASM vehicle detect traffic signs and traffic lights and comply with the corresponding traffic rules. ▪ Fellow vehicles feature a realistic physical behavior regarding their acceleration and their speed in curves. You can specify different driving styles using the traffic drivers in ModelDesk. You can assign the traffic drivers to the vehicles in a scenario. Therefore, different driving styles can be promoted for different vehicles during the simulation. For details, refer to Working with the Traffic Driver (ASM Traffic Model Description ).

Method	To specify the properties of a traffic driver 1 Open the Traffic Driver dialog: ▪ To open an existing traffic driver, go to the Environment ribbon and click Traffic Driver – Open. ▪ To create a new traffic driver, go to the Environment ribbon and click Traffic Driver – New.
---------------	--



- 2
- Specify the properties of the traffic driver on the pages of the dialog. For details of the properties, refer to [Traffic Driver Dialog](#) on page 96.
- 3
- Click Save or Save As to save the properties of the traffic driver.

Result

The properties of the traffic driver are specified.

Related topics

References

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Working with Scenarios

Introduction You can connect the scenarios you created to roads and work with them in ModelDesk.

Where to go from here **Information in this section**

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You can use scenarios together with roads in the current project or in other parameterization projects. Scenarios can also be downloaded to the simulation model for starting simulation.	
Modifying Properties of a Scenario via Alias Variables.....	69
When a scenario is created, you can modify properties using alias variables in ModelDesk testing and automation scripts.	

Using Scenarios

Introduction You can use scenarios together with roads in the current project or in other parameterization projects. Scenarios can also be downloaded to the simulation model for starting simulation.

Using scenarios in other projects All created scenarios are stored in the Pool, from which you can export them and then import them to the Pool of another project.

Export You can export scenario files into a ZIP archive in a similar way to exporting parameter files. Refer to [How to Export Parameter Files \(ModelDesk Parameterizing !\[\]\(05be7c7a8995decd503647c99211f7c2_img.jpg\)](#)) and [Export \(ModelDesk Basics !\[\]\(16cd6e1a39784ecf52b4db09f4865f40_img.jpg\)](#)).

Import You can import scenario files to your project in a similar way to importing parameter files. Refer to [How to Import Parameter Files \(ModelDesk Parameterizing !\[\]\(aa53ad6fea213b8b2226d3077e30533a_img.jpg\)](#)) and [Import \(ModelDesk Basics !\[\]\(a1c2189b125458bd8fa8822d0c2da6bc_img.jpg\)](#)). The scenario files you have imported are saved to the Scenario folder of the Pool automatically. You can link them to your experiment from there.

Changing parameters outside ModelDesk You can export a currently linked scenario to change parameters outside ModelDesk.

Generate Traffic Ini You must export the scenario INI file to the `Simulation\IniFiles\Scenario` folder of the ASM to use the scenario together with the model and to run the simulation with. For detailed information, refer to [Generate Traffic Ini](#) on page 80. The road is exported as a MAT file.

Downloading scenarios in the simulation model

If you have created a scenario, you must download it to use the specified scenario parameters together with the simulation model. The scenario which is linked to the current experiment can be downloaded to the real-time hardware, Simulink, or VEOS with ModelDesk.

You can download the complete scenario, or the maneuver and traffic content (fellows and global user signals) separately.

Tip

To visualize the traffic fellows in MotionDesk, you must update the scene. Refer to [Basics of Synchronizing the Scene in MotionDesk \(ModelDesk Scene Synchronization !\[\]\(74d4806277d7e73349d8e8c0897931e9_img.jpg\)](#). The update can be automated by selecting an option. Refer to [Customize \(ModelDesk Scene Synchronization !\[\]\(5f42d2cd7ad901bc24e5d35a38c777fd_img.jpg\)](#).

Related topics

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Download Traffic.....	79

Modifying Properties of a Scenario via Alias Variables

Introduction

When a scenario is created, you can modify properties using alias variables in ModelDesk testing and automation scripts.

Alias variables

Using alias variables is an easy method to modify properties of the active scenario. The method allows you to modify one or more properties using one alias variable.

You create alias variables in the **Scenario Editor** for each property that you want to modify. When you create an alias variable, it gets a property reference that points to the property that you want to modify. Then, you can assign several property references to the same alias variable. When you modify the value of the alias variable, you also modify the values of all assigned property references. The modification is done via ModelDesk testing or tool automation.

Related topics

Basics

Basics of the Alias Support (ModelDesk Testing 
Setting Values of Properties Using Alias Variables (ModelDesk Automation 

HowTos

[How to Create Alias Variables \(ModelDesk Testing 📖\)](#)

Controlling Maneuvers and Driving Cycles

Introduction You can use ModelDesk to start, stop, or reset maneuvers for vehicle dynamics test and driving cycles for engine tests.

Where to go from here	Information in this section
	Basics of Controlling Maneuvers and Driving Cycles..... 71 You can use ModelDesk to start, stop, and reset maneuvers and driving cycles.
	How to Control a Maneuver or a Driving Cycle..... 73 Provides instructions on how to control (start, stop, reset) a maneuver or driving cycle by using ModelDesk.

Basics of Controlling Maneuvers and Driving Cycles

Introduction You can use ModelDesk to start, stop, and reset maneuvers and driving cycles.

Basics When you test ECUs, you want to control maneuvers and driving cycles.

Maneuvers Maneuvers are used to test ECUs for vehicle dynamics. You can specify how a driver drives the ASM vehicle by using the **Scenario Editor**.

Driving cycles Driving cycles are used to test engine control units. The driving cycles are modeled with the ASMs.

ModelDesk can start, stop, or reset the maneuvers/driving cycles only if the simulation application contains the specific ASM blocks. When you use the maneuver control of ModelDesk, ModelDesk writes a pulsed value and a unique number to a parameter of the simulation model. This parameter is evaluated by the specific ASM blocks to control the maneuver.

You can use maneuver control in real-time simulation, Simulink simulation, and offline simulation using VEOS.

Extended simulation model The simulation application must have a specific code so that ModelDesk can control maneuvers and driving cycles. This code is integrated in the simulation application when you use the **ManeuverControl** blocks of the **ASM_Utils_lib** library. The appropriate commands are enabled only if ModelDesk finds such blocks in the model during a consistency check.

Block	Button	Description
MANEUVER_START		To start the maneuver or driving cycle.
MANEUVER_STOP		To stop the maneuver or driving cycle.
RESET		To reset the states of the simulation application.
MANEUVER_STATE	—	To get the state.

Multiple ManeuverControl blocks

ModelDesk can evaluate only one ManeuverControl block of the same type. If a simulation model has more than one block of the same type, a priority defines which block is evaluated. The priority is defined based on the components as follows:

1. VehicleDynamics
2. Engine
3. ElectricComponents

Supported platform

You can use maneuver control on VEOS, Simulink simulation, and real-time platforms.

Workflow

Some steps must be performed to be able to control maneuvers and driving cycles

1. The simulation model must contain the blocks for maneuver control. Refer to:
 - [Maneuver Scheduler \(ASM Environment Reference !\[\]\(4f6d8a8b127300a02d56d34d01423d15_img.jpg\)](#)) for vehicle dynamics applications
 - [Maneuver Control \(ASM Drivetrain Basic Reference !\[\]\(7e3d1ad67bf2d7a17700a66d1a313f91_img.jpg\)](#)) for engine applications
 - [Maneuver and Reset Control \(ASM User Guide !\[\]\(6aaf22e5a325c32ef2122c2939c64f9c_img.jpg\)](#))
2. Register the platform that you want to use for simulation. Refer to [Registering Platforms \(ModelDesk Platform Management !\[\]\(27421322d686d4980ae2b7f101ee89ba_img.jpg\)](#)).
3. Create a ModelDesk project and experiment. When you create an experiment, choose the simulation model that contains the ManeuverControl blocks. Refer to [Creating Projects and Experiments \(ModelDesk Project and Experiment Management !\[\]\(c45bffa8c48916027d3ccd561ed26723_img.jpg\)](#)).
4. Specify the ModelDesk experiment. If you want to control a maneuver for vehicle dynamics, create a maneuver using the Scenario Editor and/or a road by using the Road Generator.
5. Optional: Download the simulation application and the active experiment to the platform. This step is optional because the Start command can download the simulation application and the active experiment to the platform if no or another simulation application is loaded.
6. Finally, you can control the maneuver or driving cycle. Refer to [How to Control a Maneuver or a Driving Cycle](#) on page 73.

Each time you change the maneuver state, ModelDesk checks whether it is allowed. If it is not allowed, ModelDesk displays an appropriate message in the Message Viewer.

Related topics

Basics

[Overview of the Maneuver and Reset Control \(ASM User Guide !\[\]\(666e09182d4cd268646ea700ea60dcdf_img.jpg\)\)](#)

How to Control a Maneuver or a Driving Cycle

Objective

This topic provides instructions on how to control (start, stop, reset) a maneuver or driving cycle by using ModelDesk.

Preconditions

- To be able to control the maneuver, the following preconditions must be fulfilled.
- An experiment must be active.
 - A platform and a simulation model must be selected.
 - The simulation model must contain `ManeuverControl` blocks.

Method

To control a maneuver or a driving cycle

- 1 On the Home ribbon, click **Maneuver Control – Start**.
ModelDesk checks the platform and the running simulation application.
 - If no simulation application is running on the platform, you can have ModelDesk download the simulation application and the road, scenario, and parameter set of the active experiment.
 - If another simulation application is running on the platform, you can stop the running simulation application and download the simulation application matching the active experiment and the road, scenario, and parameter set of the active experiment.

If the simulation application is running with the parameters of the active experiment and the current maneuver state is suitable, the maneuver starts.
- 2 Check the **Message Viewer**. If ModelDesk was not able to start the maneuver, the **Message Viewer** displays the appropriate message.
- 3 To stop the maneuver, go to the Home ribbon and click **Maneuver Control – Stop**.
- 4 To reset the maneuver of the simulation application, go to the Home ribbon and click **Maneuver Control – Reset**.

Result

You controlled the maneuver.

Related topics

Basics

[Basics of Controlling Maneuvers and Driving Cycles..... 71](#)

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To specify scenarios.	
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ModelDesk provides commands for controlling the maneuver.	

Scenario Editor Management Commands

Introduction

ModelDesk provides various commands to manage the scenarios.

Where to go from here

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To remove a scenario from the current experiment.	
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To save the scenario.	
Save As	85
To save the current scenario under a new name and path.	

Activate

Access

You can access the command via:

Ribbon	Environment – Scenario
Context menu of	None
Shortcut key	None
Icon	

Purpose

To activate the open scenario by linking it to the current experiment.

Result

The currently open scenario is linked to the experiment and active.

Related topics

Basics

[Using Scenarios.....](#) 68

Activate from Pool

Access

You can access this command via:

Ribbon	None
Context menu of	- Project Navigator – Scenario - Configuration page (scenario)
Shortcut key	None
Icon	None

Purpose

To activate a scenario saved in the Pool by linking it to the current experiment.

Result

ModelDesk opens the Specify File Name dialog for you to select the scenario file from the Pool and activate it by linking it to the current experiment.

Description

To view or edit the active scenario, it can be opened via the Open command.

Related topics**References**

[Open.....](#) 82

Download / Download Scenario

Access

You can access this command via:

Ribbon	▪ Home – Experiment – split button ▪ Environment – Scenario
Context menu of	Project Navigator – Scenario
Shortcut key	None
Icon	
Button	▪ Configuration page (scenario) ▪ Scenario configuration page

Purpose

To download the active scenario to the simulation platform.

Description

A scenario includes the specification of the maneuver, fellows, and global user signals.

Result

The currently active scenario is downloaded.

Description

Only the active scenario can be downloaded. If the open scenario is not the active one, a message dialog appears asking you if you want to activate the open scenario.

In the real-time mode, ModelDesk generates a parameter file, which can be loaded to the real-time hardware. In the Simulink mode, the scenario parameters are downloaded to Simulink for further use.

Related topics**Basics**

[Using Scenarios.....](#) 68

Download Maneuver

Access

You can access this command via:

Ribbon	Environment – Scenario – split button
Context menu of	None
Shortcut key	None
Icon	
Button	None

Purpose

To download the maneuver of the active scenario to the simulation platform.

Result

The maneuver is downloaded.

Related topics

Basics

[Using Scenarios.....](#) 68

Download Traffic

Access

You can access this command via:

Ribbon	Environment – Scenario – split button
Context menu of	None
Shortcut key	None
Icon	
Button	None

Purpose

To download the fellows and global user signals of the active scenario to the simulation platform.

Result

The fellows and global user signals are downloaded.

Related topics	Basics
	Using Scenarios..... 68

Generate Traffic Ini

Access	You can access the command via: <table><tr><td>Ribbon</td><td>None</td></tr><tr><td>Context menu of</td><td>Project Navigator – Scenario</td></tr><tr><td>Shortcut key</td><td>None</td></tr><tr><td>Icon</td><td>None</td></tr></table>	Ribbon	None	Context menu of	Project Navigator – Scenario	Shortcut key	None	Icon	None
Ribbon	None								
Context menu of	Project Navigator – Scenario								
Shortcut key	None								
Icon	None								
Purpose	To export a currently linked scenario to change parameters outside ModelDesk.								
Result	The scenario is exported to the specified folder and named as a MAT file.								
Description	The initialization file of a scenario must be exported to Simulation\IniFiles\Scenario of the ASM to use the scenario together with the model and to run the simulation with. ModelDesk opens the Generate MAT File dialog for you to select the path and folder to save the initialization file to.								
Save As dialog	Save in Lets you select the path and folder to export the file to. File name Lets you specify the name of the file. You can also select a name from the files listed under the chosen path and folder. Save as type Lets you save the file as a MAT file. If you select a file from those listed under the chosen path and folder, you are prompted before it is overwritten.								

Related topics	Basics
	Using Scenarios..... 68

New

Access

You can access this command via:

Ribbon	Environment – Scenario
Context menu of	Project Navigator – Scenario
Shortcut key	None
Icon	

Purpose

To create a new scenario.

Description

ModelDesk opens the Scenario pane. The pane has a maneuver and if ASM Traffic Model is used, a fellow by default. You can create further fellows and global user signals in this pane by using the New – Fellow or New – User Signal command.

You can specify the properties of the scenario, such as the related road, in the Properties pane.

The name of the scenario is specified when it is saved.

Related topics

HowTos

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Traffic Scenario Properties	156

New (Traffic Driver)

Access

You can access this command via:

Ribbon	Environment – Traffic Driver
Context menu of	None
Shortcut key	None
Icon	

Purpose	To create a new traffic driver.
Result	ModelDesk opens the Traffic Driver dialog for you to specify the properties of the traffic driver.
Related topics	HowTos How to Specify the Properties of a Traffic Driver..... 66 References Traffic Driver Dialog..... 96

Open

Access	You can access this command via: <table><tr><td>Ribbon</td><td>Environment – Scenario</td></tr><tr><td>Context menu of</td><td>Project Navigator – Scenario</td></tr><tr><td>Shortcut key</td><td>None</td></tr><tr><td>Icon</td><td></td></tr></table>	Ribbon	Environment – Scenario	Context menu of	Project Navigator – Scenario	Shortcut key	None	Icon	
Ribbon	Environment – Scenario								
Context menu of	Project Navigator – Scenario								
Shortcut key	None								
Icon									
Purpose	To open a scenario linked to the current experiment.								
Result	ModelDesk opens the linked scenario in the Scenario Editor.								
Related topics	Basics Scenarios..... 25								

Open (Traffic Driver)

Access

You can access this command via:

Ribbon	Environment – Traffic Driver
Context menu of	None
Shortcut key	None
Icon	

Purpose

To open a traffic driver stored in the pool of the ModelDesk project.

Result

ModelDesk loads the properties of the traffic driver and opens the Traffic Driver dialog.

Related topics

HowTos

[How to Specify the Properties of a Traffic Driver](#)..... 66

References

[Traffic Driver Dialog](#)..... 96

Open from Pool

Access

You can access this command via:

Ribbon	Environment – Traffic
Context menu of	Project Navigator – Scenario
Shortcut key	None
Icon	

Purpose

To open a scenario from the Pool to edit it.

Result

ModelDesk opens the standard Specify File Name dialog for you to select the path, folder, file type, and name of the scenario to be opened from the ModelDesk Pool.

Description The scenario can be edited, but it is not linked to the current experiment. If you want to link it, use the **Activate** command.

Related topics**Basics**

[Scenarios.....](#) 25

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[Activate.....](#) 77

Remove

Access

You can access the command via:

Ribbon	None
Context menu of	Project Navigator – Scenario
Shortcut key	None
Icon	None

Purpose

To remove a scenario from the current experiment.

Result

The active scenario is removed from the current experiment.

Description

If no scenario is linked to the current experiment, the command is unavailable.

Related topics**Basics**

[Using Scenarios.....](#) 68

Save

Access

You can access this command via:

Ribbon	Environment – Scenario
Context menu of	Project Navigator – Scenario
Shortcut key	Ctrl+S
Icon	

Purpose

To save the scenario.

Result

The currently open scenario is saved.

Related topics

Basics

Workflow for Maneuver Creation.....	38
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------------------------------	----

Save As

Access

You can access this command via:

Ribbon	None
Context menu of	Project Navigator – Scenario
Shortcut key	None
Icon	None

Purpose

To save the current scenario under a new name and path.

Result

ModelDesk saves the scenario under a new name and path that you specified.

Description

ModelDesk opens the Specify File Name dialog to change the current entries.

Specify File Name dialog

- Save in** Lets you select the path and folder to export the file to.
- File name** Lets you specify the name of the file. You can also select a name from the files listed under the chosen path and folder.
- Save as type** Lets you save the file as an XML file.
If you select a file from those listed under the chosen path and folder, you are prompted before it is overwritten.

Related topics

Basics

Workflow for Maneuver Creation	38
Workflow for Traffic Creation	56

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Save	85
----------------------	----

Scenario Editor Dialogs and Panes

Introduction The Scenario Editor has several panes that are necessary to specify scenarios.

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	Edit Table Dialog..... 87 To specify values of table.
	Environment Configuration Dialog..... 88 To configure settings for the simulation model.
	Scenario Configuration Page..... 90 To activate, open, and download scenarios linked to the current active experiment.
	Scenario Pane..... 92 To specify a scenario.
	Scenario Properties Dialog..... 93 To configure the properties of the open scenario.
	Scenario Settings Dialog..... 96 To configure settings for the simulation model.
	Traffic Driver Dialog..... 96 To configure settings for the simulation model.

Edit Table Dialog

Access The Edit Table dialog opens when you click the Edit table button for a property.

Purpose To specify values of a table.

Description The table is used for various properties, which can be specified by a map. The x-axis and y-axis of the table are adapted to the property.

To specify the values of a table, you add columns to the table using the Append command of the context menu and modify the values in the table cells. The area above the table displays a curve with the specified values. You can add values in the area using the Insert point here and Append commands of the context menu.

To delete unnecessary columns, select and delete them with the **Delete** command.

You can copy values of the table to the Clipboard and paste them to other Windows applications, for example, Excel.

You can paste values from the Clipboard to selected columns of the table. Existing values are overwritten. It is not possible to paste more columns than the number of selected columns.

It is possible to export and import the values of the table to a MAT file. So you can reuse specified tables for other properties.

Commands	The dialog has buttons and a context menu that provides the following commands. Append Lets you append a column at the end of the table Insert point here Lets you insert a new value at the current position of the pointer to the table. Delete Lets you delete the selected column. Import Lets you load the values of a MAT file to the table. When the MAT file is loaded, existing values are overwritten. Export Lets you save the values of the table to a MAT file.
-----------------	---

Related topics	References Scenario Editor Properties..... 113
-----------------------	--

Environment Configuration Dialog

Access	The dialog opens when you select the Configure command. <table><tr><td>Ribbon</td><td>Environment – Environment</td></tr><tr><td>Context menu of</td><td>None</td></tr><tr><td>Shortcut key</td><td>None</td></tr><tr><td>Icon</td><td></td></tr></table>	Ribbon	Environment – Environment	Context menu of	None	Shortcut key	None	Icon	
Ribbon	Environment – Environment								
Context menu of	None								
Shortcut key	None								
Icon									
Purpose	To configure settings for the simulation model.								
Description	In this dialog, you can set certain parameters for the simulation model. Some parameters should be modified only when the simulation model is also modified.								

Otherwise, the download will fail. To avoid any unintended modification, you must enable the **Advanced Configuration** option for modifying these parameters.

The dialog has four pages where you can specify parameters for maneuver, road, traffic, and testing, and commands available via buttons and context menu.

Maneuver dialog settings

User Signals Displays the user signals of the maneuver.

Triggers Displays the triggers of the maneuver.

Road dialog settings

Activate OSI Road Support Lets you activate the OSI road support. Activate this option if you want to download a road which data is prepared according to the ground truth data format of the Open Simulation Interface (OSI). For details, refer to [Groundtruth](#) ([ASM Traffic Reference](#) ).

Max Lane Boundary Point Distance Lets you specify the maximum distance of the lane boundary points in meters.

Traffic dialog settings

You can modify the following properties only if the **Advanced Configuration** option is enabled.

Maximum number of fellows Lets you specify the maximum number of fellows that you want to use in the traffic.

Fellow User Signals Displays the index, name, and description of the fellow user signals.

Global User Signals Displays the index, name, and description of the global user signals.

Triggers Displays the index, name, and description of the triggers.

Testing dialog settings

You can modify the following properties only if the **Advanced Configuration** option is enabled.

The page lets you specify the signal path and signal collector of the maneuver state and maneuver time signal.

Signal Path Lets you specify the path and name of the signal containing the maneuver state or maneuver time.

Signal Collector Lets you specify the label of the signal collector from which the maneuver state or maneuver time is read. The signal collector is an ASM block that collects signals of the model to provide them to ModelDesk for plotting and testing. For details, refer to [ASMSignalInterface](#) ([ASM User Guide](#) ).

If you modify the label of the `ASMSignalInterface` block in the model, you must adapt this property and update the model in ModelDesk.

If the model is migrated from Release 2020-A or earlier, the label is `Simulation model1`.

Global dialog settings

The dialog has the following buttons.

Import Lets you import the configuration of all pages from an XML file.

Export Lets you export the configuration of all pages to an XML file.

Advanced Configuration Lets you enable or disable the advanced configuration mode.

Note

Some parameters can be modified only in the advanced configuration mode. Modifying these parameters can lead to errors during download or simulation when they are not consistent with the simulation model.

The dialog has a context menu with the following commands.

Move up Lets you move up the selected row.

Move down Lets you move down the selected row.

Delete Lets you delete the selected row.

Related topics**Basics**

[Configuring the Traffic Options \(ASM Traffic Model Description !\[\]\(f60b7a900783ac3fd531bfd9c111be6d_img.jpg\)\)](#)

HowTos

[How to Create a Fellow User Signal..... 57](#)

[How to Increase the Maximum Number of Fellows \(ASM Traffic Model Description !\[\]\(eabd9f9ababee93effadc3b380fe65fd_img.jpg\)\)](#)

[How to Increase the Number of Fellow User Signals \(ASM Traffic Model Description !\[\]\(83bbbd261710c59db0214aa27b2edc0d_img.jpg\)\)](#)

[How to Increase the Number of Global User Signals \(ASM Traffic Model Description !\[\]\(166772600a13ad0a433053f90fe45649_img.jpg\)\)](#)

[How to Increase the Number of Trigger Signals \(ASM Traffic Model Description !\[\]\(291e070cef6c4d5e78fefe4696ef53be_img.jpg\)\)](#)

References

[OSI Groundtruth Interface \(ASM Traffic Reference !\[\]\(f507db636256ac11a5525ef93ec6b8d7_img.jpg\)\)](#)

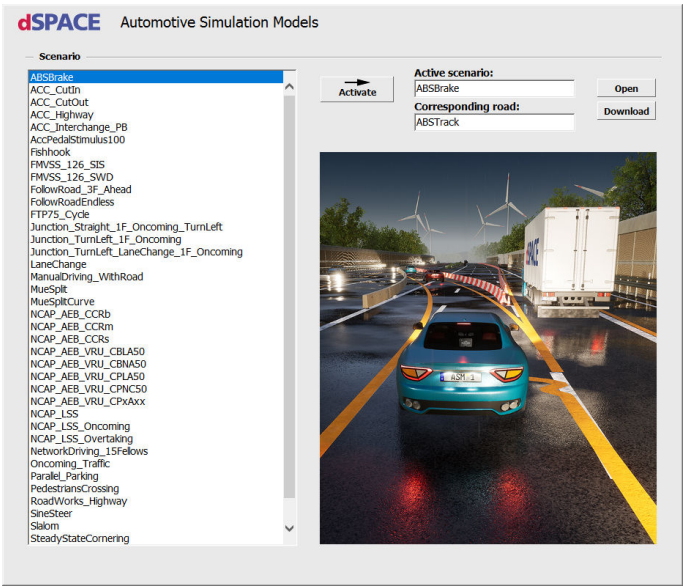
Scenario Configuration Page

Access

The Scenario Configuration page opens automatically when you select the Scenario node in the Project Navigator.

Purpose To activate, open, and download scenarios linked to the current active experiment.

Description The following illustration shows the Scenario Configuration page.



- Dialog settings**
- Scenario** Lists all the available scenarios saved in the Pool for you to choose one.
 - Activate** Lets you activate the selected scenario. Only one scenario can be active in a ModelDesk project at a time. All the others are inactive. To work with a scenario, it must be active.
 - Active scenario** Displays the current active scenario.
 - Corresponding road** Displays the road name if the current active scenario is linked to a road.
 - Open** Opens the Scenario Editor to let you edit the active scenario.
 - Download** Lets you download the selected scenario to the model. Refer to [Download / Download Scenario](#) on page 78.

Related topics	References
	Activate 77
	Download / Download Scenario 78
	Open 82

Scenario Pane

Purpose

To specify a scenario.

Description

In the pane, you can specify the maneuver, fellows and global user signals of the scenario.

Commands

The pane displays different elements. When an element is selected, a specific context menu is provided.

No selected element When no element is selected, the following commands are available.

Command	Purpose
New – Maneuver	To add a maneuver to the scenario.
New – Fellow	To add a new fellow to the scenario.
New – User Signal	To add a new global user signal to the scenario.
Layout Diagram	To arrange all the fellows and global user signals on the Scenario pane.

Fellow or global user signal When a maneuver, fellow, or a global user signal is selected, the following commands are available.

Command	Purpose
New – Maneuver	To add a maneuver to the scenario.
New – Fellow	To add a new fellow to the scenario.
New – User Signal	To add a new global user signal to the scenario.
Append Sequence	To add a new sequence to the selected element.
Copy	To copy the element to the Clipboard.
Cut	To cut the selected element and copy it to the Clipboard.
Paste	To paste an element from the Clipboard to the current position.
Delete <element>	To delete the selected element.
Collapse	To collapse the selected block.
Expand	To expand the selected block.

Start segment When a start segment is selected, the following commands are available.

Command	Purpose
Append Segment	To add a new segment at the end of the sequence.
Append Sequence	To add a new sequence to the selected element.
Copy	To copy the element to the Clipboard.

Command	Purpose
Paste	To overwrite the start segment properties with properties of the start segment in the Clipboard.
Delete Sequence	To delete the sequence starting with the selected element.

Activity When an activity is selected, the following commands are available.

Command	Purpose
Copy	To copy the element to the Clipboard.
Paste	To overwrite the element properties with properties of the element in the Clipboard.

Transition When a transition is selected, the following commands are available.

Command	Purpose
Copy	To copy the element to the Clipboard.
Paste	To overwrite the element properties with properties of the element in the Clipboard.

Segment When a segment is selected, the following commands are available.

Command	Purpose
Insert Segment - After	To insert a new segment after the selected segment.
Insert Segment - Before	To insert a new segment before the current segment.
Append Segment	To add a new segment at the end of the current sequence.
Copy	To copy the element to the Clipboard.
Cut	To cut the selected element and copy it to the Clipboard.
Paste	To overwrite the element properties with properties of the element in the Clipboard.
Delete	To delete the selected element.

Related topics

Basics

[Graphical User Interface of the Scenario Editor](#)..... 22

Scenario Properties Dialog

Access

The Scenario Properties dialog opens when you select the Properties command.

Purpose

To configure the properties of the open scenario.

Description

The property values that you specify in this dialog must be consistent to the property values in the **Model Configuration** dialog. The **Scenario Editor** checks whether the settings of both dialogs are consistent and shows the inconsistencies. To get consistent values, you can click the **Reset** button. This sets all the settings of the **Scenario Properties** dialog to the values of the **Model Configuration** dialog.

Global Properties

Lane index interpretation Lets you specify which lane is used as reference lane when you select the lane index. For details, refer to [How to Specify a Lane Change Maneuver](#) on page 48.

Property	Description
Relative to preferred lane and driving direction	The reference lane is the preferred lane. Negative lane indices indicate lanes to the right of the preferred lane in relation to the direction of travel of the traffic participant. Positive lane indices indicate lanes to the left.
Relative to center lane (absolute lane index)	The reference lane is the center lane. The center lane is specified with the Road Generator and always has the index 0. The direction of travel of the traffic participants is not considered.

Maneuver dialog settings

Lets you specify the global settings for the maneuver.

Maneuver User Signals Lets you edit a table containing the maneuver user signals. You can add a description to each user signal and modify their names.

Maneuver Trigger Lets you edit a table containing the maneuver trigger signals. You can add a description to each trigger.

Driver Lets you specify driver settings.

Property	Description
Driver type	Lets you specify standard values for a slow, normal, and fast driver. The driver settings are used when the ASM vehicle is to follow the road (refer to Lateral Type Properties (Maneuver) on page 127).
Max. longitudinal acceleration	Lets you specify the maximum acceleration in longitudinal direction in m/s^2 .
Max. lateral acceleration	Lets you specify the maximum acceleration in lateral direction in m/s^2 .
Brake acceleration threshold	Lets you specify the brake acceleration threshold in m/s^2 .

Property	Description
Curvature used for reference velocity calculation	Lets you select which curvature is used for calculating the reference velocity: ▪ Vehicle Trajectory: The vehicle trajectory uses a curve fitting of the vehicle trajectory to calculate the reference velocity. It is recommended to use this option. ▪ Road Network: The curvature of the road network is used. It is not recommended to use this option. It exists only for migration purposes.

Lateral Controller Types Lets you select the controller type. Refer to [Lateral Controller 1 \(ASM Environment Reference !\[\]\(d84e7ea36f695d92cb39ec32c307ac93_img.jpg\)](#)) and [Lateral Controller 2 \(ASM Environment Reference !\[\]\(db9b0c6fa4ac1078c53d7f74438ad75d_img.jpg\)](#)).

Traffic dialog setting

Lets you specify the global settings for the traffic.

Fellow User Signals Lets you edit a table containing the fellow user signals. You can add a description to each user signal.

To add a new user signal, click in the * Name cell and specify a name.

To delete a user signal, open its context menu and select **Delete**.

To sort the user signals, open the context menu and select **Move Up** or **Move Up**.

Fellow Trigger Lets you edit a table containing the fellow trigger signals. You can add a description to each trigger.

To add a new trigger, click in the Name cell and specify a name.

To delete a trigger, open its context menu and select **Delete**.

To sort the triggers, open the context menu and select **Move Up** or **Move Up**.

Global dialog settings

The dialog has a context menu with the following buttons commands.

Move up Lets you move up the selected row.

Move down Lets you move down the selected row.

Reset Lets you set the property values in this dialog according to the property values of the **Model Configuration** dialog.

Delete Lets you delete the selected row.

Related topics

HowTos

[How to Specify a Lane Change Maneuver.....](#) 48

References

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Activity Properties (Maneuver).....	116
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Scenario Settings Dialog

Access	The Scenario Settings dialog opens when you select the Settings command.				
Purpose	To configure settings for the simulation model.				
Dialog settings	Minimum vector size (Simulink) Lets you specify the minimum vector size for the download. Use it to avoid recompiles of the simulation model in the MATLAB accelerator mode after downloading a modified scenario.				
Related topics	HowTos <table> <tr> <td>How to Create a Fellow User Signal.....</td> <td>57</td> </tr> </table> References <table> <tr> <td>Settings.....</td> <td>112</td> </tr> </table>	How to Create a Fellow User Signal.....	57	Settings.....	112
How to Create a Fellow User Signal.....	57				
Settings.....	112				

Traffic Driver Dialog

Access	The Traffic Driver dialog opens when you select the New or Open command.
Purpose	To configure the properties of a traffic driver.
General page	Author Lets you specify the name of the author. Description Lets you specify a description of the traffic driver.
Driving Style page	Distance at standstill Lets you specify the distance of the driver's vehicle to the vehicle ahead at standstill in meters.

Distance Lets you specify the distance of the driver's vehicle to the vehicle ahead in seconds while driving. This value is used to calculate the required distance in meters.

Detection range Lets you specify the range in front of the driver's vehicle in meters within other traffic participants are detected along the road reference line.

Maximum longitudinal acceleration Lets you specify the maximum acceleration in longitudinal direction in everyday traffic in m/s^2 .

Comfortable deceleration Lets you specify the comfortable deceleration in everyday traffic in m/s^2 .

Maximum lateral acceleration Lets you specify the maximum lateral acceleration in curves in m/s^2 .

Rule-Compliant Driving page

Distance at standstill Lets you specify the distance between the driver's vehicle and a traffic light or priority sign at standstill in meters.

Speed limit offset Lets you specify the offset to the specified speed limit at which the driver's vehicle can drive slower or faster than the specified speed limit in km/h .

Sign-compliant driving start Lets you specify the offset from the position of the speed limit or no passing start sign to the position where the driver's vehicle complies with the traffic sign in meters.

Sign-compliant driving end Lets you specify the offset from the position of the speed limit or no passing end sign to the position where the driver's vehicle complies with the traffic sign in meters.

Yield speed factor Lets you specify the deceleration factor before junctions in percent. This value is used to calculate the velocity the vehicle slows down when approaching a junction where the right-of-way status must be checked.

Speed limit mode Lets you select the mode for road scenery speed limit use.

Value	Description
Ignore scenery	The speed limit specified by the road scenery is not evaluated.
Use scenery	The speed limit specified by the road scenery is evaluated.

Traffic Sign Recognition page

Distance front Lets you specify the distance for traffic sign recognition in front of the vehicle in meters.

Distance right Lets you specify the distance for traffic sign recognition to the right of the vehicle in meters.

Distance left Lets you specify the distance for traffic sign recognition to the left of the vehicle in meters.

Distance vertical Lets you specify the distance for traffic sign recognition above the vehicle in meters.

Relative sign angle Lets you specify the maximum angle between the traffic and the vector from the sign to the vehicle main point in degrees.

Related topics

HowTos

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Scenario Editor Creation Commands

Introduction

The Scenario Editor provides various commands to specify scenarios. They are accessible via the ribbon and context menus.

Where to go from here

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To add a new segment after the last segment of the sequence.	
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To collapse all the blocks.	
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To paste a sequence from the Clipboard to the current position.	
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To configure the properties of the open scenario.	
Settings.....	112
To specify settings of the Scenario Editor.	

Append Segment

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – Segment
Shortcut key	None
Icon	None

Purpose

To add a new segment after the last segment of the sequence.

Result

A new segment is added to the sequence.

Related topics

HowTos

How to Add Sequences and Segments for the Maneuver.....	41
How to Initialize and Specify the Movements of a Fellow.....	60
How to Specify a Global User Signal.....	64

Append Sequence

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – block

Shortcut key	None
Icon	None

Purpose To add a new sequence to the selected element.

Result A new sequence is added to the maneuver, fellow, or global user signal.

Description When you add a new sequence, it is created with two segments. The first segment should be used to initialize the element, especially the maneuver or a fellow. To specify the elements, select them and specify the properties in the Properties pane.

You can add further segments by using the **Insert Segment - Before** and **Insert Segment - After** commands.

Related topics

HowTos

How to Add Sequences and Segments for the Maneuver	41
How to Initialize and Specify the Movements of a Fellow	60

References

Delete Sequence	104
Insert Segment - After	106
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Collapse

Access You can access this command via:

Ribbon	None
Context menu of	Scenario pane – block – Change View
Shortcut key	None
Icon	None
Others	Double-click on an expanded block

Purpose To collapse the selected block.

Result The block collapsed.

Related topics

References

Collapse All	102
Expand	105
Expand All	105

Collapse All

Access

You can access this command via:

Ribbon	None
Context menu of	None
Shortcut key	None
Icon	 on the Scenario pane

Purpose

To collapse all the blocks.

Result

The blocks are collapsed.

Related topics

Basics

Filtering, Collapsing, or Expanding Blocks in the Scenario Editor	24
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References

Collapse	101
Expand	105
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Copy

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – Maneuver Scenario pane – Fellow Scenario pane – Segment Scenario pane – Activity Scenario pane – Transition
Shortcut key	Ctrl+C
Icon	None

Purpose

To copy an element to the Clipboard.

Result

The selected element is copied to the Clipboard. You can paste it to another position.

Related topics

References

[Paste.....](#) 110

Cut

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – Maneuver Scenario pane – Fellow Scenario pane – Segment
Shortcut key	Ctrl+X
Icon	None

Purpose

To cut an element and copy it to the Clipboard.

Result

The selected element is deleted from the current position and copied to the Clipboard. You can paste it to a new position.

Related topics**References**

[Paste](#)..... 110

Delete <Element>

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – Maneuver Scenario pane – Fellow Scenario pane – Global user signal Scenario pane – Segment
Shortcut key	Del
Icon	None

Purpose

To delete an element.

Result

The selected element is deleted.

Related topics**References**

[Cut](#)..... 103

Delete Sequence

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – Sequence
Shortcut key	None
Icon	None

Purpose

To delete a sequence.

Result The selected sequence is deleted.

Related topics

Basics

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Expand

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – block – Change View
Shortcut key	None
Icon	None
Others	Double-click on a collapsed block

Purpose

To expand a collapsed block.

Result

The block is expanded.

Related topics

References

Collapse.....	101
Collapse All.....	102
Expand All.....	105

Expand All

Access

You can access this command via:

Ribbon	None
Context menu of	None
Shortcut key	None
Icon	 on the Scenario pane

Purpose To expand all the blocks.

Result The blocks are expanded.

Related topics

Basics

[Filtering, Collapsing, or Expanding Blocks in the Scenario Editor.....](#) 24

References

[Collapse.....](#) 101
[Collapse All.....](#) 102
[Expand.....](#) 105

Insert Segment - After

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – Segment
Shortcut key	None
Icon	None

Purpose To add a new segment after the selected segment.

Result A new segment is added to the sequence.

Related topics

HowTos

[How to Add Sequences and Segments for the Maneuver.....](#) 41
[How to Initialize and Specify the Movements of a Fellow.....](#) 60
[How to Specify a Global User Signal.....](#) 64

Insert Segment - Before

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – segment
Shortcut key	None
Icon	None

Purpose

To add a new segment before the current segment.

Result

A new segment is added to the sequence.

Related topics

HowTos

How to Add Sequences and Segments for the Maneuver.....	41
How to Initialize and Specify the Movements of a Fellow.....	60
How to Specify a Global User Signal.....	64

Layout Diagram

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane
Shortcut key	None
Icon	None

Purpose

To arrange all the maneuver, fellow, and global user signal blocks on the Scenario pane.

Result

ModelDesk arranges all the blocks of the active Scenario pane and resets the zoom level.

Related topics**Basics**

Filtering, Collapsing, or Expanding Blocks in the Scenario Editor	24
Graphical User Interface of the Scenario Editor	22

New – Fellow

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane
Shortcut key	None
Icon	None

Purpose

To add a new fellow to the scenario.

Result

A new fellow is added to the scenario and can be specified.

Description

When you add a new fellow, it is created with two segments. The first segment should be used to initialize the fellow. To specify the segments, select them and specify their properties in the Properties pane.

You can add further segments by using the **Insert Segment - Before** and **Insert Segment - After** commands.

Related topics**HowTos**

How to Create and Specify a Fellow	59
--	----

References

Fellow Properties	120
Insert Segment - After	106
Insert Segment - Before	107
Start Properties (Fellow)	153

New – Maneuver

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane
Shortcut key	None
Icon	None

Purpose

To add a maneuver to the scenario.

Result

A maneuver is added to the scenario and can be specified.

Description

The command is disabled if the scenario already contains a maneuver. A scenario must contain only one maneuver.

When you add a maneuver, it is created with one segment. To specify the segment, select it and specify their properties in the **Properties** pane.

You can add further segments by using the **Insert Segment - Before** and **Insert Segment - After** commands.

Related topics

HowTos

[How to Start Scenario Creation](#)..... 39

References

[Insert Segment - After](#)..... 106
[Insert Segment - Before](#)..... 107
[Start Properties \(Maneuver\)](#)..... 155

New – User Signal

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane
Shortcut key	None
Icon	None

Purpose	To add a new global user signal to the scenario.
Result	A new global user signal is added to the scenario.
Description	When you add a new global user signal, it is created with one segment. To specify the user signal, select the segment and specify its properties in the Properties pane.
Related topics	HowTos How to Specify a Global User Signal..... 64 References Start Properties (Global User Signal)..... 154

Paste

Access	You can access this command via: <table> <tr> <td>Ribbon</td><td>None</td></tr> <tr> <td>Context menu of</td><td> Scenario pane – Fellow Scenario pane – Segment Scenario pane – Activity Scenario pane – Transition </td></tr> <tr> <td>Shortcut key</td><td>Ctrl+V</td></tr> <tr> <td>Icon</td><td>None</td></tr> </table>	Ribbon	None	Context menu of	Scenario pane – Fellow Scenario pane – Segment Scenario pane – Activity Scenario pane – Transition	Shortcut key	Ctrl+V	Icon	None
Ribbon	None								
Context menu of	Scenario pane – Fellow Scenario pane – Segment Scenario pane – Activity Scenario pane – Transition								
Shortcut key	Ctrl+V								
Icon	None								
Purpose	To paste an element from the Clipboard to the current position.								
Result	The element is pasted to the current position.								
Related topics	References Copy..... 103 Cut..... 103								

Paste Sequence

Access

You can access this command via:

Ribbon	None
Context menu of	Scenario pane – Sequence
Shortcut key	None
Icon	None

Purpose

To paste a sequence from the Clipboard to the current position.

Result

The sequence is pasted to the current position.

Related topics

References

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Properties

Access

You can access this command via:

Ribbon	Environment – Scenario
Context menu of	None
Shortcut key	None
Icon	

Purpose

To configure the properties of the open scenario.

Result

The Scenario Properties dialog opens.

Related topics

References

Scenario Properties Dialog.....	93
---	----

Settings

Access

You can access this command via:

Ribbon	Environment – Scenario
Context menu of	None
Shortcut key	None
Icon	

Purpose

To specify settings of the Scenario Editor.

Description

The Scenario Settings Dialog dialog opens.

Related topics

HowTos

How to Create a Fellow User Signal.....	57
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References

Scenario Settings Dialog.....	96
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Scenario Editor Properties

Overview

The Scenario Editor provides a Properties pane to specify the properties of the active scenario.

Where to go from here

Information in this section

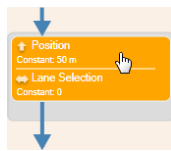
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To specify the properties of an activity of a global user signal.	
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To specify the properties of an activity of a maneuver.	
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To specify the properties of the clutch and gear in an activity for an ASM vehicle with manual transmission.	
Fellow Properties	120
To specify the properties of a fellow.	
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To specify the properties of a global user signal.	
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To specify the properties of an activity in the lateral direction of the ASM vehicle.	
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To specify the properties of a maneuver.	
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To specify the properties of a transition.	
Source Properties.....	162
The source properties are used in several properties to specify the kind of source.	

Activity Properties (Fellow)

Purpose To specify the properties of an activity of a fellow.

Access To access the properties, click the activity on the Scenario pane.



Properties

Consistency Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given.

Description Lets you specify a description for the activity.

Fellow User Signals Lets you select and specify a fellow user signals. The fellow user signals must have been created before in the [Scenario Properties Dialog](#) on page 93.

Longitudinal Lets you specify the activity in the longitudinal direction. For details of the properties, refer to [Longitudinal Type Properties \(Fellow\)](#) on page 135.

Lateral Lets you specify the activity in the lateral direction. For details of the properties, refer to [Lateral Type Properties \(Fellow\)](#) on page 123.

Related topics

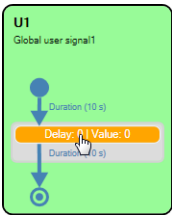
Basics

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---	--------------------

Activity Properties (Global User Signal)

Purpose To specify the properties of an activity of a global user signal.

Access To access the properties, click the activity on the Traffic Scenario pane.



Properties

Consistency Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given.

Delay Lets you specify a time delay after which the specified value is applied after the start of the segment in seconds.

Description Lets you specify a description for the activity.

Parameters Lets you specify the parameters of the source type that is selected for the global user signal.

Type	Property	Description
Constant	Constant	Lets you specify a constant value.
Ramp	Start with current value	Lets you specify to start with the current value.
	Start	Lets you specify the start value.
	End	Lets you specify the end value.
	Extent	Lets you specify the time or distance that is required to change the user signal from the start value to the end value.
	Interpretation type	Lets you specify whether the value of Extent is specified in seconds or meters.
Pulse	Value	Lets you specify the maximum value of the pulse.
	Step 1 start	Lets you specify the time or distance when the pulse starts.
	Step 1 end	Lets you specify the time or distance when the pulse reaches the specified value.
	Step 2 start	Lets you specify the time or distance when the starts falling.
	Step 2 end	Lets you specify the time or distance when the value is zero.

Type	Property	Description
	Interpretation type	Lets you specify whether the start and end values are specified as time in seconds or distance in meters.
Table	Edit table	Lets you specify values in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify whether the values of the table are related to the time or distance.

Type Lets you specify the source type for the global user signal.

Related topics

Basics

[Basics of Activities](#)..... 30

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[Edit Table Dialog](#)..... 87

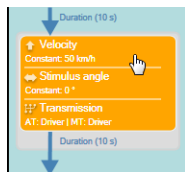
Activity Properties (Maneuver)

Purpose

To specify the properties of an activity of a maneuver.

Access

To access the properties, click the activity of the maneuver on the Scenario pane.



Properties

Clutch and gear (MT) Lets you specify the settings of the clutch and gear for an ASM vehicle that has a manual transmission. For details of the properties, refer to [Clutch and Gear \(MT\) Properties](#) on page 117.

Consistency Displays the consistency state. If the specified properties are inconsistent, the reasons for the inconsistency are given.

- Description** Lets you specify a description for the activity.
- Longitudinal** Lets you specify the activity in the longitudinal direction. For details of the properties, refer to [Longitudinal Type Properties \(Maneuver\)](#) on page 143.
- Lateral** Lets you specify the activity in the lateral direction. For details of the properties, refer to [Lateral Type Properties \(Maneuver\)](#) on page 127.
- Maneuver User Signals** Lets you select and specify maneuver user signals. The maneuver user signals must have been created before in the [Scenario Properties Dialog](#) on page 93.
- Selector lever (AT)** Lets you specify the selector lever for an ASM vehicle that has an automatic transmission. For details of the properties, refer to [Selector Lever \(AT\) Properties](#) on page 151.

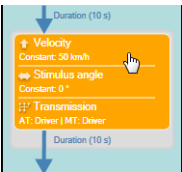
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How to Follow a Road Automatically.....	46
How to Select and Specify Maneuver User Signals.....	53
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Clutch and Gear (MT) Properties

Purpose To specify the properties of the clutch and gear in an activity for an ASM vehicle with manual transmission.

Access To access the properties, click the activity of the maneuver on the Scenario pane.



Properties

Clutch and gear type Lets you specify the type of the clutch and gear settings of the maneuver.

Clutch and Gear Type	Description
Open clutch	Lets you specify an opened clutch. Refer to Open Clutch properties on page 118.
Driver	Lets you specify that the driver model specifies the clutch and gear. You can saturate the gear. Refer to Driver properties on page 118.
Stimulus	Lets you specify the clutch and gear settings are specified by properties in ModelDesk. Refer to Stimulus properties on page 118.
Reference gear	Lets you specify a specific gear used during the activity. Refer to Reference Gear properties on page 120.

Open Clutch properties

An Open Clutch type has no properties.

Driver properties

Saturate gear Lets you enable the saturation of the gear.

Gear Lets you enter a number specifying the highest gear the driver model can use.

Stimulus properties

Gear source type Lets you specify the gear source type.

Type	Description
Constant	Lets you specify a constant value for the gear. You can also use the gear used in the previous segment.
Extern	Lets you control the gear via Simulink model.
Manual	Lets you control the gear manually via experiment software.
Table	Lets you enter values for gears shift over time [s] or distance [m].

Gear source properties Lets you specify the properties of the selected gear source type.

Type	Property	Description
Constant	Hold current value	Lets you specify to use the gear of the previous segment.
	Gear	Lets you specify a specific gear.
Extern	–	Parameters are not required.
Manual	–	Parameters are not required.
Table	Edit table	Lets you edit a table that contains the values for stimulation. Refer to Edit Table Dialog on page 87.

Type	Property	Description
	Interpretation type	Lets you specify whether the values of the table are related to the time or distance.

Clutch source type Lets you specify the clutch source type.

Type	Description
Constant	Lets you specify a constant value for the clutch position or you can use the clutch position of the previous segment.
Ramp	Lets you specify a linear change of the clutch position.
Extern	Lets you control the clutch position via Simulink model.
Manual	Lets you control the clutch position manually via experiment software.
Table	Lets you enter values for clutch position over time [s] or distance [m].

Clutch source properties Lets you specify the properties of the selected clutch source type.

Type	Property	Description
Constant	Hold current value	Lets you specify to use the clutch position of the previous segment.
	Position	Lets you specify a specific clutch position.
Ramp	Start with current value	Lets you specify to start with the clutch position of the previous segment.
	Start value	Lets you specify the start value of the clutch position.
	End value	Lets you specify the end value of the clutch position.
	Extent	Lets you specify the time or distance that is required to change the clutch position from the start value to the end value.
	Interpretation type	Lets you specify whether the value of Extent is specified as time or distance.
Extern	–	Parameters are not required.
Manual	–	Parameters are not required.
Table	Edit table	Lets you edit a table that contains the values for the position in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify whether the values of the table are related to the time or distance.

Reference Gear properties**Gear source type** Lets you specify the gear source type.

Type	Description
Constant	Lets you enter a number to indicate one gear, for example, the number 1 for the first gear.
Extern	Lets you control the gear via Simulink model.
Manual	Lets you control the gear manually via experiment software.
Table	Lets you enter values for gears shift over time [s] or distance [m].

Gear source properties Lets you specify the properties of the selected gear source type.

Type	Property	Description
Constant	Hold current value	Lets you specify to use the gear of the previous segment.
	Gear	Lets you specify a specific gear.
Extern	–	Parameters are not required.
Manual	–	Parameters are not required.
Table	Edit table	Lets you edit a table that contains the values for the gear in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify whether the values of the table are related to the time or distance.

Related topics**HowTos**

[How to Shift Gears in a Maneuver](#)..... 44

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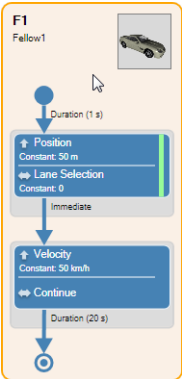
Fellow Properties

Purpose

To specify the properties of a fellow.

Access

To access the properties, click the fellow on the Scenario pane.



Properties

Consistency Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given.

Description Lets you specify a description for the fellow.

Image Displays the selected traffic object.

Name Lets you specify the fellow's name.

Open Traffic Driver Opens the Traffic Driver dialog for you to edit the traffic driver. Refer to [Traffic Driver Dialog](#) on page 96.

Relative path Displays the relative path to the file of the traffic object.

Simulation ID Lets you specify the fellow's index. The index sets the order of the fellows in the traffic vector of the simulation model. Refer to [Traffic Scheduler \(ASM Traffic Reference\)](#).

Traffic Driver Lets you select the traffic driver that is used for the fellow. You can edit the object in a dialog, refer to [How to Specify the Properties of a Traffic Driver](#) on page 66.

Traffic Object Lets you select a traffic object to be used as the geometry for the fellow. When you click the button, the Traffic Object Browser opens to let you select a traffic object.

Tip

If the available traffic objects are not suitable for your fellow, you can create further traffic objects. Refer to [How to Create a Traffic Object \(ModelDesk Traffic Object Management\)](#).

Related topics

Basics

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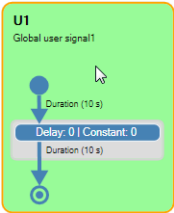
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Manage Traffic Objects (ModelDesk Traffic Object Management )

Global User Signal Properties

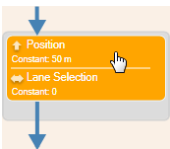
Purpose	To specify the properties of a global user signal.
Access	To access the properties, click the fellow on the Scenario pane. 
Properties	Description Lets you specify a description for the global user signal. Model Configuration Name Displays the name of the global user signal specified in the Model Configuration dialog. Refer to Environment Configuration Dialog on page 88. Name Lets you specify the name of the global user signal. Simulation ID Lets you specify the global user signal's index.

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Lateral Type Properties (Fellow)

Purpose To specify the properties of an activity in the lateral direction.

Access To access the properties, click the activity on the Scenario pane.



Properties **Lateral type** Lets you specify the longitudinal movement of a fellow in the activity.

Lateral Type	Description
Not used	Lets you specify that the fellow is not used in the specified activity. Refer to Not used properties on page 123.
Continue	Lets you specify that the fellow continues with the settings of the previous activity. Refer to Continue properties on page 123.
Lateral deviation	Lets you specify the lateral deviation of the fellow on the road. Refer to Lateral deviation properties on page 123.
Lane selection	Lets you specify the lane selection of a fellow. Refer to Lane selection properties on page 125.

Not used properties A Not Used lateral type has no properties.

Continue properties A Continue lateral type has no properties.

Lateral deviation properties

Point on reference object Lets you select the point on the reference object when the Reference Type property is set to Relative. The lateral deviation is the distance between this point and the point selected by the Reference point property.

Reference point Lets you select a reference point of the ego-vehicle when the Reference Type property is set to Relative. The lateral deviation is the distance between this point and the point selected by the Point on reference object property.

Reference type Lets you select whether the fellow's velocity is specified with absolute values or relative to a vehicle.

Reference object Lets you select the reference object when the Reference type property is set to Relative.

Type Lets you select the profile type for the fellow's lateral deviation profile. The following types are possible:

- **Constant:** Stimulus with constant values
- **Extern:** Stimulus with a value defined by an external signal
- **Final:** Stimulus with a final value at the end of the current segment
- **Ramp:** Stimulus with a ramp profile
- **Smooth lane change:** Stimulus with a smooth sine profile
- **Table:** Stimulus with values of a table

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you select whether the fellow's lateral deviation is kept from the end of the last segment until the end of the active segment.
	Constant	Lets you specify the fellow's lateral deviation.
Extern	–	It is not necessary to specify a property for a stimulus with a value defined by an external signal.
Final	Final deviation	Lets you specify the target lateral deviation from the reference line. It can happen that the final deviation is not reached at the end of the segment. This depends on the lateral velocity and the duration of the segment.
	Lateral velocity	Lets you specify the fellow's maximum lateral velocity for the lateral deviation.
Ramp	Start deviation	Lets you specify the ramp profile's initial value for the fellow's lateral deviation.
	End deviation	Lets you specify the ramp profile's final value for the fellow's lateral deviation.
	Extent	Lets you specify the time for the fellow to change from the initial lateral deviation to the final lateral deviation.
Smooth lane change	Start deviation	Lets you specify the profile's initial value for the fellow's lateral deviation.
	End deviation	Lets you specify the profile's final value for the fellow's lateral deviation.
	Duration	Lets you specify the time for the fellow to change from the initial lateral deviation to the final lateral deviation.
Table	Edit table	Lets you specify values for the lateral deviation in relation to the time or distance. Refer to Edit Table Dialog on page 87.

Type	Property	Description
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Lane selection properties

Force to road Lets you enable/disable that the nearest available lane index is used if the defined lane index is not available in the current road lane section.

Reference type Lets you select whether the fellow's lane selection is specified relative to a lane or to an object.

Reference object Lets you select the reference object when the Reference type property is Relative to object.

Type Lets you select the type for the fellow's lane selection. The following types are possible:

- **Constant:** The fellow constantly uses the same lane (either a specified lane or the lane of the previous segment).
- **Lane change:** The fellow changes lanes linearly within a specified time.
- **Smooth lane change:** The fellow changes lanes smoothly within a specified time.
- **Final:** Lane change from the lane used in the previous segment to a specified final lane. You can specify the lateral velocity.
- **Extern:** Lane selection with a value defined by an external signal. Non-integer values are allowed. For their interpretation, see [Road \(ASM Environment Reference\)](#).
- **Table:** Stimulus with values of a table.

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you select whether the lane index is held from the end of the last segment until the end of the active segment.
	Lane index	Lets you specify the lane index used by the fellow.
Lane change	Start lane index	Lets you specify the index of the lane used by the fellow at the start of the segment.
	End lane index	Lets you specify the index of the lane used by the fellow at the end of the segment.

Type	Property	Description
	Extent	Lets you specify the time for the fellow to change from the initial lane to the final lane.
Smooth lane change	Start lane index	Lets you specify the index of the lane used by the fellow at the start of the segment.
	End lane index	Lets you specify the index of the lane used by the fellow at the end of the segment.
	Extent	Lets you specify the time for the fellow to change from the initial lane to the final lane.
Final	Lateral velocity	Lets you specify the fellow's maximum lateral velocity necessary to reach the specified lane at the end of the current segment.
	Final lane index	Lets you specify the index of the lane used by the fellow at the end of the current segment.
Extern	–	It is not necessary to specify a property for a stimulus with a value defined by an external signal.
Table	Edit table	Lets you specify values for the lane selection in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Related topics

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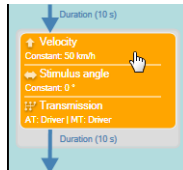
References

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Lateral Type Properties (Maneuver)

Purpose To specify the properties of an activity in the lateral direction of the ASM vehicle.

Access To access the properties, click the activity of the maneuver on the **Scenario** pane.



Properties **Lateral type** Lets you specify the movement of the ASM vehicle in lateral direction in the activity.

Lateral Type	Description
Basic road	Lets you specify that the ASM vehicle drives straight ahead or circular. Refer to Basic road properties on page 132.
Follow road	Lets you specify that the ASM vehicle follows the road. Refer to Follow road properties on page 132.
Not used	Lets you specify that the lateral movement is not changed in the activity. Refer to Not used properties on page 127.
Stimulus angle	Lets you specify that lateral movement is stimulated by specifying the steering angle. Refer to Stimulus angle properties on page 127.
Stimulus torque	Lets you specify that the lateral movement is stimulated by specifying the torque at the steering wheel. Refer to Stimulus torque properties on page 130.

Not used properties A Not used lateral type has no properties.

Stimulus angle properties **Type** Lets you specify the stimulus type.

Type	Description
Constant	Lets you specify a constant value for the steering angle. You can also continue with the steering angle of the previous segment.
Ramp	Lets you specify a linear change of the steering angle.
Extern	Lets you control the steering angle via Simulink model.
Manual	Lets you control the steering angle manually via experiment software.
Fixed	Lets you specify a fixed steering angle.

Type	Description
Step	Lets you specify stepwise change of the steering angle.
Pulse	Lets you specify a pulsed change of the steering angle.
Sine with dwell	Lets you specify a sine with dwell change of the steering angle.
Table	Lets you enter values for the steering angle over time or distance.

Parameters Lets you specify the parameters of the source type that is selected for the angle stimulus.

Type	Property	Description
Constant	Hold current value	Lets you select whether the steering angle is kept from the end of the previous segment.
	Angle	Lets you specify a constant steering angle in degrees.
Ramp	Start with current value	Lets you specify to start with the steering angle of the previous segment.
	Start angle	Lets you specify the start value of the steering angle in degrees.
	End angle	Lets you specify the end value of the steering angle in degrees.
	Extent	Lets you specify the duration or distance that is required to change the steering angle from the start value to the end value.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Extern	–	Parameters are not required.
Sine	Start amplitude	Lets you specify the amplitude of the steering angle at the start in degrees.
	End amplitude	Lets you specify the amplitude of the steering angle at the end in degrees.
	Start frequency	Lets you specify the frequency of the steering angle at the start in Hz.
	End frequency	Lets you specify the frequency of the steering angle at the end in Hz.

Type	Property	Description
	Offset	Lets you specify an offset value in degrees.
	Duration	Lets you specify the duration in seconds.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Manual	–	Parameters are not required.
Fixed	–	Parameters are not required.
Step	Direction Angle Step start Step end Interpretation type	Lets you specify the direction. Lets you specify the steering angle in degrees. Lets you specify the start value. Lets you specify the end value. Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Pulse	Direction Angle Step 1 start Step 1 end Step 2 start Step 2 end Interpretation type	Lets you specify the direction. Lets you specify the maximum steering angle. Lets you specify the start value. Lets you specify the end value (when the steering angle is the maximum angle). Lets you specify the start value. Lets you specify the end value (when the steering value is zero). Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds.

Type	Property	Description
		▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Sine with dwell	Direction Angle Frequency Dwell Interpretation type	Lets you specify the direction. Lets you specify the amplitude of the steering angle in degrees. Lets you specify the frequency in Hz. Lets you specify the length of the dwell. Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Table	Edit table Interpretation type	Lets you specify values for the steering angle in relation to the time or distance. Refer to Edit Table Dialog on page 87. Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Stimulus torque properties**Type** Lets you specify the stimulus type.

Type	Description
Constant	Lets you specify a constant value for the torque. You can also continue with the torque of the previous segment.
Ramp	Lets you specify a linear change of the torque.
Extern	Lets you control the torque via Simulink model.

Type	Description
Manual	Lets you control the torque manually via experiment software.
Table	Lets you enter values for the torque over time or distance.

Parameters Lets you specify the parameters of the source type that is selected for the torque stimulus.

Type	Property	Description
Constant	Hold current value	Lets you select whether the torque is kept from the end of the previous segment.
	Angle	Lets you specify a constant torque in Nm.
Ramp	Start with current value	Lets you specify to start with the torque of the previous segment.
	Start angle	Lets you specify the start value of the torque in Nm.
	End angle	Lets you specify the end value of the torque in Nm.
	Extent	Lets you specify the time or distance that is required to change the torque from the start value to the end value.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Extern	–	Parameters are not required.
Manual	–	Parameters are not required.
Table	Edit table	Lets you specify values for the torque in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Basic road properties**Steering type** Lets you specify the steering type.

Type	Description
Circle	Lets you specify that the ASM vehicle drives a circle with a constant radius.
Straight	Lets you specify that the ASM vehicle drives straight ahead.

Parameters Lets you specify the parameters of the source type that is selected for the steering type.

Type	Property	Description
Circle	Radius	Lets you specify the radius of the circle to drive in meters.
	Direction	Lets you specify the direction.
	Control mode	Lets you specify the control mode: - Position: This is suitable for all maneuvers without road (i.e., with a basic road). - Yaw rate: This control is without preview and is intended only for the circle lateral type, in particular for steady-state maneuvers. For more information, refer to Lateral Controller 2 (ASM Environment Reference ).
Straight	–	Parameters are not required.

Follow road properties**Adapt velocity to road** Lets you specify that the driver adapts the velocity to the road.**Driver type** Lets you select the driver type. You can use a predefined driver type (refer to [Scenario Properties Dialog](#) on page 93) or specify a user-defined driver.**Maximum longitudinal acceleration** Lets you specify the maximum acceleration in longitudinal direction in m/s^2 .**Maximum lateral acceleration** Lets you specify the maximum acceleration in lateral direction in m/s^2 .**Brake acceleration threshold** Lets you specify the brake acceleration threshold in m/s^2 .**Enable lateral driver tolerance** Lets you enable the lateral driver tolerance.**Lateral driver tolerance** Lets you specify the lateral driver tolerance in meters. For more information, refer to [Lateral Controller 1 \(ASM Environment Reference !\[\]\(626ce8ac21792b9405bfddfea8e0c96a_img.jpg\)](#)) and [Lateral Controller 2 \(ASM Environment Reference !\[\]\(2b752d244c1fc411d86684a042d55b85_img.jpg\)](#)).

Reference type Lets you specify how the lane index is interpreted. For details, refer to [How to Specify a Lane Change Maneuver](#) on page 48.

Reference Type	Description
Relative to lane	The lane index is interpreted as specified in the global properties in the scenario settings. Refer to Scenario Properties Dialog on page 93.
Relative to object	The lane index is interpreted relative to the lane index of another reference object. The route and the direction of travel of the reference object and the fellow vehicle must be identical. Lanes to the right of the fellow vehicle in relation to its direction of travel have negative indices. Lane to the left have positive indices.

Reference object Displays the reference object to which the lane index is related. It is limited to the ASM vehicle.

Lane index type Lets you specify the lane index type.

Lane Index Type	Description
Constant	Lets you specify a lane index where the ASM vehicle drives or specify that the ASM vehicle stays on the current lane.
Extern	Lets you specify the lane index via Simulink model.
Final	Lets you specify a lane change by specifying the target lane and a lateral velocity.
Manual	Lets you specify the lane index via experiment software.
Table	Lets you enter values for the lane index over time or distance.

Parameters (lane index) Lets you specify the parameters of the source type that is used as lane index type. Non-integer values are allowed as lane index. For their interpretation, refer to [Road \(ASM Environment Reference\)](#).

Type	Property	Description
Constant	Hold current value	Lets you select that the ASM vehicle stays on the current lane.
	Lane index	Lets you specify the lane where the ASM vehicle drives.
Extern	–	Parameters are not required.
Final	Final lane index	Lets you specify the target lane.
	Lateral velocity	Lets you specify the lateral velocity of the ASM vehicle when it changes the lane.
Manual	–	Parameters are not required.
Table	Edit table	Lets you specify values for the lane index in relation to the time or distance. Refer to Edit Table Dialog on page 87.

Type	Property	Description
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Force to road Lets you select whether the ASM vehicle is forced to stay on the road. If the option is set and a lane outside the road is selected, the ASM vehicle remains on the road and drives on the outermost available lane. Otherwise, the vehicle drives besides the road.

Lateral offset type Lets you specify the lateral offset type.

Lane Index Type	Description
Constant	Lets you specify a constant distance of the ASM vehicle to the lane in meters. You can specify a specific value or use the value of the previous segment.
Ramp	Lets you specify a linear change of the distance of the ASM vehicle to the lane.
Extern	Lets you specify the distance of the ASM vehicle to the lane via Simulink model.
Manual	Lets you specify the distance of the ASM vehicle to the lane via experiment software.
Table	Lets you enter values for the distance of the ASM vehicle to the lane over time or distance.

Parameters (lateral offset) Lets you specify the parameters of the source type that is used as lateral offset type.

Type	Property	Description
Constant	Hold current value	Lets you select that the ASM vehicle stays on the current lane.
	Lateral offset	Lets you specify the lane where the ASM vehicle drives.
Ramp	Start with current value	Lets you specify to use the distance of the ASM vehicle to the lane of the previous lane.
	Start offset	Lets you specify the distance of the ASM vehicle to the lane at the start.
	End offset	Lets you specify the distance of the ASM vehicle to the lane at the end.
	Extent	Lets you specify the time or distance that is required to change the offset from the start value to the end value.

Type	Property	Description
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Extern	–	Parameters are not required.
Manual	–	Parameters are not required.
Table	Edit table	Lets you specify values for the lane index in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Related topics

HowTos

How to Follow a Road Automatically	46
How to Specify a Lane Change Maneuver	48
How to Steer in a Maneuver	49

References

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Lateral Controller 1 (ASM Environment Reference )	
Lateral Controller 2 (ASM Environment Reference )	

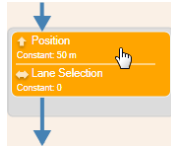
Longitudinal Type Properties (Fellow)

Purpose

To specify the properties of an activity in the longitudinal direction.

Access

To access the properties, click the activity on the Scenario pane.

**Properties**

Longitudinal type Lets you specify the longitudinal movement of a fellow in the activity.

Longitudinal Type	Description
Not used	Lets you specify that the fellow is not used in the specified activity. Refer to Not used properties on page 136.
Continue	Lets you specify that the fellow continues with the settings of the previous activity. Refer to Continue properties on page 136.
Acceleration	Lets you specify the acceleration of the fellow. Refer to Acceleration properties on page 137.
Distance [m]	Lets you specify the fellow's distance in meters relative to another fellow or the ASM vehicle. Refer to Distance m properties on page 137.
Distance [s]	Lets you specify the fellow's distance in seconds relative to another fellow or the ASM vehicle. Refer to Distance s properties on page 138.
Position	Lets you specify the longitudinal position of the fellow from the road start position. Refer to Position properties on page 139.
Traffic Driver	Lets you use a traffic driver for the fellow. Refer to Traffic Driver properties on page 140.
Velocity	Lets you specify the velocity of the fellow. Refer to Velocity properties on page 140.
Stop	Lets you specify the fellow's deceleration until stopping within a specified distance. Refer to Stop properties on page 142.

Not used properties

A Not Used longitudinal type has no properties.

Continue properties

A Continue longitudinal type has no properties.

Acceleration properties

Fellow dynamics Lets you select the behavior of the fellow for realistic behavior in curves and acceleration:

- Activated: The fellow behavior is specified with the properties of the selected traffic object.
- Deactivated: The fellow drives exactly with the specified acceleration.

Type Lets you select the profile type for the fellow's acceleration profile. The following types are possible:

- Constant: Stimulus with constant values
- Ramp: Stimulus with a ramp profile
- Table: Stimulus with values of a table

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Acceleration	Lets you specify the acceleration in m/s^2 .
	Start acceleration	Lets you specify the ramp profile's initial value for acceleration in the longitudinal direction in m/s^2 .
	Duration	Lets you specify for how long the fellow accelerates in the longitudinal direction in seconds.
Ramp	End acceleration	Lets you specify the ramp profile's final value for acceleration in the longitudinal direction in m/s^2 .
	Edit table	Lets you specify values for the acceleration. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Valid only Time (Segment): The values are a time duration in seconds.
Table		

Distance m properties

Reference point Lets you select the reference point of the selected fellow.

Reference object Lets you select the object, for example, a vehicle, relative to which the fellow's distance is specified.

Point on reference object Lets you select the reference point of the referenced object that is used in the calculation of the distance.

Type Lets you select the profile type for the fellow's distance profile. The following types are possible:

- Constant: Stimulus with constant values
- Ramp: Stimulus with a ramp profile
- Extern: Stimulus with a value defined by an external signal
- Final: Stimulus with a final value at the end of the current segment
- Table: Stimulus with values of a table

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you select whether the fellow's distance in meters relative to another vehicle is held from the end of the last segment until the end of the active segment.
	Distance	Lets you specify the fellow's distance in meters relative to another vehicle.
Extern	–	It is not necessary to specify a property for a stimulus with a value defined by an external signal.
Final	Distance	Lets you specify the fellow's distance in meters relative to another vehicle.
	Relative velocity	Lets you specify the fellow's velocity relative to another vehicle at the end of the current segment.
Ramp	Start distance	Lets you specify the ramp profile's initial value for the fellow's distance in meters relative to another vehicle.
	End distance	Lets you specify the ramp profile's final value for the fellow's distance in meters relative to another vehicle.
	Duration	Lets you specify the time for the fellow to move from the start distance to the end distance.
Table	Edit table	Lets you specify values for the distance in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Distance s properties

Reference vehicle Lets you select the vehicle relative to which the fellow's distance is specified.

Type Lets you select the profile type for the fellow's distance profile. The following types are possible:

- **Constant:** Stimulus with constant values
- **Ramp:** Stimulus with a ramp profile
- **Extern:** Stimulus with a value defined by an external signal

- **Final:** Stimulus with a final value at the end of the current segment
- **Table:** Stimulus with values of a table

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you select whether the fellow's distance in seconds relative to another vehicle is held from the end of the last segment until the end of the active segment.
	Distance	Lets you specify the fellow's distance in seconds relative to another vehicle.
Extern	–	It is not necessary to specify a property for a stimulus with a value defined by an external signal.
Final	Distance	Lets you specify the fellow's distance in seconds relative to another vehicle.
	Relative velocity	Lets you specify the fellow's velocity relative to another vehicle at the end of the current segment.
Ramp	Start distance	Lets you specify the ramp profile's initial value for the fellow's distance in seconds relative to another vehicle.
	End distance	Lets you specify the ramp profile's final value for the fellow's distance in seconds relative to another vehicle.
	Duration	Lets you specify the time for the fellow to move from the start distance to the end distance.
Table	Edit table	Lets you specify values for the distance in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Position properties

Type Lets you select the profile type for the fellow's position profile. The following types are possible:

- **Constant:** Stimulus with constant values
- **Ramp:** Stimulus with a ramp profile

- Extern: Stimulus with a value defined by an external signal
- Table: Stimulus with values of a table

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you select whether the fellow's longitudinal position is held from the end of the last segment until the end of the active segment.
	Position	Lets you specify the fellow's longitudinal position from the road start position.
	Yaw angle	Lets you specify the fellow's initial orientation relative to the direction of the road reference line.
Extern	–	It is not necessary to specify a property for a stimulus with a value defined by an external signal.
Ramp	Start position	Lets you specify the ramp profile's initial value for the fellow's longitudinal position.
	End position	Lets you specify the ramp profile's final value for the fellow's longitudinal position.
	Duration	Lets you specify the time for the fellow to move from the start position to the end position.
Table	Edit table	Lets you specify values for the distance in relation to the time. Refer to Edit Table Dialog on page 87.

Traffic Driver properties

Type Lets you select the profile type for the fellow's distance profile. The following types are possible:

- Constant: Stimulus with constant values

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you select whether the fellow's velocity is held from the end of the last segment until the end of the active segment.
	Distance	Lets you specify the fellow's velocity in km/h.

Velocity properties

Fellow dynamics Lets you select the behavior of the fellow for realistic behavior in curves and acceleration:

- Activated: The fellow behavior is specified with the properties of the selected traffic object.
- Deactivated: The fellow drives exactly with the specified velocity.

Reference type Lets you select whether the fellow's velocity is specified with absolute values or relative to a vehicle.

Reference vehicle Lets you select the reference vehicle when the Reference Type property is set to Relative.

Type Lets you select the profile type for the fellow's velocity profile. The following types are possible:

- **Constant:** Stimulus with constant values
- **Ramp:** Stimulus with a ramp profile
- **Extern:** Stimulus with a value defined by an external signal
- **Final:** Stimulus with a final value at the end of the current segment
- **Sine:** Stimulus with a sinusoidal form in the current segment
- **Table:** Stimulus with values of a table

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you select whether the fellow's velocity is held from the end of the last segment until the end of the active segment.
	Constant	Lets you specify the fellow's velocity.
Extern	–	Parameters are not required.
Final	Final velocity	Lets you specify the fellow's velocity at the end of the current segment.
	Positive acceleration	Lets you specify the acceleration value for reaching the final value.
	Negative acceleration	Lets you specify the deceleration value for reaching the final value.
Ramp	Start velocity	Lets you specify the ramp profile's initial value for the fellow's velocity.
	End velocity	Lets you specify the ramp profile's final value for the fellow's velocity.
	Extent	Lets you specify the time for the fellow to accelerate or decelerate from the start velocity to the end velocity.
Sine	Amplitude	Lets you specify the amplitude of the velocity in km/h.
	Start frequency	Lets you specify the initial frequency of the sine wave in Hz.
	End frequency	Lets you specify the final frequency of the sine wave in Hz.
	Offset	Lets you specify an offset value.
	Extent	Lets you specify the extent (duration or distance) for the fellow to accelerate and brake in a sinusoidal form from the start frequency to the end frequency.

Type	Property	Description
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Table	Edit table	Lets you specify values for the velocity in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Stop properties

Type (read-only) Displays the constant profile as the acceleration stimulus type.

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Stop within distance	Hold current value	Lets you specify to use the velocity of the previous segment for the fellow. If you enable the option, you do not have to specify the Velocity property.
	Velocity	Lets you specify the velocity of the fellow at the beginning of the segment in km/h.
	Deceleration	Lets you specify a deceleration for the braking process in m/s ² .
	Stop distance	Lets you specify the distance in meters in which the fellow must stop.

Related topics

HowTos

[How to Create and Specify a Fellow.....](#) 59

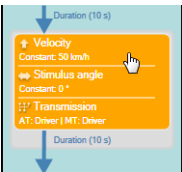
References

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Longitudinal Type Properties (Maneuver)

Purpose To specify the properties of a maneuver's activity in the longitudinal direction.

Access To access the properties, click the activity of the maneuver on the Scenario pane.



Properties **Longitudinal type** Lets you specify the longitudinal movement of the ASM vehicle in the activity.

Longitudinal Type	Description
Lateral acceleration	Lets you specify the lateral acceleration of the ASM vehicle. All the longitudinal definitions of the current segment are ignored. The longitudinal velocity depends on the lateral acceleration. This type can be used only if the lateral type is Basic road and the steering type is Circle . Refer to Lateral acceleration properties on page 144.
Pedal stimulus	Lets you specify the stimulation of the acceleration pedal and brake pedal. Refer to Pedal stimulus properties on page 144.
Standstill	Lets you specify the standstill of the ASM vehicle. Refer to Standstill properties on page 147.
Stop	Lets you specify to stop the ASM vehicle within a specified distance or brake until stop. Refer to Stop properties on page 147.
Traffic Driver	Lets you use the selected traffic driver for the ASM vehicle. Refer to Traffic Driver properties on page 147.
Velocity	Lets you specify the velocity of the ASM vehicle. Refer to Velocity properties on page 147.

Lateral acceleration properties

Type Lets you select the profile type for the lateral acceleration. The following types are possible:

- **Constant:** Specifies the lateral acceleration using a constant values
- **Ramp:** Specifies the lateral acceleration using a ramp profile
- **Table:** Specifies the lateral acceleration using a table

Parameters Lets you specify the parameters of the source type that is selected for the lateral acceleration.

Type	Property	Description
Constant	Lateral acceleration	Lets you specify a constant value for the lateral acceleration in m/s^2 .
Ramp	Start acceleration	Lets you specify the start value of the ramp profile in m/s^2 .
	End acceleration	Lets you specify the end value of the ramp profile in m/s^2 .
	Extent	Lets you specify the distance or duration that is used to change from the start value to the end value.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Table	Edit table	Lets you specify values for lateral acceleration in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Pedal stimulus properties

Acceleration pedal type Lets you select the profile type for the stimulation of the acceleration pedal. The following types are possible:

- **Constant:** Stimulates the pedal with constant values
- **Ramp:** Stimulates the pedal with a ramp profile
- **Extern:** Stimulates the pedal with a value defined by the simulation model

- **Manual:** Stimulates the pedal with a value defined by the experiment software
- **Table:** Stimulates the pedal with values defined in a table

Acceleration pedal parameters Lets you specify the parameters of the source type that is selected for the acceleration pedal.

Type	Property	Description
Constant	Hold current value	Lets you select to use the pedal position of the end of the previous segment.
	Position	Lets you specify a constant value for the pedal position in percents.
Ramp	Start with current value	Lets you specify to start the ramp profile with the pedal position of the end of the previous segment.
	Start value	Lets you specify the start value of the ramp profile in percents.
	End value	Lets you specify the end value of the ramp profile in percents.
	Extent	Lets you specify the distance or duration that is used to change from the start value to the end value.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Extern	–	Parameters are not required.
Manual	–	Parameters are not required.
Table	Edit table	Lets you specify values for pedal positions in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Brake pedal type Lets you select the profile type for the stimulation of the brake pedal. The following types are possible:

- **Constant:** Stimulates the pedal with constant values
- **Ramp:** Stimulates the pedal with a ramp profile
- **Extern:** Stimulates the pedal with a value defined by the simulation model
- **Manual:** Stimulates the pedal with a value defined by the experiment software
- **Table:** Stimulates the pedal with values defined in a table

Brake pedal parameters Lets you specify the parameters of the source type that is selected for the brake pedal.

Type	Property	Description
Constant	Hold current value	Lets you select to use the pedal position of the end of the previous segment.
	Position	Lets you specify a constant value for the pedal position in percents.
Ramp	Start with current value	Lets you specify to start the ramp profile with the pedal position of the end of the previous segment.
	Start value	Lets you specify the start value of the ramp profile in percents.
	End value	Lets you specify the end value of the ramp profile in percents.
	Extent	Lets you specify the distance or duration that is used to change from the start value to the end value.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Extern	–	Parameters are not required.
Manual	–	Parameters are not required.
Table	Edit table	Lets you specify values for pedal positions in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters.

Type	Property	Description
		Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Standstill properties

A Standstill longitudinal type has no properties.

Stop properties

Type Lets you select the type of stop. The following types are possible:

- Stop within distance: Stops the ASM vehicle within a specified distance.
- Brake until stop: Brakes the ASM vehicle until it stops.

Parameters Lets you specify the parameters of the source type that is selected for the distance m.

Type	Property	Description
Stop within distance	Hold current value	Lets you select whether the velocity of the previous segment is used at the beginning of the segment.
	Velocity	Lets you specify the velocity at the beginning of the segment.
	Deceleration	Lets you specify the deceleration of the ASM vehicle in m/s ² .
	Stop distance	Lets you specify the distance at which the ASM vehicle must stop in meters.
Brake until stop	–	Parameters are not required.

Traffic Driver properties

Type Lets you select the profile type for the fellow's distance profile. The following types are possible:

- Constant: Stimulus with constant values

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you select whether the ASM vehicle's velocity is held from the end of the last segment until the end of the active segment.
	Distance	Lets you specify the ASM vehicle's velocity in km/h.

Velocity properties

Reference type Lets you specify whether the velocity is specified absolutely or relatively to a vehicle.

Reference vehicle Lets you select the vehicle if the velocity is relative to a vehicle.

Type Lets you select the type for the velocity. The following types are possible:

- **Constant:** Lets you specify a constant value as velocity
- **Ramp:** Lets you specify a ramp profile for the velocity
- **Extern:** Lets you specify the velocity via simulation model
- **Sine:** Lets you specify a sine value as velocity
- **Final:** Lets you specify a final value at the end of the current segment
- **Manual:** Lets you control the velocity manually via experiment software.
- **Table:** Lets you specify the velocity using values of a table

Parameters Lets you specify the parameters of the source type that is selected for the velocity.

Type	Property	Description
Constant	Hold current value	Lets you select whether the velocity of the previous segment is hold.
	Velocity	Lets you specify a constant velocity.
Ramp	Start with current value	Lets you specify to start the ramp using the velocity at the end of the previous segment.
	Start velocity	Lets you specify the start value for the velocity.
	End velocity	Lets you specify the end value of the velocity.
	Extent	Lets you specify the distance or duration that is used to change from the start value to the end value.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Extern	–	Parameters are not required.
Sine	Start amplitude	Lets you specify the amplitude of the velocity at the start in km/h.
	End amplitude	Lets you specify the amplitude of the velocity at the end in km/h.
	Start frequency	Lets you specify the frequency of the velocity at the start in Hz.
	End frequency	Lets you specify the frequency of the velocity at the end in Hz.
	Offset	Lets you specify an offset to the velocity in km/h.

Type	Property	Description
	Extent	Lets you specify the distance or duration that is used to change from the start value to the end value.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.
Final	Final velocity	Lets you specify the velocity at the end of the segment.
	Positive acceleration	Lets you specify the acceleration in m/s^2
	Negative acceleration	Lets you specify the deceleration in m/s^2
Manual	–	Parameters are not required.
Table	Edit table	Lets you specify values for the velocity in relation to the time or distance. Refer to Edit Table Dialog on page 87.
	Interpretation type	Lets you specify how the property values are interpreted: ▪ Time (Segment): The values are a time duration in seconds. ▪ Driven Distance (Segment): The values are the distance driven by the vehicle in meters. ▪ Distance (Route): The values are the distance in relation to the reference line of the road (s coordinate) in meters.

Related topics

HowTos

[How to Accelerate and Brake in a Maneuver](#)..... 43

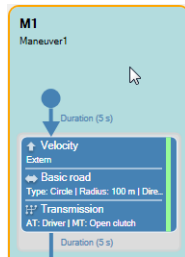
References

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Maneuver Properties

Purpose To specify the properties of a maneuver.

Access To access the properties, click the maneuver on the Scenario pane.



Properties **Consistency** Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given.

Description Lets you specify a description for the maneuver.

Name Lets you specify the name of the maneuver.

Open Traffic Driver Opens the Traffic Driver dialog for you to specify a new traffic driver. Refer to [Traffic Driver Dialog](#) on page 96.

Relative Path Displays the relative path to the traffic object that is used as the contour line object.

Traffic Driver Lets you select the traffic driver that is used for the ASM vehicle in the scenario. You can edit the object in a dialog, refer to [How to Specify the Properties of a Traffic Driver](#) on page 66.

Vehicle Object Lets you select a traffic object whose contour line is used as the contour line for the ASM vehicle.

You can create traffic objects using the Traffic Object Manager. Refer to [Basics of Traffic Objects \(ModelDesk Traffic Object Management !\[\]\(ab4e2b3fc7e7887b7a72f548aa6f5e60_img.jpg\)](#)).

Related topics

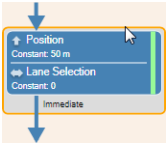
Basics

[Basics of Maneuvers](#)..... 26

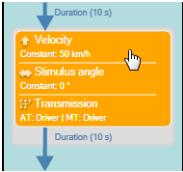
HowTos

[How to Start Scenario Creation](#)..... 39

Segment Properties

Purpose	To specify the properties of a segment.
Access	To access the properties, click the segment on the Traffic Scenario pane. 
Properties	Consistency Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given. Description Lets you enter a description for the segment. Index Displays the index of the segment. Name Lets you specify the name of the segment.
Related topics	HowTos How to Create and Specify a Fellow..... 59 How to Specify a Global User Signal..... 64

Selector Lever (AT) Properties

Purpose	To specify the properties of selector lever in an activity for an ASM vehicle with automatic transmission.
Access	To access the properties, click the activity of the maneuver on the Scenario pane. 

Properties

Selector lever type Lets you specify the selector lever type of the ASM vehicle in the activity.

Selector Lever Type	Description
Driver	Lets you specify that the driver model specifies the selector lever. Refer to Driver properties on page 152.
Stimulus	Lets you specify the selector lever settings are specified by properties in ModelDesk. Refer to Stimulus properties on page 152.

Driver properties

A Driver selector lever type has no properties.

Stimulus properties

Type Lets you specify the stimulus type.

Type	Description
Constant	Lets you specify a constant value for the position. You can also use the position of the previous segment.
Ramp	Lets you specify a linear change of the position.
Extern	Lets you control the selector lever via Simulink model.
Manual	Lets you control the selector lever manually via experiment software.
Table	Lets you enter values for position shift over time [s] or distance [m].

Parameters Lets you specify the properties of the selected type.

Type	Property	Description
Constant	Hold current value	Lets you specify to use the position of the previous segment.
	Position	Lets you specify a specific position.
Ramp	Start with current value	Lets you specify to start with the position of the previous segment.
	Start value	Lets you specify the start value for the position.
	End value	Lets you specify the end value for the position.
	Extent	Lets you specify the time or distance that is required to change the position from the start value to the end value.
	Interpretation type	Lets you specify whether the value of Extent is related to the time or distance.
Extern	–	Parameters are not required.
Manual	–	Parameters are not required.

Type	Property	Description
Type	Edit table	Lets you edit a table that contains the values for the position in relation to the time or distance.
	Interpretation type	Lets you specify whether the values of the table are related to the time or distance.

Related topics

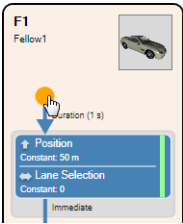
HowTos

[How to Shift Gears in a Maneuver.....](#) 44

Start Properties (Fellow)

Purpose To specify the start properties of a fellow.

Access To access the properties, click the start of a fellow on the Scenario pane.



- Properties**

Consistency Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given.

Description Lets you specify a description for the start segment.

Name Lets you specify the name of the start segment.

Reverse direction Lets you specify that the fellow drives the route in the reverse direction. The route must be suitable for the reverse direction.

Reverse possible Displays whether it is possible to drive the route in the reverse direction.

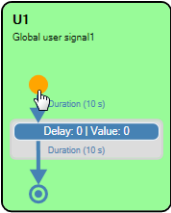
Route Lets you select the route that the fellow drives. Usually, the entries display the indices and names of the routes. If only the index is displayed, the previously selected route is no longer available.
Routes are specified in the Road Generator. For more information, refer to [How to Specify Routes on Road Networks \(ModelDesk Road Creation\)](#).

- Route index** Lets you select the route that the fellow drives.
The **Route index** specifies the same route as the **Route** property but you can create an alias for **Route index**. You can therefore use the alias feature with this property.
- Use external** Lets you specify to use external route definitions.

Related topics

Basics	
Basics of Fellows.....	28

Start Properties (Global User Signal)

Purpose	To specify the start properties of a global user signal.
Access	To access the properties, click the start of a global user signal on the Traffic Scenario pane. 
Properties	Consistency Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given. Description Lets you specify a description for the start segment. Name Lets you specify the name of the start segment.

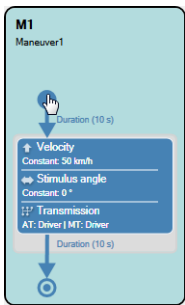
Related topics

Basics	
Basics of Global User Signals.....	29

Start Properties (Maneuver)

Purpose To specify the start properties of a maneuver.

Access To access the properties, click the start of a maneuver on the Scenario pane.



Start properties

Description Lets you specify a description for the start segment.

Name Lets you specify the name of the start segment.

Reverse direction Lets you specify that the fellow drives the route in the reverse direction. The route must be suitable for the reverse direction.

Reverse possible Displays whether it is possible to drive the route in the reverse direction.

Route Lets you select the route that the ASM vehicle drives. Usually, the entries display the indices and names of the routes. If only the index is displayed, the previously selected route is no longer available. Routes are specified in the Road Generator. For details, refer to [How to Specify Routes on Road Networks \(ModelDesk Road Creation\)](#).

Route index Lets you select the route that the ASM vehicle drives. The Route index specifies the same route as the Route property but you can create an alias for Route index. You can therefore use the alias feature with this property.

Initial vehicle position

Additional lateral offset Lets you specify an additional lateral offset of the ASM vehicle position on the road in meters. This is a d coordinate in the coordinate system related to the selected lane.

Initial lane index Lets you specify the initial lane index.

Start position on route Lets you specify the start position of the ASM vehicle on the road in meters.

Vehicle orientation Lets you specify an initial relative yaw angle in degrees.

Initial velocity **Initial longitudinal velocity** Lets you specify an initial velocity in longitudinal direction.

Initial vehicle height **Initial height mode** Lets you select the reference point for specifying the initial height.

Option	Description
Relative to road	Lets you specify the initial vehicle height relative to the road.
Offset to tire radius	Lets you specify the initial vehicle height as an offset to the tire radius.

Initial height value Lets you specify the initial height of the ASM vehicle in meters. This value is relative to the road or tire depending on the setting of the Initial height mode property.

Consistency **Valid** Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given.

Related topics

Basics

[Basics of Maneuvers.....](#) 26

HowTos

[How to Start Scenario Creation.....](#) 39

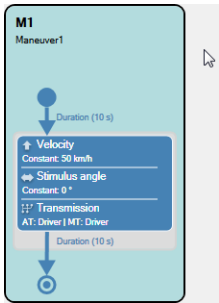
References

[Traffic Scenario Properties.....](#) 156

Traffic Scenario Properties

Purpose To specify the properties of a scenario.

Access To access the properties, click the workspace on the Scenario pane outside any element.



Traffic scenario properties

Author Lets you specify the name of the author.

Creation date Displays the date on which the scenario has been created.

Description Lets you specify a description for the scenario.

Name Displays the name of the scenario. The name is specified when you save the scenario by using the **Save As** command.

References properties

Change road Lets you select another road for the scenario.

Reload road Lets you reload the referenced road. Click this button to update the routes and position markers specified for the selected road.

Remove road Lets you remove the reference to the road.

Road Displays the name of the road that is selected for the scenario.

Consistency **Consistency** Displays the consistency state.

Related topics

HowTos

How to Start Scenario Creation.....	39
-------------------------------------	----

References

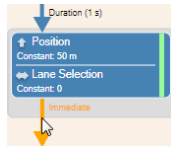
Save As.....	85
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Transition Properties

Purpose To specify the properties of a transition.

Access

To access the properties, click the transition (name or arrow) on the Scenario pane.

**Transition properties**

Add a New Condition Lets you add a new end condition to the transition. A condition type can only be used once in a transition. The following end condition types are possible:

Duration: The next segment is started after a specified duration.

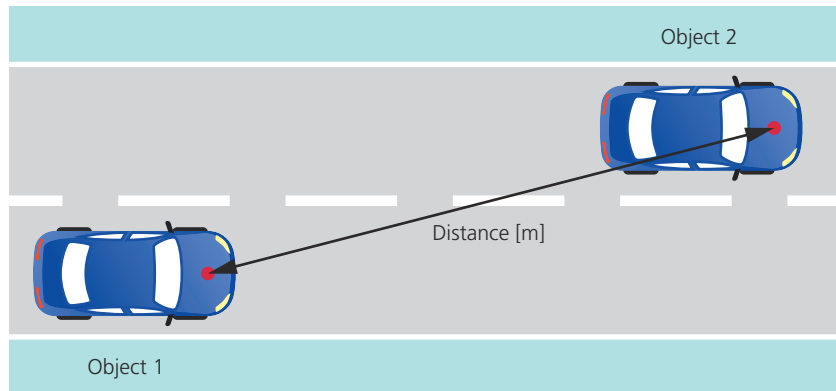
Endless: The active segment is executed endlessly.

Immediate: The next segment is started immediately.

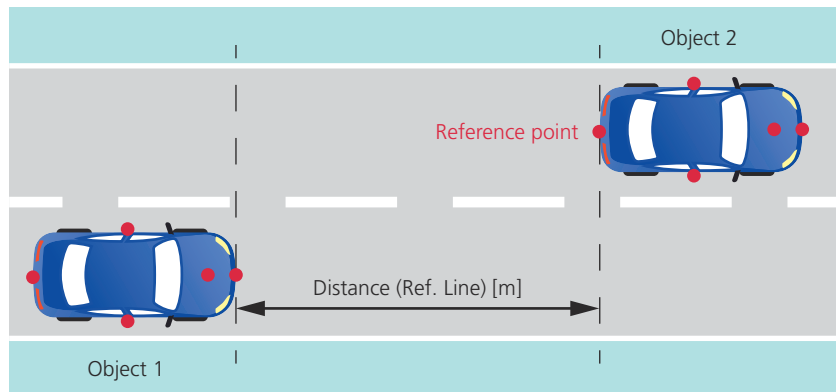
Position: The next segment is started when a specific specified vehicle has a specified position.

Driven Distance: The next segment is started when a specified vehicle has traveled a specified distance.

Distance (between main points): The next segment is started when the distance between two specified objects (vehicles or position markers) is greater or smaller than a specified value. The distance is measured between the main points of the objects.



Distance (along road reference line): The next segment is started when the distance between two specified objects (vehicles or position markers) is greater or smaller than a specified value. The distance is measured between selectable reference points of the objects along the reference line.



Lateral acceleration: (Only for a maneuver) The next segment is started when the lateral acceleration of the ASM vehicle is greater or smaller than a specified value.

Time Gap: (Only for fellows and global user signals) The next segment is started when the time gap between two specified objects (vehicles or position markers) is greater or smaller than a specified value.

Velocity: The next segment is started when a specified vehicle has reached or exceeded a specified velocity.

Trigger: The next segment is started when an event is triggered externally.

Conditions Lets you specify the properties of the end conditions.

Condition Type	Property	Description
Duration	Duration	Lets you specify the duration of the active segment in seconds.
Endless	–	The Endless type has no properties.
Immediate	–	The Immediate type has no properties.
Position	Position	Lets you enter the position value that must be reached or exceeded to end the active segment and start the next one.
	Reference vehicle	Lets you select the ASM vehicle or a fellow as the vehicle that has to meet the specified position condition.
Driven Distance	Distance	Lets you enter the driven distance value that must be reached or exceeded to end the active segment and start the next one.
	Reference vehicle	Lets you select the ASM vehicle or a fellow as the vehicle that has to meet the specified driven distance condition.

Condition Type	Property	Description
Distance (between main points)	Approximation Type	Lets you select whether the distance between two objects must fall below or exceed the specified value to end the active segment and start the next one.
	Value	Lets you enter the value that the distance between two objects must exceed or fall below to end the active segment and start the next one.
	Object 1	Lets you select the ASM vehicle or a fellow that has to meet the specified distance condition.
	Object 2	Lets you select the vehicle (ASM vehicle, fellow) or a position marker that has to meet the specified distance condition relative to object 1. A position marker is a trigger point specified on a road or junction element. Refer to Position Markers (ModelDesk Road Creation ).
	Object 2 (Type)	Lets you select the type of object 2.
Distance (along road reference line)	Value	Lets you enter the value that the distance between two objects must exceed or fall below to end the active segment and start the next one.
	Object 1	Lets you select the ASM vehicle or a fellow that has to meet the specified distance condition.
	Object 1 (reference point)	Lets you select the reference point of object 1 that is used for the condition.
	Object 2	Lets you select the vehicle (ASM vehicle, fellow) that has to meet the specified distance condition relative to object 1.
	Object 2 (reference point)	Lets you select the reference point of object 2 that is used for the condition.
	Approximation type	Lets you select whether the distance between two objects must fall below or exceed the specified value to end the active segment and start the next one.

Condition Type	Property	Description
Lateral acceleration	Value	Lets you enter the value that the velocity must exceed or fall below to end the active segment and start the next one.
	Compare	Lets you select whether the velocity must exceed or fall below the specified value to end the active segment and start the next one.
Time Gap	Approximation Type	Lets you select whether the time gap between two objects must fall below or exceed the specified value to end the active segment and start the next one.
	Object 1	Lets you select the ASM vehicle or a fellow that has to meet the specified time gap condition.
	Object 2	Lets you select the vehicle (ASM vehicle or fellow) or a position marker that has to meet the specified time gap condition relative to object 1. A position marker is a trigger point specified on a road or junction element. Refer to Position Markers (ModelDesk Road Creation ).
	Object 2 (Type)	Lets you select the type of object 2.
	Value	Lets you enter the value that the distance between two objects must exceed or fall below to end the active segment and start the next one.
Velocity	Velocity	Lets you enter the value that the velocity must reach or exceed or fall below to end the active segment and start the next one.
	Compare	Lets you select whether the velocity must exceed or fall below the specified value to end the active segment and start the next one.
	Reference vehicle	Lets you select the ASM vehicle or a fellow as the vehicle that has to meet the specified velocity condition.

Condition Type	Property	Description
Trigger	Trigger	Lets you select an external trigger to end the active segment and start the next one.

Condition delay Lets you specify a value for time delay. You must first select the **Enable delay** property.

Consistency Displays the consistency state. If the specified properties are not consistent, the reasons for the inconsistency are given.

Description Lets you specify a description for the transition.

Enable delay Lets you enable a delay for the transition. When the property is enabled, the next segment is started with the delay specified in the **Condition delay** property.

Evaluated Displays whether the appropriate condition will be evaluated in the simulation. If this property is selected, the appropriate condition is evaluated and can end the activity.

Related topics

Basics

[Basics of Transitions.....](#) 34

HowTos

[How to Specify a Transition for a Maneuver.....](#) 51

Source Properties

Introduction

The following properties are used in several other properties to specify the kind of source.

Circle

To specify a circle source.

Radius Lets you specify the end radius of the circle source.

Direction Lets you specify the direction.

Control Mode Lets you specify the control mode.

Constant

To specify a constant source.

Constant Lets you specify the constant value.

Hold current value Lets you specify to keep the final value from the previous segment until the end of the current segment.

Pulse	To specify a pulse. Direction Lets you specify the direction. Steering wheel angle Lets you specify the steering wheel angle. Step 1 start Lets you specify the start value of step 1. Step 1 end Lets you specify the end value of step 1. Step 2 start Lets you specify the start value of step 2. Step 2 end Lets you specify the end value of step 2. Interpretation type Lets you select the interpretation type.
Ramp	To specify a ramp source. Duration Lets you specify the total time it takes to apply the ramp signal. End value Lets you specify the end value. Start value Lets you specify the start value. Interpretation type Lets you select the interpretation type. Start with current value Lets you specify to start the segment with the current value.
Sine	To specify a sine source. Start amplitude Lets you specify the amplitude at the beginning of the segment. Duration Lets you specify the total time it takes to apply the sine signal. End frequency Lets you specify the frequency at the end of the segment. Start frequency Lets you specify the frequency at the beginning of the segment. End amplitude Lets you specify the amplitude at the end of the segment. Offset Lets you specify an offset. Interpretation type Lets you select the interpretation type.
Sine with dwell	To specify a sin with dwell source. Direction Lets you specify the direction. Steering wheel angle Lets you specify the steering wheel angle. Frequency Lets you specify the frequency. Dwell Lets you specify the dwell. Interpretation type Lets you specify the interpretation type.

Step

To specify a step source.

Direction Lets you specify the direction.

Interpretation type Lets you specify the interpretation type.

Steering wheel angle Lets you specify the steering wheel angle.

Value 1 Lets you specify the first value.

Value 2 Lets you specify the second value.

Table

To specify a table source.

Edit table Lets you edit the table.

Interpretation type Lets you specify the interpretation type.

User signal

To specify a user signal.

Name Displays the name of the user signal. You can specify the name in the Scenario Settings dialog.

Index Displays the index of the user signal.

Valid Displays the information whether the user signal is valid in the scenario.

Delay Lets you specify the duration after which the value is applied.

Type Lets you select the type of the user signal.

Parameters Lets you specify the parameters of the selected source type.

Related topics**References**

Clutch and Gear (MT) Properties.....	117
Lateral Type Properties (Maneuver).....	127
Longitudinal Type Properties (Maneuver).....	143
Scenario Settings Dialog.....	96

Maneuver Control Commands

Introduction ModelDesk provides commands for controlling the maneuver.

Where to go from here	Information in this section
	Reset..... 165 To reset the active maneuver.
	Start..... 166 To start the active maneuver.
	Stop..... 167 To stop the running maneuver.

Reset

Access	You can access this command via:	
	Ribbon	Home – Maneuver Control
	Context menu of	None
	Shortcut key	None
	Icon	

Purpose To reset the active maneuver.

Result The active maneuver is reset.

Description The command is available only if the simulation model contains the RESET block of the ASM_Utils_lib library.

Related topics	Basics
	Basics of Controlling Maneuvers and Driving Cycles..... 71
	HowTos
	How to Control a Maneuver or a Driving Cycle..... 73

References

Start	166
Stop	167

Start

Access

You can access this command via:

Ribbon	Home – Maneuver Control
Context menu of	None
Shortcut key	None
Icon	

Purpose

To start the active maneuver.

Result

The active maneuver is started.

Description

The command is available only if the simulation model contains the MANEUVER_START block of the ASM_Utls_lib library.

Related topics

Basics

Basics of Controlling Maneuvers and Driving Cycles	71
--	----

HowTos

How to Control a Maneuver or a Driving Cycle	73
--	----

References

Reset	165
Stop	167

Stop

Access	You can access this command via:	
	Ribbon	Home – Maneuver Control
	Context menu of	None
	Shortcut key	None
	Icon	

Purpose	To stop the running maneuver.
---------	-------------------------------

Result	The maneuver is stopped.
--------	--------------------------

Description	The command is available only if the simulation model contains the MANEUVER_STOP block of the ASM_Utils_lib library.
-------------	--

Related topics	Basics
	Basics of Controlling Maneuvers and Driving Cycles..... 71
	HowTos
	How to Control a Maneuver or a Driving Cycle..... 73
	References
	Reset..... 165
	Start..... 166

Automation

Where to go from here

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Programming ModelDesk Automation

Where to go from here

Information in this section

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Automated Maneuver Control in Python.....	174
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Overview of the Classes for Creating Scenarios.....	177
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Overview of the Object Model for Model Configuration.....	181
The overview gives you a quick overview of the classes for the setting the model configuration.	
Overview of the Object Model for Configuring Traffic Drivers.....	182
This gives you a quick overview of the classes for for configuring traffic drivers.	

Automating Scenarios in Python

Introduction

You can change the parameters of the fellow vehicles in a scenario.

Compatibility of the automation interface

You can use scripts that use the tool automation of previous ModelDesk versions. In ModelDesk 4.7, the Traffic Editor has been renamed to Scenario Editor because it also supports maneuvers. However, scripts that use the tool automation of ModelDesk 4.1 up to ModelDesk 4.6 can still be used.

Accessing the experiment

The following example shows how you can open a project and activate an experiment. Replace the project name and path by your own project in the script below.

```
from win32com.client import Dispatch

# Start ModelDesk application
Application = Dispatch("ModelDesk.Application")
Application.Visible = True

# Load project and activate experiment
MyProject = Application.OpenProject(r"C:\ExamplePath\Example_001\Example_001.CDP")
MyExperiments = MyProject.Experiments
MyExperiment = MyExperiments.Item(0)
MyActiveExperiment = MyExperiment.Activate(False)
```

A project is opened and an experiment is accessed in this part. For a description, refer to [Handling Projects and Experiments in Python \(ModelDesk Project and Experiment Management !\[\]\(1d3a1175dd4902218e694b9c098adb83_img.jpg\)](#)).

Tip

If you use the interpreter in ModelDesk, an **Application** interface is already created. You can use this interface as starting point for your Python scripts.

Creating a new scenario

The following listing shows how to create a scenario.

```
import dspace.com
Enums = dspace.com.Enums(Application)
# To create a new scenario file
MyProject.Pool.CreateFile(Enums.ContentTypes.Scenario, 'MyNewScenario', True)
```

The scenario is created in the pool of the project. To work with the scenario, you must activate it.

Activating a scenario

The following listing shows how to activate a scenario.

```
# Activate the scenario
MyActiveExperiment.ActivateTrafficScenario("Junction_TurnLeft_1F_Crossing")
# Get an object for handling the scenario
MyScenario = MyActiveExperiment.TrafficScenario
# Print information on the scenario
print("Name: ", MyScenario.Name)
print("Comment: ", MyScenario.Comment)
print("Author: ", MyScenario.Author)
print("Corresponding road: ", MyScenario.Road)
```

This part activates a scenario. The **MyScenario** object is created for accessing the scenario. Information about the scenario is printed.

Selecting a road

The following listing shows how to select a road for the scenario.

```
MyScenario.Road = 'NCAP_LSS'
```

The road with the name **NCAP_LSS** is used in **MyScenario**. If a road with the specified name does not exist in the pool of the ModelDesk project, an exception occurs.

Accessing and specifying the maneuver

The following listing shows how to access the maneuver of a scenario.

```
# Get object for the maneuver
MyManeuver = MyScenario.Maneuver.Item(0)
print("Name of the maneuver: ", MyManeuver.CompleteName)
```

The following part shows how you can add a new sequence to the maneuver. It modifies the first segment that is added by default.

```
# Add a new sequence
MySequence = MyManeuver.Sequences.Add()
# Add an Immediate condition
MySequence.StartCondition.Conditions.Add("Immediate")
# Remove the first condition, it is not evaluated
MySequence.StartCondition.Conditions.Remove(0)
# Set first activity to the Standstill Longitudinal profile
MySegment = MySequence.Segments.Item(0)
MySegment.Activity.LongitudinalType.Activate("Standstill")
MySegment.Activity.LateralType.Activate("NotUsed")
MySegment.Transition.Conditions.Item(0).Duration = 3
```

The following part shows how you can stimulate a pedal. The acceleration pedal is stimulated by a ramp first. Afterwards, the pedal position is held for 10 seconds.

```
# Activity to stimulate the accelerator pedal
# Pedal is set from 0 to 70% in 1 second
MySegment = MySequence.Segments.Append()
MyLongType = MySegment.Activity.LongitudinalType
MyLongType.Activate("PedalStimulus")
# Activate the Ramp source type
MyPedalSet = MyLongType.ActiveElement.AccelerationPedal.Activate("Ramp")
MyPedalSet.Start = 0
MyPedalSet.End = 70
MyPedalSet.Extent = 1
MySegment.Transition.Conditions.Item(0).Duration = 1
# Activity to hold the acceleration pedal
MySegment = MySequence.Segments.Append()
MyLongType = MySegment.Activity.LongitudinalType
MyLongType.Activate("PedalStimulus")
# Activate the Constant source type
MyPedalSet = MyLongType.ActiveElement.AccelerationPedal.Activate("Constant")
MyPedalSet.HoldCurrentValue = True
MySegment.Transition.Conditions.Item(0).Duration = 10
```

Adding and specifying fellows

The following listing shows how to access the Fellows collection.

```
# Get object for scenario fellows
MyFellows = MyScenario.Fellows
print("Number of scenario fellows: ", MyFellows.Count)
```

The **MyFellows** object is created for accessing the fellows. The number of fellows is printed.

```
# Add a new fellow to the scenario
MyFellow = ActiveTraffic.Fellows.Add()
# Specify a name for the fellow
MyFellow.Name = 'MyNewFellow'
# Select a traffic object for the fellow
MyFellow.TrafficObjectType.Activate('Vehicles\\SimpleCarBrown.tro')
```

A fellow is created with one sequence containing two segments. The first segment is used to initiate the fellow's position, i.e. the activity has the **Position** type and the transition has the **Immediate** type. The following listing shows how to access the first segment and specify a new position value.

```
# Access the fellow's 1st sequence and segment
MySequence = MyFellow.Sequences.Item(0)
MySegment = MySequence.Segments.Item(0)
# Access the activity of the 1st segment
ActiveLongitudinal = MySegment.Activity.LongitudinalType.ActiveElement
# Specify the position value
ActiveLongitudinal.SourceType.ActiveElement.Constant = 42
```

The following listing demonstrates how to append a segment to the fellow's sequence and specify the segment's activity and transition.

```
# Add a segment
MySegment = MySequence.Segments.Append()
# Specify the activity of the segment
ActiveLongitudinal = MySegment.Activity.LongitudinalType.Activate("StopWithin")
ActiveLongitudinal.SourceType.ActiveElement.HoldCurrentValue = True
ActiveLongitudinal.SourceType.ActiveElement.StopDistance = 42
ActiveLateral = MySegment.Activity.LateralType.Activate("Continue")
# Access the 1st condition. It is created by default.
Transition = MySegment.Transition.Conditions.Item(0)
Transition.Duration = 8
# Add a Position condition and specify its properties
Transition = MySegment.Transition.Conditions.Add(4)
Transition.Position = 54
```

Adding and specifying global user signals

A user signal is created with one sequence and segment. In the activity of the segment a constant value is set, the transition has the `Duration` type by default. The following listing shows how to add a global user signal to a scenario and specify the value and duration.

```
# Add a user signal to the collection
MyUserSignal = ActiveTraffic.UserSignals.Add()
# Specify a name for the global user signal
MyUserSignal.Name = 'MyUserSignal'
# Access the 1st sequence and segment of the user signal
MySequence = MyUserSignal.Sequences.Item(0)
MySegment = MySequence.Segments.Item(0)
# Specify a value and duration time
MySegment.Activity.ActiveElement.Constant = 42
MySegment.Transition.Conditions.Item(0).Duration = 8
```

Configuring traffic drivers

The following listing shows how you can create a new traffic driver in the pool.

```
import dspace.com
Enums = dspace.com.Enums(Application)
# To create a new traffic driver
MyProject.Pool.CreateFile(Enums.ContentTypes.TrafficDriver, 'MyNewTrafficDriver', True)
```

The following listing shows how you can configure a traffic driver.

```
# Access the traffic drivers of the current project
MyTrafficDrivers = MyProject.TrafficDrivers
# Access a specific traffic driver
MyTrafficDriver = MyTrafficDrivers.Item(0)
# Set properties
MyTrafficDriver.DrivingStyle.DetectionRange = 110.0
MyTrafficDriver.RuleCompliantDriving = DistanceAtStandstill = 0.5
MyTrafficDriver.TrafficSignRecognition.DistanceFront = 42.0
```

```
# Save the traffic driver
MyTrafficDriver.SaveAs('MyTrafficDriver.trd',0)
```

The following listing shows how you can use the traffic driver for the maneuver.

```
# Activate the traffic driver for the maneuver
MyManeuver.TrafficDriverType.Activate('MyTrafficDriver.trd')
```

Saving and downloading the scenario

The following listing shows how to save and download the scenario.

```
# Save the scenario
MyScenario.Save()
# Check the scenarios consistency
MyScenario.CheckConsistency()
# Download the scenario
MyScenario.Download()
```

After saving the scenario, the scenario is check for consistency.

Related topics

Basics

[Handling Projects and Experiments in Python \(ModelDesk Project and Experiment Management !\[\]\(d5d7044e5caf6907399af2dced8d6ff8_img.jpg\)](#))

References

Classes for Configuring Fellows.....	228
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Classes for Configuring the Maneuver.....	201
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Automated Maneuver Control in Python

Introduction

You can start, stop, and reset maneuvers.

Accessing the experiment

The following example shows how you can open a project and activate an experiment. Replace the project name and path by your own project in the script below.

```

from win32com.client import Dispatch

# Start ModelDesk application
Application = Dispatch("ModelDesk.Application")
Application.Visible = True

# Load project and activate experiment
MyProject = Application.OpenProject(r"C:\ExamplePath\Example_001\Example_001.CDP")
MyExperiments = MyProject.Experiments
MyExperiment = MyExperiments.Item(0)
MyActiveExperiment = MyExperiment.Activate(False)

```

A project is opened and an experiment is accessed in this part. For a description, refer to [Handling Projects and Experiments in Python \(ModelDesk Project and Experiment Management\)](#).

Tip

If you use the interpreter in ModelDesk, an **Application** interface is already created. You can use this interface as starting point for your Python scripts.

Activating a maneuver

The following listing shows how to activate a maneuver.

```

# Activate the maneuver
MyActiveExperiment.ActivateManeuver("FollowRoad")
# Get an object for handling the road
MyManeuver = MyActiveExperiment.Maneuver
# Print information on the maneuver
print("Name of the maneuver:", MyManeuver.Name)
print("Comment of the maneuver:", MyManeuver.Comment)
print("Author of the maneuver:", MyManeuver.Author)

```

This part activates a maneuver. The **MyManeuver** object is created for accessing the maneuver. The name, comment and author of the maneuver are printed.

Starting the maneuver

The following listing shows how to start the maneuver.

```

# Start the maneuver
MyActiveExperiment.ManeuverControl.Start()

```

Example

The following example shows how you can start two maneuvers one after the other. Both maneuvers are available in the standard installation of ModelDesk. When you use the Python script, you must only replace the **ModelDeskProjectFile** and **ExperimentName** variables by your project and experiment name. Before the Python script is started, you must have access to the platform where the simulation of the appropriate simulation model should be started. When you want to monitor the maneuver, you can use **MotionDesk**.

```

import sys, os
import time
import win32com.client
ModelDeskProjectFile = r"C:\ExamplePath\ExampleProjectName.CDP"
ExperimentName = r"ExampleExperimentName"

```

```

ManeuverNames          = ["MueSplit", "LaneChange"]
gManeuverStateMapping = {2: 'Init',
                        3: 'Run',
                        4: 'Stopping',
                        5: 'Wait',
                        6: 'Init Manual',
                        7: 'Manual'}

# Timeout for wait time maneuver running, stopping in seconds
Timeout = 120
if __name__ == "__main__":
    ModelDeskApplication = None
    print("Start demo: %s.\n" % time.asctime())
    try:
        #-----
        # Access ModelDesk Application
        #-----
        ModelDeskApplication = win32com.client.Dispatch("ModelDesk.Application")
        ModelDeskApplication.Visible = True
        #-----
        # Load and activate project
        #-----
        print("Initializing ModelDesk...\n")
        SaveActiveProjectFlag = True
        ModelDeskApplication.OpenProject(ModelDeskProjectFile, "", SaveActiveProjectFlag)
        # Activate first experiment
        MyActiveManeuver = ModelDeskApplication.ActiveProject.Experiments[0].Activate(False)
        for ManeuverName in ManeuverNames:
            print("About to activate maneuver: '%s'" % ManeuverName)
            MyActiveManeuver.ActivateManeuver(ManeuverName)
            print("Maneuver '%s' is active. Download maneuver." % MyActiveManeuver.Maneuver.Name)
            MyActiveManeuver.Maneuver.Download()
            # Reset vehicle
            print("Reset vehicle...")
            MyActiveManeuver.ManeuverControl.Reset()
            # Start maneuver using ModelDesk tool automation
            print("Start maneuver...")
            MyActiveManeuver.ManeuverControl.Start(True)
            # Wait for maneuver starts running.
            ManeuverRunning = False
            StartTime = time.time()
            while(True):
                if time.time() - StartTime >= Timeout:
                    print("TIMEOUT REACHED! Break polling loop to wait for maneuver start ..")
                    # Stop maneuver
                    MyActiveManeuver.ManeuverControl.Stop()
                    break
                if gManeuverStateMapping[MyActiveManeuver.ManeuverControl.ManeuverState] == "Run":
                    print("Maneuver is running...")
                    ManeuverRunning = True
                    break
            # Wait 5 seconds
            time.sleep(5)
            # Wait for end of maneuver run.
            StartTime = time.time()
            if ManeuverRunning:
                while(True):
                    if time.time() - StartTime >= Timeout:
                        print("TIMEOUT REACHED. Break polling loop and stop maneuver..")
                        # Stop maneuver
                        MyActiveManeuver.ManeuverControl.Stop()
                        break

```



```

        if gManeuverStateMapping[MyActiveManeuver.ManeuverControl.ManeuverState] != "Run":
            print("Maneuver is no longer running...")
            break
        # Wait 1 second
        time.sleep(1)

    print
    print("Demo successfully finished: %s.\n" % time.asctime())
finally:
    #-----
    # make sure to shutdown ModelDesk
    #-----
    if ModelDeskApplication != None:
        ModelDeskApplication.Quit(False)
        ModelDeskApplication = None

```

Related topics

Basics

[Handling Projects and Experiments in Python \(ModelDesk Project and Experiment Management 📖\)](#)

References

[Overview of the Object Model for Accessing ModelDesk Experiments \(ModelDesk Project and Experiment Management 📖\)](#)

Overview of the Classes for Creating Scenarios

Classes

The following tables give you a quick overview of the classes for creating scenarios.

Class	Level
ActiveTrafficScenario on page 188	
Maneuvers on page 210	
Maneuver on page 209	
ManeuverSequences on page 219	
ManeuverSequence on page 218	
ManeuverSegments on page 214	
ManeuverSegment on page 213	
UserOutputs on page 249	
UserOutput on page 250	
UserSignalValue on page 265	
ManeuverActivity on page 213	
LongitudinalProfile on page 207	
Velocity on page 362	

Class	Level
Stop on page 354	
PedalStimulus on page 331	
Standstill on page 352	
LateralAcceleration on page 321	
LateralProfile on page 205	8
NotUsed on page 330	9
StimulusAngle on page 353	
StimulusTorque on page 353	
BasicRoad on page 311	
FollowRoad on page 320	
GearProfile on page 203	8
SelectorProfile on page 225	
ManeuverTransition on page 222	7
TransitionsConditions on page 278	8
ConditionDuration on page 271	9
ConditionEndless on page 272	
ConditionImmediate on page 272	
ConditionPosition on page 274	
ConditionDrivenDistance on page 270	
ConditionDistance on page 269	
ConditionVelocity on page 276	
ConditionTrigger on page 275	
ConditionLateralAcceleration on page 273	
Fellows on page 230	1
Fellow on page 229	2
FellowSequences on page 239	3
FellowSequence on page 238	4
FellowSegments on page 234	5
FellowSegment on page 233	6
UserOutputs on page 249	7
UserOutput on page 250	8
UserSignalValue on page 265	9
FellowActivity on page 230	7
LongitudinalProfile on page 245	8
NotUsed on page 331	9
Continue on page 277	
Acceleration on page 307	
AccelerationSource on page 308	10
DistanceMeter on page 315	9

Class	Level
DistanceMeterSource on page 316	10
DistanceTime on page 317	9
DistanceTimeSource on page 318	10
Position on page 332	9
PositionSource on page 333	10
Velocity on page 362	9
VelocitySource on page 363	10
StopWithin on page 355	9
StopWithinSource on page 356	10
LateralProfile on page 243	8
Continue on page 277	9
NotUsed on page 330	
LateralDeviation on page 322	
LateralDeviationSource on page 322	10
LaneSelection on page 324	9
LaneSelectionSource on page 325	10
FellowTransition on page 242	7
TransitionsConditions on page 278	8
ConditionDistance on page 269	9
ConditionDrivenDistance on page 270	
ConditionDuration on page 271	
ConditionEndless on page 272	
ConditionImmediate on page 272	
ConditionPosition on page 274	
ConditionTimegap on page 275	
ConditionTrigger on page 275	
ConditionVelocity on page 276	
FellowTransition on page 242 (StartCondition)	4
RouteSelection on page 339	
Route on page 339	5
TrafficObjectTypes on page 247	3
TrafficObjectType on page 248	4
GlobalUserSignals on page 252	1
GlobalUserSignal on page 255	2
UserSignalSequences on page 261	3
UserSignalSequence on page 264	4
UserSignalSegments on page 256	5
UserSignalSegment on page 260	6

Class	Level
UserSignalValue on page 265	
UserSignalTransition on page 265	
UserSignalSegment on page 260 (Successor)	
UserSignalTransition on page 265 (StartCondition)	

The following tables give you a quick overview of the miscellaneous classes for creating scenarios.

Class	Level
ApproximationOperators on page 309	
ApproximationOperator on page 311	
DependencyTypes on page 312	
DependencyTypeAbsolute on page 314	
DependencyTypeRelative on page 314	
RelativeOperators on page 335	
RelativeOperators on page 334	
RelativeVehicleSelection on page 337	
RelativeObject on page 336	
SourceTypes on page 341	
Constant on page 343	
Extern on page 343	
FinalDeviation on page 344	
FinalDistance on page 344	
FinalLane on page 345	
FinalVelocity on page 345	
LaneChange on page 346	
Ramp on page 347	
Sine on page 348	
SmoothLaneChange on page 349	
StopWithin on page 350	
SimpleConstant on page 348	
YawAngleConstant on page 351	
Table on page 357	
Step on page 349	
Pulse on page 347	
SineWithDwell on page 348	
Circle on page 343	
UntilStop on page 351	
Fixed on page 345	
Straight on page 350	

Related topics

Basics

[Automating Scenarios in Python](#)..... 170

References

[Classes for Configuring Fellows](#)..... 228
[Classes for Configuring Global User Signals](#)..... 252
[Classes for Configuring the Maneuver](#)..... 201
[Classes for Creating Scenarios](#)..... 188
[Enumerations for Scenarios](#)..... 366
[Miscellaneous Classes for Scenarios](#)..... 305

Overview of the Object Model for Model Configuration

Classes

The following tables give you a quick overview of the classes for the setting the model configuration.

Classes	Level
ActiveProject	
ModelConfiguration on page 288	
ManeuverConfiguration on page 288	
ReadOnlyListElements on page 290	
Triggers	
UserSignals	
ListElement on page 287	
ScenarioConfiguration on page 292	
ListElements on page 283	
FellowUserSignals	
GlobalUserSignals	
Triggers	
ListElement on page 287	

Related topics

References

[Environment Configuration Dialog](#)..... 88
[Project \(ModelDesk Project and Experiment Management !\[\]\(bd3b31712ad9bab5a241210fa6925cdd_img.jpg\)](#))

Overview of the Object Model for Configuring Traffic Drivers

Classes

The following tables give you a quick overview of the classes for configuring traffic drivers.

Classes	Level
ActiveProject	
TrafficDrivers on page 296	
TrafficDriver on page 300	
DrivingStyle on page 294	
RuleCompliantDriving on page 295	
TrafficSignRecognition on page 303	

Related topics

References

Project (ModelDesk Project and Experiment Management )	
Traffic Driver Dialog.....	96

Classes for Maneuver Control

Where to go from here

Information in this section

[ManeuverControl..... 183](#)
To control the active maneuver.

Information in other sections

[Overview of the Classes for Creating Scenarios..... 177](#)
The overview gives you a quick overview of the classes for the Scenario Editor.

[Automating Scenarios in Python..... 170](#)
You can change the parameters of the scenario.

ManeuverControl

Purpose

To control the active maneuver.

Where to go from here

Information in this section

[Class Description \(ManeuverControl\)..... 184](#)
To describe the class and its attributes.

[Start..... 185](#)
To start the active maneuver.

[Reset..... 186](#)
To reset the active maneuver.

[Stop..... 186](#)
To stop the running maneuver.

Information in other sections

[Automated Maneuver Control in Python..... 174](#)
You can start, stop, and reset maneuver.

Class Description (ManeuverControl)

Syntax

```
ManeuverControl = Experiment.ManeuverControl
```

Purpose

To control the active maneuver.

Description

So that you can change the state of the maneuver, the maneuver must be in specific states. These states depend on the kind of model. Refer to the state diagram of the maneuver.

Vehicle dynamics: [Maneuver Scheduler \(ASM Environment Reference !\[\]\(e8fb589d58dad1692debababa5e928b6_img.jpg\)](#))

Engine: [Maneuver Control \(ASM Drivetrain Basic Reference !\[\]\(f95dab70c751fda7d824b8b03650f7aa_img.jpg\)](#))

You can read the current state with the **ManeuverState** attribute.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ManeuverState	Long	To get the maneuver state. The meanings of the values depend on the kind of model. Maneuver state for vehicle dynamics (refer to Maneuver Scheduler (ASM Environment Reference )): ▪ 1: Maneuver starting ▪ 2: Maneuver initializing ▪ 3: Maneuver running ▪ 4: Maneuver braking ▪ 5: Maneuver waiting ▪ 6: Manual initializing ▪ 7: Manual driving Maneuver state for engine (refer to Maneuver Control (ASM Drivetrain Basic Reference )): ▪ 1: Maneuver stopped ▪ 2: Maneuver paused ▪ 3: Maneuver running ▪ 4: Manual state

Methods

The class contains the following methods:

Method	Purpose
Reset	To reset the active maneuver. Refer to Reset on page 186.
Start	To start the active maneuver. Refer to Start on page 185.
Stop	To stop the running maneuver. Refer to Stop on page 186.

Related topics**Basics**

Automated Maneuver Control in Python.....	174
Basics of Controlling Maneuvers and Driving Cycles.....	71

References

[ActiveExperiment \(ModelDesk Project and Experiment Management !\[\]\(d66ff64371a51729ac8c1cdaa685ba6f_img.jpg\)\)](#)

Start

Class

ManeuverControl

Syntax

```
RetVal = ManeuverControl.Start(DownloadExperimentIfApplicationNotRunning)
```

Purpose

To start the active maneuver.

Parameters

The method uses the following parameters:

Parameter	Type	Description
DownloadExperimentIfApplicationNotRunning	Boolean	If True: If no or another simulation application is loaded on the simulation platform, the simulation application of the ModelDesk experiment and the active experiment is loaded. The active experiment contains the scenario, road, and parameter set.

Return value

The method returns an object of the following type:

Type	Description
Boolean	True if the maneuver is successfully started

Related topics**References**

Class Description (ManeuverControl).....	184
--	-----

Reset

Class ManeuverControl

Syntax `RetVal = ManeuverControl.Reset()`

Purpose To reset the active maneuver.

Parameters –

Return value The method returns an object of the following type:

Type	Description
Boolean	True if the maneuver is successfully reset

Related topics

References

[Class Description \(ManeuverControl\)](#)..... 184

Stop

Class ManeuverControl

Syntax `RetVal = ManeuverControl.Stop()`

Purpose To stop the running maneuver.

Parameters –

Return value The method returns an object of the following type:

Type	Description
Boolean	True if the maneuver is successfully stopped

Related topics

References

Class Description (ManeuverControl)	184
---	-----

Classes for Creating Scenarios

Where to go from here

Information in this section

ActiveTrafficScenario.....	188
The scenario currently linked to the experiment.	
DriverProperties.....	196
To get the properties of the driver.	
ManeuverProperties.....	197
To get the maneuver properties.	
ScenarioProperties.....	197
To get information of the properties of a scenario.	
TrafficProperties.....	199
To get the traffic properties.	

Information in other sections

Overview of the Classes for Creating Scenarios.....	177
The overview gives you a quick overview of the classes for the Scenario Editor.	
Automating Scenarios in Python.....	170
You can change the parameters of the scenario.	

ActiveTrafficScenario

Purpose

This is the scenario currently linked to the experiment.

Where to go from here

Information in this section

Class Description (ActiveTrafficScenario).....	189
To describe the class and its attributes.	
CheckConsistency.....	190
To check the complete scenario for consistency.	
Download.....	191
To download the active scenario to Simulink or the real-time hardware.	
DownloadContent.....	191
To download the maneuver and/or traffic to the real-time hardware or Simulink.	

GenerateModelIni.....	192
To generate the model initialization file.	
ReloadRoad.....	193
To reload the road assigned to the scenario.	
RemoveRoad.....	193
To remove the road connection from the scenario.	
ReplaceContent.....	194
To replace a part of the scenario, for example, the settings of the maneuver of another scenario.	
Save.....	195
To save the active scenario.	
SaveAs.....	195
To save the active scenario under a new name.	

Class Description (ActiveTrafficScenario)

Syntax

```
TrafficScenario = ActiveExperiment.TrafficScenario
```

Purpose

To modify the contents of the active scenario.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Author	String	To get/set the author of the scenario.
Comment	String	To get/set the comment of the scenario.
Fellows	Fellows ¹⁾	To get the Fellows collection object of the scenario.
Maneuver	Maneuvers ²⁾	To get the maneuver of the scenario.
Name	String	To get the name of the scenario.
Properties	ScenarioProperties ³⁾	To get the properties of the scenario.
Road	String	To get/set the name of the road assigned to the scenario.
UserSignals	GlobalUserSignals ⁴⁾	To get the GlobalUserSignals collection object of the scenario.

¹⁾ Refer to [Fellows](#) on page 230.

²⁾ Refer to [Maneuvers](#) on page 210.

³⁾ Refer to [ScenarioProperties](#) on page 197.

⁴⁾ Refer to [GlobalUserSignals](#) on page 252.

Methods

The class contains the following methods:

Method	Purpose
CheckConsistency	To check the consistency of the complete scenario. Refer to CheckConsistency on page 190.
Download	To download the active scenario to the real-time hardware or Simulink. Refer to Download on page 191.
DownloadContent	To download the maneuver and/or traffic to the real-time hardware or Simulink. Refer to DownloadContent on page 191.
GenerateModelIni	To generate the model initialization file. Refer to GenerateModelIni on page 192.
ReloadRoad	To reload the road assigned to the scenario. Refer to ReloadRoad on page 193.
RemoveRoad	To remove the road connection from the scenario. Refer to RemoveRoad on page 193.
ReplaceContent	To replace a part of the scenario, for example, the settings of the maneuver of another scenario. Refer to ReplaceContent on page 194.
Save	To save the active scenario with its current name. Refer to Save on page 195.
SaveAs	To save the active scenario with a specified name. Refer to SaveAs on page 195.

Related topics**Basics**

[Automating Scenarios in Python](#)..... 170

References

[ActiveExperiment \(ModelDesk Project and Experiment Management\)](#)

CheckConsistency

Class

ActiveTrafficScenario

Syntax

```
ActiveTrafficScenario.CheckConsistency()
```

Purpose

To check the complete scenario for consistency.

Parameters

—

Return value

The method returns a value of the following type:

Type	Description
Boolean	True if the scenario is consistent, otherwise false.

Related topics**References**

[Class Description \(ActiveTrafficScenario\)](#)..... 189

Download

Class

ActiveTrafficScenario

Syntax

```
ActiveTrafficScenario.Download()
```

Purpose

To download the active scenario to Simulink or the real-time hardware.

Parameters

—

Return value

The method returns a value of the following type:

Type	Description
Boolean	True if successful, otherwise false.

Related topics**References**

[Class Description \(ActiveTrafficScenario\)](#)..... 189

DownloadContent

Class

ActiveTrafficScenario

Syntax

```
ActiveTrafficScenario.DownloadContent(DownloadSelectionTypes Type)
```

Purpose To download the maneuver and/or traffic to the real-time hardware or Simulink.

Parameters The method uses the following parameters:

Parameter	Type	Description
Type	DownloadSelectionTypes ¹⁾	Lets you specify whether the maneuver, the traffic, or the whole scenario is downloaded.

¹⁾ Refer to [DownloadSelectionTypes](#) on page 367.

Return value The method returns a value of the following type:

Type	Description
Boolean	True if successful, otherwise false.

Related topics

References

[Class Description \(ActiveTrafficScenario\)](#)..... 189

GenerateModelIni

Class ActiveTrafficScenario

Syntax `ActiveTrafficScenario.GenerateModelIni(string FilePath)`

Purpose To generate the model initialization file.

Parameters The method uses the following parameters:

Parameter	Type	Description
FilePath	String	The target file path of the model initialization file.

Return value The method returns a value of the following type:

Type	Description
Boolean	True if the file could be created, otherwise false.

Related topics**References**

[Class Description \(ActiveTrafficScenario\).....](#) 189

ReloadRoad

Class

ActiveTrafficScenario

Syntax

```
RetVal = ActiveTrafficScenario.ReloadRoad()
```

Purpose

To reload the road assigned to the scenario.

Parameters

—

Return value

The method returns a value of the following type:

Type	Description
Boolean	True if successful, otherwise false.

Related topics**References**

[Class Description \(ActiveTrafficScenario\).....](#) 189

RemoveRoad

Class

ActiveTrafficScenario

Syntax

```
RetVal = ActiveTrafficScenario.RemoveRoad()
```

Purpose

To remove the road connection from the scenario.

Parameters

—

Return value

The method returns a value of the following type:

Type	Description
Boolean	True if successful, otherwise false.

Related topics**References**

[Class Description \(ActiveTrafficScenario\)](#)..... 189

ReplaceContent

Class

ActiveTrafficScenario

Syntax

```
ActiveTrafficScenario.ReplaceContent(FilePath, Type)
```

Purpose

To replace a part of the scenario, for example, the settings of the maneuver of another scenario.

Parameters

The method uses the following parameters:

Parameter	Type	Description
FilePath	String	The scenario file that is used for the replacement.
Type	ReplacementTypes ¹⁾	The part of the scenario that is replaced.

¹⁾ Refer to [ReplacementTypes](#) on page 370.

Return value

The method returns a value of the following type:

Type	Description
Boolean	True if successful, otherwise false.

Related topics**References**

[Class Description \(ActiveTrafficScenario\)](#)..... 189

Save

Class ActiveTrafficScenario

Syntax `ActiveTrafficScenario.Save()`

Purpose To save the active scenario.

Parameters –

Return value –

Related topics References

[Class Description \(ActiveTrafficScenario\).....](#) 189

SaveAs

Class ActiveTrafficScenario

Syntax `ActiveTrafficScenario.SaveAs(FileName, OverwritingExisting)`

Purpose To save the active scenario under a new name.

Parameters The method uses the following parameters:

Parameter	Type	Description
FileName	String	The name under which the scenario is saved.
OverwriteExisting	Boolean	Saving the active scenario overwrites an existing file called FileName if true.

Return value -

Related topics**References**

[Class Description \(ActiveTrafficScenario\)](#)..... 189

DriverProperties

Purpose

To get the properties of the driver.

Class Description (DriverProperties)

Syntax

```
DriverProperties = ManeuverProperties.Driver
```

Purpose

To get the properties of the driver.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Slow	DriverType ¹⁾	To get the properties of the slow driver.
Normal	DriverType ¹⁾	To get the properties of the normal driver.
Fast	DriverType ¹⁾	To get the properties of the fast driver.

¹⁾ Refer to [DriverType](#) on page 203.

Methods

—

Related topics**References**

[ManeuverProperties](#)..... 197
[Scenario Properties Dialog](#)..... 93

ManeuverProperties

Purpose To get the maneuver properties.

Class Description (ManeuverProperties)

Syntax `ManeuverProperties = ScenarioProperties.Maneuver`

Purpose To get the maneuver properties.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
UserSignals	NamedItemList ¹⁾	To get the maneuver user signals.
Trigger	NamedItemList ¹⁾	To get the triggers.
Driver	DriverProperties ²⁾	To get the driver properties.
LateralController	LateralControllerTypes ³⁾	To get/set the lateral controller.

¹⁾ Refer to [NamedItemList](#) on page 327.

²⁾ Refer to [DriverProperties](#) on page 196.

³⁾ Refer to [LateralControllerTypes](#) on page 368.

Methods —

Related topics

References

Scenario Properties Dialog	93
ScenarioProperties	197

ScenarioProperties

Purpose To get information of the properties of a scenario.

Where to go from here**Information in this section**

Class Description (ScenarioProperties)	198
To describe the class and its attributes.	
Reset	199
To reset the scenario properties.	

Class Description (ScenarioProperties)

Syntax

```
ScenarioProperties = ActiveTrafficScenario.Properties
```

Purpose

To get information of the properties of a scenario.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
GlobalProperties	GlobalPropertiesLaneIndexInterpretationTypes ¹⁾	To get/set the lane index interpretation (relative to preferred lane and driving direction = 0 or relative to center lane = 1).
Maneuver	ManeuverProperties ²⁾	To get the properties of the maneuver.
Traffic	TrafficProperties ³⁾	To get the properties of the traffic.

¹⁾ Refer to [GlobalPropertiesLaneIndexInterpretationTypes](#) on page 368.

²⁾ Refer to [ManeuverProperties](#) on page 197.

³⁾ Refer to [TrafficProperties](#) on page 199.

Methods

The class contains the following methods:

Method	Purpose
Reset	To reset the scenario properties. Refer to Reset on page 199.

Related topics**References**

Class Description (ActiveTrafficScenario)	189
Scenario Properties Dialog	93

Reset

Class	ScenarioProperties
--------------	--------------------

Syntax	<code>ScenarioProperties.Reset()</code>
---------------	---

Purpose	To reset the scenario properties.
----------------	-----------------------------------

Parameters	—
-------------------	---

Return value	—
---------------------	---

Related topics**References**

Class Description (ScenarioProperties)	198
--	-----

TrafficProperties

Purpose	To get the traffic properties.
----------------	--------------------------------

Class Description (TrafficProperties)

Syntax	<code>TrafficProperties = ScenarioProperties.Traffic</code>
---------------	---

Purpose	To get the properties of the traffic.
----------------	---------------------------------------

Attributes	The class contains the following attributes:		
	Attributes	Type	Purpose
	UserSignals	NamedItemList ¹⁾	To get the fellow user signals.
	Trigger	NamedItemList ¹⁾	To get the triggers.

¹⁾ Refer to [NamedItemList](#) on page 327.

Methods	—
---------	---

Related topics	References
	Scenario Properties Dialog..... ScenarioProperties..... 93 197

Classes for Configuring the Maneuver

Where to go from here

Information in this section

Driver.....	202
To specify the Driver type.	
DriverType.....	203
To specify the values of the driver type.	
GearProfile.....	203
To get the clutch and gear type.	
LateralProfile.....	205
To specify the lateral profile of a maneuver's activity.	
LongitudinalProfile.....	207
To specify the longitudinal profile of a maneuver's activity.	
Maneuver.....	209
To get a maneuver.	
Maneuvers.....	210
To manage the maneuvers of a scenario.	
ManeuverActivity.....	213
To get a maneuver activity.	
ManeuverSegment.....	213
To get a maneuver segment.	
ManeuverSegments.....	214
To get the maneuver segments.	
ManeuverSequence.....	218
To get a sequence of a maneuver.	
ManeuverSequences.....	219
To get all the sequences of a maneuver.	
ManeuverTransition.....	222
To get the transition of a maneuver.	
OpenClutch.....	223
To specify the OpenClutch type.	
ReferenceGear.....	224
To specify a ReferenceGear type.	
SelectorProfile.....	225
To get the selector lever type.	
Stimulus.....	226
To specify a stimulus type.	

Information in other sections

[Overview of the Classes for Creating Scenarios..... 177](#)

The overview gives you a quick overview of the classes for the Scenario Editor.

[Automating Scenarios in Python..... 170](#)

You can change the parameters of the scenario.

Driver

Purpose To specify the Driver type.

Class Description (Driver)

Syntax `Driver = GearProfile.ActiveElement`

Purpose To specify the Driver type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the element.
Type	ClutchAndGearTypes ¹⁾	To get the type of the element.
SaturateGear	Boolean	To get/set the saturation of the gear.
Gear	Integer	To get/set the gear.

¹⁾ Refer to [ClutchAndGearTypes](#) on page 366.

Methods —

Related topics**References**

[GearProfile..... 203](#)

DriverType

Purpose To specify the values of the driver type.

Class Description (DriverType)

Syntax `DriverType = FollowRoad.DriverTypeValues`

Purpose To specify the values of the driver type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
BrakeThreshold	Double	To get/set the brake acceleration threshold.
LateralAcceleration	Double	To get/set the maximum lateral acceleration.
LongitudinalAcceleration	Double	To get/set the maximum longitudinal acceleration.

Methods —

Related topics

References

FollowRoad.....	320
Scenario Properties Dialog.....	93

GearProfile

Purpose To get the clutch and gear type.

Where to go from here

Information in this section

Class Description (GearProfile).....	204
To describe the class and its attributes.	

Activate.....	204
To activate an element for the clutch and gear profile.	

Class Description (GearProfile)

Syntax `GearProfile = ManeuverActivity.ClutchAndGearType`

Purpose To get the clutch and gear type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	OpenClutch, Driver, Stimulus, or ReferenceGear ¹⁾	To get the active element.
AvailableElements	String[]	To get a list of all the available elements.

¹⁾ Refer to [OpenClutch](#) on page 223, [Driver](#) on page 202, [Stimulus](#) on page 226, or [ReferenceGear](#) on page 224.

Methods The class contains the following methods:

Method	Purpose
Activate	To activate an element for the clutch and gear profile. Refer to Activate on page 204.

Related topics

References

ManeuverActivity.....	213
---------------------------------------	-----

Activate

Class GearProfile

Syntax `ActivatedElement = GearProfile.Activate(Type)`

Purpose To activate an element for the clutch and gear profile.

Parameters The method uses the following parameters:

Parameter	Type	Description
Type	OpenClutch, Driver, Stimulus, or ReferenceGear ¹⁾	The type to be activated.

¹⁾ Refer to [OpenClutch](#) on page 223, [Driver](#) on page 202, [Stimulus](#) on page 226, or [ReferenceGear](#) on page 224.

Return value The method returns the following parameter:

Type	Description
OpenClutch, Driver, Stimulus, or ReferenceGear ¹⁾	The activated element.

¹⁾ Refer to [OpenClutch](#) on page 223, [Driver](#) on page 202, [Stimulus](#) on page 226, or [ReferenceGear](#) on page 224.

Related topics

References

[Class Description \(GearProfile\)](#)..... 204

LateralProfile

Purpose To specify the lateral profile of a maneuver's activity.

Where to go from here

Information in this section

[Class Description \(LateralProfile\)](#)..... 205
To describe the class and its attributes.

[Activate](#)..... 206
To get a specific lateral type.

Class Description (LateralProfile)

Syntax

```
LateralProfile = ManeuverActivity.LateralType
```

Purpose To specify the lateral profile of a maneuver's activity.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	StimulusAngle, StimulusTorque, BasicRoad, FollowRoad, or NotUsed ¹⁾	To get the active lateral type object.
AvailableElements	String[]	To get the names of all available lateral types.

¹⁾ Refer to [StimulusAngle](#) on page 353, [StimulusTorque](#) on page 353, [BasicRoad](#) on page 311, [FollowRoad](#) on page 320, or [NotUsed](#) on page 330.

Methods The class contains the following methods:

Method	Purpose
Activate	To get a specific lateral type. Refer to Activate on page 206.

Related topics

References

[ManeuverActivity](#)..... 213

Activate

Class LateralProfile

Syntax `LateralType = LateralProfileType.Activate(Type)`

Purpose To get a specific lateral type.

Parameters The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific lateral type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
StimulusAngle, StimulusTorque, BasicRoad, FollowRoad, or NotUsed ¹⁾	A lateral type object according to the given parameter.

¹⁾ Refer to [StimulusAngle](#) on page 353, [StimulusTorque](#) on page 353, [BasicRoad](#) on page 311, [FollowRoad](#) on page 320, or [NotUsed](#) on page 330.

Related topics**References**

[Class Description \(LateralProfile\)](#)..... 205

LongitudinalProfile

Purpose

To specify the longitudinal profile of a maneuver's activity.

Where to go from here**Information in this section**

[Class Description \(LongitudinalProfile \(Maneuver\)\)](#)..... 207
To describe the class and its attributes.

[Activate](#)..... 208
To get a specific longitudinal type.

Class Description (LongitudinalProfile (Maneuver))

Syntax

```
LongitudinalProfile = ManeuverActivity.LongitudinalType
```

Purpose

To specify the longitudinal profile of a maneuver's activity.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Velocity, Stop, PedalStimulus, Standstill, or LateralAcceleration ¹⁾	To get the active longitudinal type object.

Attributes	Type	Purpose
AvailableElements	String[]	To get the names of all available longitudinal types.

¹⁾ Refer to [Velocity](#) on page 362, [Stop](#) on page 354, [PedalStimulus](#) on page 331, [Standstill](#) on page 352, or [LateralAcceleration](#) on page 321.

Methods

The class contains the following methods:

Method	Purpose
Activate	To get a specific longitudinal type. Refer to Activate on page 208.

Related topics

References

[ManeuverActivity](#)..... 213

Activate

Class

LongitudinalType

Syntax

```
LongitudinalType = LongitudinalProfile.Activate(Type)
```

Purpose

To get a specific longitudinal type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific longitudinal type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
Velocity, Stop, PedalStimulus, Standstill, or LateralAcceleration ¹⁾	A longitudinal type object according to the given parameter.

¹⁾ Refer to [Velocity](#) on page 362, [Stop](#) on page 354, [PedalStimulus](#) on page 331, [Standstill](#) on page 352, or [LateralAcceleration](#) on page 321.

Related topics**References**

[Class Description \(LongitudinalProfile \(Maneuver\)\)](#)..... 207

Maneuver

Purpose

To get a maneuver.

Class Description (Maneuver)

Syntax

```
Maneuver = Maneuvers.Item(Index)
Maneuver = Maneuvers.Add()
```

Purpose

To get a maneuver.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
CompleteName	String	To get the complete name.
Comment	String	To get/set a comment of the maneuver.
Index	Integer	To get/set the index of the maneuver.
Name	String	To get/set the maneuver name.
Sequences	ManeuverSequences ¹⁾	To get the sequences of the maneuver.
TrafficDriverType	TrafficDriverTypes ²⁾	To get the traffic driver.
Valid	Boolean	To get the information whether the maneuver is valid.

¹⁾ Refer to [ManeuverSequences](#) on page 219.

²⁾ Refer to [TrafficDriverTypes](#) on page 298.

Methods

—

Related topics**References**

[Maneuvers](#)..... 210

Maneuvers

Purpose To manage the maneuvers of a scenario.

Where to go from here

Information in this section

Class Description (Maneuvers)	210
To describe the class and its attributes.	
Add	211
To add a maneuver.	
Item	211
To get a specific maneuver.	
Remove	212
To remove a maneuver.	

Class Description (Maneuvers)

Syntax

```
Maneuvers = ActiveTrafficScenario.Maneuver
```

Purpose To manage the maneuvers of a scenario.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of maneuvers.

Methods

The class contains the following methods:

Method	Purpose
Add	To add a maneuver. Refer to Add on page 211.
Item	To get a specific maneuver. Refer to Item on page 211.
Remove	To remove a maneuver. Refer to Remove on page 212.

Related topics**References**

[ActiveTrafficScenario](#)..... 188

Add

Class

Maneuvers

Syntax

```
Maneuver = Maneuvers.Add()
```

Purpose

To add a maneuver.

Parameters

—

Return value

The method returns the following parameter:

Type	Description
Maneuver ¹⁾	The added maneuver.

¹⁾ Refer to [Maneuver](#) on page 209.

Related topics**References**

[Class Description \(Maneuvers\)](#)..... 210

Item

Class

Maneuvers

Syntax

```
Maneuver = Maneuvers.Item(Index)
```

Purpose

To get a specific maneuver.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Object	The index of the maneuver.

Return value

The method returns the following parameter:

Type	Description
Maneuver ¹⁾	The specific maneuver.

¹⁾ Refer to [Maneuver](#) on page 209.

Related topics**References**

[Class Description \(Maneuvers\)](#)..... 210

Remove

Class

Maneuvers

Syntax

```
RetVal = Maneuvers.Remove(Index)
```

Purpose

To remove a maneuver.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	object	The maneuver to be removed.

Return value

The method returns the following parameter:

Type	Description
Boolean	True if successful.

Related topics**References**

[Class Description \(Maneuvers\)](#)..... 210

ManeuverActivity

Purpose To get a maneuver activity.

Class Description (ManeuverActivity)

Syntax `ManeuverActivity = ManeuverSegment.Activity`

Purpose To get a maneuver activity.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
LongitudinalType	LongitudinalProfile ¹⁾	To get the longitudinal type.
LateralType	LateralProfile ²⁾	To get the lateral type.
ClutchAndGearType	GearProfile ³⁾	To get the clutch and gear type.
SelectorLeverType	SelectorProfile ⁴⁾	To get the selector lever type.

¹⁾ Refer to [LongitudinalProfile](#) on page 245.

²⁾ Refer to [LateralProfile](#) on page 243.

³⁾ Refer to [GearProfile](#) on page 203.

⁴⁾ Refer to [SelectorProfile](#) on page 225.

Methods —

Related topics

References

[ManeuverSegment](#)..... 213

ManeuverSegment

Purpose To get a maneuver segment.

Class Description (ManeuverSegment)

Syntax

```
ManeuverSegment = ManeuverSegments.Item(ID)
ManeuverSegment = ManeuverSegments.Append()
ManeuverSegment = ManeuverSegments.Insert(ID, Type)
```

Purpose

To get a maneuver segment.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Activity	ManeuverActivity ¹⁾	To get the activity of the segment.
Comment	String	To get/set a comment for the segment.
Name	String	To get/set a name for the segment.
Successor	ManeuverSegment ²⁾	To get the succeeding segment.
Transition	ManeuverTransition ³⁾	To get the transition of the segment.
UserOutputs	UserOutputs ⁴⁾	To get the user signals.
Valid	Boolean	To get the information whether the segment is valid.

¹⁾ Refer to [ManeuverActivity](#) on page 213.

²⁾ Refer to [ManeuverSegment](#) on page 213.

³⁾ Refer to [ManeuverTransition](#) on page 222.

⁴⁾ Refer to [UserOutputs](#) on page 249.

Methods

—

Related topics

References

[ManeuverSegments](#)..... 214

ManeuverSegments

Purpose

To get the maneuver segments.

Where to go from here

Information in this section

Class Description (ManeuverSegments)	215
To describe the class and its attributes.	
Append	216
To append a new segment at the end of the segments.	
Insert	216
To insert a new segment after or before a specific segment.	
Item	217
To get a specific segment.	
Remove	218
To remove a segment.	

Class Description (ManeuverSegments)

Syntax

```
ManeuverSegments = ManeuverSequence.Segments
```

Purpose

To get the maneuver segments.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of segments.

Methods

The class contains the following methods:

Method	Purpose
Append	To append a new segment at the end of the segments. Refer to Append on page 216.
Insert	To insert a new segment after or before a specific segment. Refer to Insert on page 216.
Item	To get a specific segment. Refer to Item on page 217.
Remove	To remove a segment. Refer to Remove on page 218.

Related topics

References

ManeuverSequence	218
--	-----

Append

Class ManeuverSegments

Syntax `ManeuverSegment = ManeuverSegments.Append()`

Purpose To append a new segment at the end of the segments.

Parameters –

Return value The method returns the following parameter:

Type	Description
ManeuverSegment ¹⁾	The new segment.

¹⁾ Refer to [ManeuverSegment](#) on page 213.

Related topics

References

[Class Description \(ManeuverSegments\)](#)..... 215

Insert

Class ManeuverSegments

Syntax `ManeuverSegment = ManeuverSegments.Insert(ID, Type)`

Purpose To insert a new segment after or before a specific segment.

Parameters The method uses the following parameters:

Parameter	Type	Description
ID	object	The identifier of the segment after or before which the new segment is inserted.
Type	InsertTypes ¹⁾	The insertion type.

¹⁾ Refer to [InsertTypes](#) on page 368.

Return value

The method returns the following parameter:

Type	Description
ManeuverSegment ¹⁾	The new segment.

¹⁾ Refer to [ManeuverSegment](#) on page 213.

Related topics**References**

[Class Description \(ManeuverSegments\)](#)..... 215

Item

Class

ManeuverSegments

Syntax

```
ManeuverSegment = ManeuverSegments.Insert(ID)
```

Purpose

To get a specific segment.

Parameters

The method uses the following parameters:

Parameter	Type	Description
ID	Object	The identifier of the specific segment.

Return value

The method returns the following parameter:

Type	Description
ManeuverSegment ¹⁾	The specific segment.

¹⁾ Refer to [ManeuverSegment](#) on page 213.

Related topics**References**

[Class Description \(ManeuverSegments\)](#)..... 215

Remove

Class ManeuverSegments

Syntax `RetVal = ManeuverSegments.Remove(ID)`

Purpose To remove a segment.

Parameters The method uses the following parameters:

Parameter	Type	Description
ID	object	The identifier of the segment to be removed.

Return value The method returns the following parameter:

Type	Description
Boolean	True if successful.

Related topics

References

[Class Description \(ManeuverSegments\)](#)..... 215

ManeuverSequence

Purpose To get a sequence of a maneuver.

Class Description (ManeuverSequence)

Syntax `Sequence = ManeuverSequences.Item(Index)`
`Sequence = ManeuverSequences.Add()`

Purpose To get a sequence of a maneuver.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
AdditionalLateralOffset	Double	To get/set an additional lateral offset.
InitialHeight	Double	To get/set the initial height.
InitialHeightMode	HeightModeTypes ¹⁾	To get/set the mode for specifying the initial height.
InitialLaneIndex	Integer	To get/set the initial lane index.
InitialLongitudinalVelocity	Double	To get/set the initial velocity in longitudinal direction.
Route	RouteSelection ²⁾	To get the route.
Segments	ManeuverSegments ³⁾	To get the segments of the maneuver sequence
StartCondition	ManeuverTransition ⁴⁾	To get the start condition.
StartPosition	Double	To get/set the start position.
VehicleOrientation	Double	To get/set the vehicle orientation.

¹⁾ Refer to [HeightModeTypes](#) on page 368.

²⁾ Refer to [RouteSelection](#) on page 339.

³⁾ Refer to [ManeuverSegments](#) on page 214.

⁴⁾ Refer to [ManeuverTransition](#) on page 222.

Methods

—

Related topics**References**

[ManeuverSequences](#)..... 219

ManeuverSequences

Purpose

To get all the sequences of a maneuver.

Where to go from here**Information in this section**

[Class Description \(ManeuverSequences\)](#)..... 220

To describe the class and its attributes.

[Add](#)..... 220

To add a sequence.

Item.....	221
To get a specific sequence.	
Remove.....	222
To remove a sequence.	

Class Description (ManeuverSequences)

Syntax

```
ManeuverSequences = Maneuver.Sequences
```

Purpose

To get all the sequences of a maneuver.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of sequences.

Methods

The class contains the following methods:

Method	Purpose
Add	To add a sequence. Refer to Add on page 220.
Item	To get a specific sequence. Refer to Item on page 221.
Remove	To remove a sequence. Refer to Remove on page 222.

Related topics

References

Maneuver.....	209
-------------------------------	---------------------

Add

Class

ManeuverSequences

Syntax

```
Sequence = ManeuverSequences.Add()
```

Purpose	To add a sequence.				
Parameters	–				
Return value	The method returns the following parameter: <table> <tr> <th>Type</th><th>Description</th></tr> <tr> <td>ManeuverSequence¹⁾</td><td>The added sequence.</td></tr> </table> ¹⁾ Refer to ManeuverSequence on page 218.	Type	Description	ManeuverSequence ¹⁾	The added sequence.
Type	Description				
ManeuverSequence ¹⁾	The added sequence.				
Related topics	References <table> <tr> <td>Class Description (ManeuverSequences).....</td><td>220</td></tr> </table>	Class Description (ManeuverSequences)	220		
Class Description (ManeuverSequences)	220				

Item

Class

ManeuverSequences

Syntax

Sequence = ManeuverSequences.Item(Index)

Purpose

To get a specific sequence.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Object	The index of the sequence.

Return value

The method returns the following parameter:

Type	Description
ManeuverSequence ¹⁾	The specific sequence.

¹⁾ Refer to [ManeuverSequence](#) on page 218.

Related topics**References**

[Class Description \(ManeuverSequences\)](#)..... 220

Remove

Class

ManeuverSequences

Syntax

```
RetVal = ManeuverSequences.Remove(Index)
```

Purpose

To remove a sequence.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	object	The index of the sequences to be removed.

Return value

The method returns the following parameter:

Type	Description
Boolean	True if successful

Related topics**References**

[Class Description \(ManeuverSequences\)](#)..... 220

ManeuverTransition

Purpose

To get the transition of a maneuver.

Class Description (ManeuverTransition)

Syntax

```
ManeuverTransition = ManeuverSegment.Transition
```

Purpose

To get the transition of a maneuver.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get/set the transition name.
Comment	String	To get/set the comment/description of the transition.
Delay	Double	To get/set the delay value.
Valid	Boolean	To get the information whether the transition is valid.
Conditions	TransitionsConditions ¹⁾	To get the condition types.

¹⁾ Refer to [TransitionsConditions](#) on page 278.

Methods

—

Related topics

References

[ManeuverSegment](#)..... 213

OpenClutch

Purpose

To specify the OpenClutch type.

Class Description (OpenClutch)

Syntax

```
OpenClutch = GearProfile.ActiveElement
```

Purpose

To specify the OpenClutch type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the element.
Type	ClutchAndGearTypes ¹⁾	To get the type of the element.

¹⁾ Refer to [ClutchAndGearTypes](#) on page 366.

Methods

—

Related topics**References**

[GearProfile](#)..... 203

ReferenceGear

Purpose

To specify a ReferenceGear type.

Class Description (ReferenceGear)

Syntax

```
ReferenceGear = GearProfile.ActiveElement
```

Purpose

To specify a ReferenceGear type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the element.
Type	ClutchAndGearTypes ¹⁾	To get the type of the element.
GearSource	Constant, Table, Manual, or Extern ²⁾	To get the source type of the gear source.

¹⁾ Refer to [ClutchAndGearTypes](#) on page 366.

²⁾ Refer to [Constant](#) on page 343, [Table](#) on page 351, [Manual](#) on page 346, or [Extern](#) on page 343.

Methods

—

Related topics

References

[GearProfile](#)..... 203

SelectorProfile

Purpose

To get the selector lever type.

Where to go from here

Information in this section

[Class Description \(SelectorProfile\)](#)..... 225
To describe the class and its attributes.

[Activate](#)..... 226
To activate an element for the selector profile.

Class Description (SelectorProfile)

Syntax

```
SelectorProfile = ManeuverActivity.SelectorLeverType
```

Purpose

To get the selector lever type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Driver or Stimulus ¹⁾	To get the active element.
AvailableElements	String[]	To get the names of all the available elements.

¹⁾ Refer to [Driver](#) on page 202 or [Stimulus](#) on page 226.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate an element. Refer to Activate on page 226.

Related topics

References

[ManeuverActivity](#)..... 213

Activate

Class

SelectorProfile

Syntax

```
ActivatedElement = SelectorProfile.Activate(Type)
```

Purpose

To activate an element for the selector profile.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Driver or Stimulus ¹⁾	The type to be activated.

¹⁾ Refer to [Driver](#) on page 202 or [Stimulus](#) on page 226.

Return value

The method returns the following parameter:

Type	Description
Driver or Stimulus ¹⁾	The activated element.

¹⁾ Refer to [Driver](#) on page 202 or [Stimulus](#) on page 226.

Related topics

References

[Class Description \(SelectorProfile\)](#)..... 225

Stimulus

Purpose

To specify a stimulus type.

Class Description (Stimulus)

Syntax

```
Stimulus = GearProfile.ActiveElement
```

Purpose

To specify a stimulus type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the element.
Type	ClutchAndGearTypes ¹⁾	To get the clutch and gear type.
GearSource	Constant, Table, Manual, or Extern ²⁾	To get the source type of the gear source.
ClutchSource	Constant, Table, Manual, Extern, or Ramp ³⁾	To get the source type of the clutch source.

¹⁾ Refer to [ClutchAndGearTypes](#) on page 366.

²⁾ Refer to [Constant](#) on page 343, [Table](#) on page 351, [Manual](#) on page 346, or [Extern](#) on page 343.

³⁾ Refer to [Constant](#) on page 343, [Table](#) on page 351, [Manual](#) on page 346, [Extern](#) on page 343, or [Ramp](#) on page 347.

Methods

—

Related topics

References

[GearProfile](#)..... 203

Classes for Configuring Fellows

Where to go from here

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To get information about a fellow.	
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To get information about the activity of a fellow.	
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Containing all fellows of the active scenario.	
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Containing the information of a single segment of a fellow in a scenario (including activity, transition, and fellow user signal).	
FellowSegments	234
This is the collection of all segments of a fellow definition.	
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To get the sequence of a fellow.	
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This is the collection of all sequences of a fellow definition.	
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To get the transition of a fellow.	
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To get lateral types of a fellow definition.	
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To get longitudinal types of an activity.	
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To assign or get information about the assigned traffic object type and get a collection of all the traffic objects.	
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The overview gives you a quick overview of the classes for the Scenario Editor.	

[Automating Scenarios in Python.....](#) 170
 You can change the parameters of the scenario.

Fellow

Purpose To get information about a fellow.

Class Description (Fellow)

Syntax `Fellow = Fellows.Add()`

Purpose To get information about a fellow.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get/set the fellow name.
Comment	String	To get/set the comment/description of the fellow.
CompleteName	String	To get the complete fellow name, for example, 'F7: Blue Roadster'.
Index	Integer	To get the matrix index in the model of the fellow.
Sequences	FellowSequences ¹⁾	To get the FellowSequences object.
TrafficDriverType	TrafficDriverTypes ²⁾	To get the traffic driver.
TrafficObjectType	TrafficObjectTypes ³⁾	To get the assigned traffic object type of this fellow.

¹⁾ Refer to [FellowSequences](#) on page 239.

²⁾ Refer to [TrafficDriverTypes](#) on page 298.

³⁾ Refer to [TrafficObjectTypes](#) on page 247.

Related topics

References

[Fellows.....](#) 230

FellowActivity

Purpose To get information about the activity of a fellow.

Class Description (FellowActivity)

Syntax No direct creation

Purpose To get information about the activity of a fellow.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
LongitudinalType	LongitudinalProfile ¹⁾	To get the longitudinal type definition of the activity.
LateralType	LateralProfile ²⁾	To get the lateral type definition of the activity.

¹⁾ Refer to [LongitudinalProfile](#) on page 245.

²⁾ Refer to [LateralProfile](#) on page 243.

Fellows

Purpose The class contains all fellows of the active scenario.

Where to go from here

Information in this section

Class Description (Fellows)	231
To describe the class and its attributes.	
Add	231
To add a fellow to the fellows collection.	
Item	232
To get a specific fellow from the fellows collection.	
Remove	233
To remove a specific fellow from the fellows collection.	

Class Description (Fellows)

Syntax Is not created directly

Purpose The class contains all information and parameters about the fellows of the active scenario.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of fellows in the scenario.

Methods The class contains the following methods:

Method	Purpose
Add	To add a fellow to the fellows collection. Refer to Add on page 231.
Item	To get a specific fellow from the fellows collection. Refer to Item on page 232.
Remove	To remove a specific fellow from the fellows collection. Refer to Remove on page 233.

Related topics

HowTos

[How to Create and Specify a Fellow](#)..... 59

References

[Fellow Properties](#)..... 120

Add

Class Fellows

Syntax `Fellow = Fellows.Add()`

Purpose To add a fellow to the fellows collection.

Parameters

—

Return value

The method returns an object of the following type:

Type	Description
Fellow ¹⁾	The new fellow object.

¹⁾ Refer to [Fellow](#) on page 229.

Related topics**References**

[Class Description \(Fellows\)](#)..... 231

Item

Class

Fellows

Syntax

```
Fellow = Fellows.Item(Index)
```

Purpose

To get a specific fellow from the fellows collection.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The index of the specific fellow. Strings (fellow name) and integers (fellow index) are supported. Note The index can differ from the index that is displayed in ModelDesk if the simulation ID has been modified. Use the simulation ID to clearly identify a fellow.

Return value

The method returns an object of the following type:

Type	Description
Fellow ¹⁾	The specific fellow object.

¹⁾ Refer to [Fellow](#) on page 229.

Related topics**References**

[Class Description \(Fellows\)](#)..... 231

Remove

Class

Fellows

Syntax

```
Fellow = Fellows.Remove(Index)
```

Purpose

To remove a specific fellow from the fellows collection.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The index of the specific fellow. Strings (fellow name), integers (fellow index) and the fellow itself are supported.

Return value

The method returns an object of the following type:

Type	Description
Boolean	True if successful

Related topics**References**

[Class Description \(Fellows\)](#)..... 231

FellowSegment

Purpose

The class contains the information of a single segment of a fellow in a scenario (including activity, transition, and fellow user signal).

Class Description (FellowSegment)

Syntax No direct creation

Purpose To get information of a single segment of a fellow in a scenario (including activity, transition, and fellow user signal).

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Activity	FellowActivity ¹⁾	To get the FellowActivity object.
Transition	FellowTransition ²⁾	To get the FellowTransition object.
UserOutputs	UserOutputs ³⁾	To get the UserOutputs object.
Comment	String	To get/set the comment of the segment.
Name	String	To get/set the segment name.
Valid	Boolean	To get the valid state of the segment.

¹⁾ Refer to [FellowActivity](#) on page 230.

²⁾ Refer to [FellowTransition](#) on page 242.

³⁾ Refer to [UserOutputs](#) on page 249.

FellowSegments

Purpose This is the collection of all segments of a fellow definition.

Where to go from here

Information in this section

Class Description (FellowSegments)	235
To describe the class and its attributes.	
Append	235
To append a new FellowSegment object to the FellowSegments collection.	
Insert	236
To insert a new FellowSegment object to the FellowSegments collection.	
Item	237
To get a FellowSegment object specified by the given index.	
Remove	237
To remove a specific FellowSegment object from the FellowSegments collection.	

Class Description (FellowSegments)

Syntax

```
FellowSegments = FellowSequence.Segments
```

Purpose

To handle a collection of segments of a fellow.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of FellowSegment objects.

Methods

The class contains the following methods:

Method	Purpose
Append	To append a new FellowSegment object to the FellowSegments collection. Refer to Append on page 235.
Insert	To insert a new FellowSegment object to the FellowSegments collection. Refer to Insert on page 236.
Item	To get a specific FellowSegment object from the collection. Refer to Item on page 237.
Remove	To remove a specific FellowSegment object from the FellowSegments collection. Refer to Remove on page 237.

Related topics

References

[FellowSequence](#)..... 238

Append

Class

FellowSegments

Syntax

```
FellowSegment = FellowSegments.Append()
```

Purpose

To append a new FellowSegment object to the FellowSegments collection.

Parameters

—

Return value

The method returns an object of the following type:

Type	Description
FellowSegment ¹⁾	The new fellow segment.

¹⁾ Refer to [FellowSegment](#) on page 233.

Related topics**References**

[Class Description \(FellowSegments\)](#)..... 235

Insert

Class

FellowSegments

Syntax

```
FellowSegment = FellowSegments.Insert(Index, Type)
```

Purpose

To insert a new FellowSegment object to the FellowSegments collection.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	Index of a specific FellowSegment object to which the new FellowSegment object is inserted before or after. The value can be specified as object (FellowSegment) or integer (index).
Type	InsertTypes ¹⁾	Specifies whether the new FellowSegment object is inserted before or after the specified FellowSegment object.

¹⁾ Refer to [InsertTypes](#) on page 368.

Return value

The method returns an object of the following type:

Type	Description
FellowSegment ¹⁾	The new FellowSegment object.

¹⁾ Refer to [FellowSegment](#) on page 233.

Related topics**References**

[Class Description \(FellowSegments\)](#)..... 235

Item

Class FellowSegments

Syntax `FellowSegment = FellowSegments.Item(Index)`

Purpose To get a specific FellowSegment object from the FellowSegments collection.

Parameters The method uses the following parameters:

Parameter	Type	Description
Index	Variant	Index of the specific fellow segment. The value can be specified as string (name) or integer (index).

Return value The method returns an object of the following type:

Type	Description
FellowSegment ¹⁾	The specified segment.

¹⁾ Refer to [FellowSegment](#) on page 233.

Related topics

References

[Class Description \(FellowSegments\)](#)..... 235

Remove

Class FellowSegments

Syntax `Result = FellowSegments.Remove(Index)`

Purpose To remove a specific FellowSegment object from the FellowSegments collection.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	Index of the FellowSegment object to be removed. You can specify the value as string (name) or integer (index).

Return value

The method returns an object of the following type:

Type	Description
Boolean	True if successful.

Related topics**References**

[Class Description \(FellowSegments\)](#)..... 235

FellowSequence

Purpose

To get the sequence of a fellow.

Class Description (FellowSequence)

Syntax

```
FellowSequence = FellowSequences.Item(Index)
```

Purpose

To get the sequence of a fellow.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Segments	FellowSegments ¹⁾	To get the fellow segments.
StartCondition	FellowTransition ²⁾	To get the start transition.
Route	RouteSelection ³⁾	To get the route for the fellow.

¹⁾ Refer to [FellowSegments](#) on page 234.

²⁾ Refer to [FellowTransition](#) on page 242.

³⁾ Refer to [RouteSelection](#) on page 339.

Related topics**References**

[FellowSequences](#)..... 239

FellowSequences

Purpose

This is the collection of all sequences of a fellow definition.

Where to go from here**Information in this section**

Class Description (FellowSequences (Collection))	239
To describe the class and its attributes.	
Add	240
To add a new FellowSequence object to the collection.	
Item	241
To get a FellowSequence object from the FellowSequences collection.	
Remove	241
To remove a specific FellowSequence object from the collection.	

Class Description (FellowSequences (Collection))

Syntax

```
FellowSequences = Fellow.Sequences
```

Purpose

To handle a collection of sequences of a fellow.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of FellowSequence objects.

Methods

The class contains the following methods:

Method	Purpose
Add	To add a new FellowSequence object to the collection. Refer to Add on page 240.
Item	To get a specific FellowSequence object from the collection. Refer to Item on page 241.
Remove	To remove a specific FellowSequence object from the collection. Refer to Remove on page 241.

Related topics**References**

[Fellow](#)..... 229

Add

Class

FellowSequences

Syntax

```
Result = FellowSegments.Remove(Index)
```

Purpose

To add a new FellowSequence object to the collection.

Parameters

—

Return value

The method returns an object of the following type:

Type	Description
FellowSequence ¹⁾	The new FellowSequence object.

¹⁾ Refer to [FellowSequence](#) on page 238.

Related topics**References**

[FellowSequences](#)..... 239

Item

Class FellowSequences

Syntax `FellowSequence = FellowSequences.Item(Index)`

Purpose To get a FellowSequence object from the FellowSequences collection.

Parameters The method uses the following parameters:

Parameter	Type	Description
Index	Variant	Index of a specific FellowSequence object. The value can be specified as string (name) or integer (index).

Return value The method returns an object of the following type:

Type	Description
FellowSequence ¹⁾	The specified FellowSequence object.

¹⁾ Refer to [FellowSequence](#) on page 238.

Related topics

References

[FellowSequences](#)..... 239

Remove

Class FellowSequences

Syntax `Result = FellowSequences.Remove(Index)`

Purpose To remove a specific FellowSequence object from the collection.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	Index of the FellowSequence object to be removed. The value can be specified as FellowSequence object or integer (index).

Return value

The method returns an object of the following type:

Type	Description
Bool	True if successful

Related topics**References**

[FellowSequences.....](#) 239

FellowTransition

Purpose

To get the transition of a fellow.

Class Description (FellowTransition)

Syntax

```
FellowTransition = FellowSequence.StartCondition
FellowTransition = FellowSegment.Transition
```

Purpose

To get the transition of a fellow.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get/set the transition name.
Comment	String	To get/set the comment/description of the transition.
Delay	Double	To get/set the delay value.
Valid	Boolean	To get the information whether the transition is valid.

Attributes	Type	Purpose
Conditions	TransitionsConditions ¹⁾	To get the condition types.

¹⁾ Refer to [TransitionsConditions](#) on page 278.

Related topics

References

FellowSegment	233
FellowSequence	238

LateralProfile

Purpose

The class contains lateral types of a fellow definition.

Where to go from here

Information in this section

Class Description (LateralProfile)	243
To describe the class and its attributes.	
Activate	244
To get a specific lateral type.	

Class Description (LateralProfile)

Syntax

```
LateralProfile = FellowActivity.LateralType
```

Purpose

To get lateral types of an activity.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	NotUsed, Continue, LateralDeviation, or LaneSelection ¹⁾	To get the active lateral type object.
AvailableElements	Strings (collection)	To get the names of all available lateral types.

¹⁾ Refer to [NotUsed](#) on page 330, [Continue](#) on page 277, [LateralDeviation](#) on page 322, or [LaneSelection](#) on page 324.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a lateral type. Refer to Activate on page 244.

Related topics

References

[FellowActivity](#)..... 230

Activate

Class

LateralProfile

Syntax

```
LateralType = LateralProfileType.Activate(Type)
```

Purpose

To get a specific lateral type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific lateral type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
NotUsed, Continue, LateralDeviation, or LaneSelection ¹⁾	A lateral type object according to the given parameter.

¹⁾ Refer to [NotUsed](#) on page 330, [Continue](#) on page 277, [LateralDeviation](#) on page 322, or [LaneSelection](#) on page 324.

Related topics

References

[Class Description \(LateralProfile\)](#)..... 243

LongitudinalProfile

Purpose The class contains longitudinal types of an activity.

Where to go from here

Information in this section

Class Description (LongitudinalProfile).....	245
To describe the class and its attributes.	
Activate.....	246
To get a specific longitudinal type.	

Class Description (LongitudinalProfile)

Syntax

```
LongitudinalProfile = FellowActivity.LongitudinalType
```

Purpose

To get the longitudinal type of a fellow's activity.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	NotUsed, Continue, Acceleration, DistanceMeter, DistanceTime, Position, Velocity, or StopWithin ¹⁾	To get the active longitudinal type object.
AvailableElements	Strings (collection)	To get the names of all available longitudinal types.

¹⁾ Refer to [NotUsed](#) on page 331, [Continue](#) on page 277, [Acceleration](#) on page 307, [DistanceMeter](#) on page 315, [DistanceTime](#) on page 317, [Position](#) on page 332, [Position](#) on page 332, [Velocity](#) on page 362, or [StopWithin](#) on page 355.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a longitudinal type. Refer to Activate on page 246.

Related topics

References

[FellowActivity](#)..... 230

Activate

Class

LongitudinalType

Syntax

```
LongitudinalType = LongitudinalProfile.Activate(Type)
```

Purpose

To get a specific longitudinal type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific longitudinal type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
NotUsed, Continue, Acceleration, DistanceMeter, DistanceTime, Position, Velocity, or StopWithin ¹⁾	A longitudinal type object according to the given parameter.

¹⁾ Refer to [NotUsed](#) on page 331, [Continue](#) on page 277, [Acceleration](#) on page 307, [DistanceMeter](#) on page 315, [DistanceTime](#) on page 317, [Position](#) on page 332, [Velocity](#) on page 362, or [StopWithin](#) on page 355.

Related topics

References

[Class Description \(LongitudinalProfile\)](#)..... 245

TrafficObjectTypes

Purpose To assign or get information about the assigned traffic object type and get a collection of all the traffic objects.

Where to go from here Information in this section

Class Description (TrafficObjectTypes)	247
To describe the class and its attributes.	
Activate	248
To activate a traffic object type.	

Class Description (TrafficObjectTypes)

Syntax `TrafficObjectTypes = Fellow.TrafficObjectTypes`

Purpose To assign or get information about the assigned traffic object type and get a collection of all the traffic objects.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	TrafficObjectType ¹⁾	To get the active TrafficObjectType object.
AvailableElements	Strings (collection)	To get the names of all available fellow types.

¹⁾ Refer to [TrafficObjectType](#) on page 248.

Methods The class contains the following methods:

Method	Purpose
Activate	To activate a traffic object type. Refer to Activate on page 248.

Related topics

References

Fellow	229
------------------------------	-----

Activate

Class TrafficObjectTypes

Syntax `TrafficObjectType = TrafficObjectTypes.Activate(Item)`

Purpose To activate a traffic object type.

Parameters The method uses the following parameters:

Parameter	Type	Description
Item	Variant	The identifier of the traffic object type. Only traffic object type names with or without file name extension (string) are supported.

Return value The method returns an object of the following type:

Type	Description
TrafficObjectType ¹⁾	The activated TrafficObjectType object.

¹⁾ Refer to [TrafficObjectType](#) on page 248.

Related topics

References

[Class Description \(TrafficObjectTypes\)](#)..... 247

TrafficObjectType

Purpose To get information about a traffic object type.

Class Description (TrafficObjectType)

Syntax No direct creation

Purpose To get information about a traffic object type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Path	String	To get the path including the file name and extension of the traffic object type.
Comment	String	To get the comment/description of the traffic object type.

UserOutputs

Purpose

To get information about all available fellow and maneuver user signals of an activity.

Where to go from here**Information in this section**

Class Description (UserOutputs (Collection)).....	249
To describe the class and its attributes.	
Item.....	250
To get a specific user signal from the collection.	

Class Description (UserOutputs (Collection))

Syntax

```
UserOutputs = FellowSegment.UserOutputs
UserOutputs = ManeuverSegment.UserOutputs
```

Purpose

To get information about all available fellow or maneuver user signals of an activity.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of all available user outputs in the segment.

Methods

The class contains the following methods:

Method	Purpose
Item	To get a specific user signal from the collection. Refer to Item on page 250.

Related topics**References**

FellowSegment	233
ManeuverSegment	213

Item

Class

UserOutputs

Syntax`UserOutput = UserOutputs.Item(Index)`**Purpose**

To get a specific user signal from the collection.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The index of the specific user signal. Integers (index) are supported.

Return value

The method returns an object of the following type:

Type	Description
UserOutput ¹⁾	The user signal.

¹⁾ Refer to [UserOutput](#) on page 250.**Related topics****References**

Class Description (UserOutputs (Collection))	249
--	-----

UserOutput

Purpose

To get information about a fellow user signal.

Class Description (UserOutput)

Syntax

```
UserOutput = UserOutputs.Item(Index)
```

Purpose

To get information about a user signal.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the alias of the user signal.
Value	UserSignalValue ¹⁾	To get the value of the user signal if isUsed is True.
IsUsed	Boolean	To get/set the state whether the user signal is used.
Valid	Boolean	To get the valid state.
Comment	String	To get the comment/description of the user signal.

¹⁾ Refer to [UserSignalValue](#) on page 265.

Related topics**References**

[UserOutputs](#)..... 249

Classes for Configuring Global User Signals

Where to go from here

Information in this section

GlobalUserSignals	252
To administrate the global user signals.	
GlobalUserSignal	255
To access a global user signal.	
UserSignalSegments	256
To administrate the segments of a global user signal.	
UserSignalSegment	260
To access a segment of a global user signal.	
UserSignalSequences	261
To administrate the sequences of a global user signal.	
UserSignalSequence	264
To access a sequence of a global user signal.	
UserSignalTransition	265
To get information on the transition of a segment.	
UserSignalValue	265
To get the value of the global user signal.	

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Overview of the Classes for Creating Scenarios	177
The overview gives you a quick overview of the classes for the Scenario Editor.	
Automating Scenarios in Python	170
You can change the parameters of the scenario.	

GlobalUserSignals

Purpose

To administrate the global user signals.

Where to go from here

Information in this section

Class Description (GlobalUserSignals)	253
To describe the class and its attributes.	

Add.....	253
To add a GlobalUserSignal object to the collection.	
Item.....	254
To get a specific GlobalUserSignal object.	
Remove.....	255
To remove a specific GlobalUserSignal object.	

Class Description (GlobalUserSignals)

Syntax

```
GlobalUserSignals = ActiveTrafficScenario.GlobalUserSignals
```

Purpose

To administrate the global user signals.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of GlobalUserSignal objects.

Methods

The class contains the following methods:

Method	Purpose
Add	To add a GlobalUserSignal object to the collection. Refer to Add on page 253.
Item	To get a specific GlobalUserSignal object. Refer to Item on page 254.
Remove	To remove a specific GlobalUserSignal object. Refer to Remove on page 255.

Related topics

References

ActiveTrafficScenario.....	188
--	---------------------

Add

Class

GlobalUserSignals

Syntax

```
GlobalUserSignal = GlobalUserSignals.Add()
```

Purpose

To add a GlobalUserSignal object to the collection.

Parameters

–

Return value

The method returns an object of the following type:

Type	Description
GlobalUserSignal ¹⁾	The new GlobalUserSignal object.

¹⁾ Refer to [GlobalUserSignal](#) on page 255.

Related topics**References**

[Class Description \(GlobalUserSignals\)](#)..... 253

Item

Class

GlobalUserSignals

Syntax

```
GlobalUserSignal = GlobalUserSignals.Item(Index)
```

Purpose

To get a specific GlobalUserSignal object.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The index of the specific GlobalUserSignal object. The value can be specified as string (name) or integer (index).

Return value

The method returns an object of the following type:

Type	Description
GlobalUserSignal ¹⁾	The specific GlobalUserSignal object.

¹⁾ Refer to [GlobalUserSignal](#) on page 255.

Related topics**References**[Class Description \(GlobalUserSignals\)](#)..... 253

Remove

Class

GlobalUserSignals

Syntax`Result = GlobalUserSignals.Remove(Index)`**Purpose**

To remove a specific GlobalUserSignal object.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The index of the specific GlobalUserSignal object. The value can be specified as string (name), integer (index), or GlobalUserSignal object.

Return value

The method returns an object of the following type:

Type	Description
Boolean	True if successful

Related topics**References**[Class Description \(GlobalUserSignals\)](#)..... 253

GlobalUserSignal

Purpose

To access a global user signal.

Class Description (GlobalUserSignal)

Syntax

```
GlobalUserSignal = GlobalUserSignals.Item(Index)
```

Purpose

To access a global user signal.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get/set the name of the global user signal.
Comment	String	To get/set a comment for the global user signal.
CompleteName	String	To get the complete name such as "U2: My UserSignal", of the global user signal.
Valid	Boolean	To get the valid state.
Index	Integer	To get the simulation ID.
Sequences	UserSignalSequences ¹⁾	To get the UserSignalSequences object.

¹⁾ Refer to [UserSignalSequences](#) on page 261.

Related topics

References

[GlobalUserSignals](#)..... 252

UserSignalSegments

Purpose

To administrate the segments of a global user signal.

Where to go from here

Information in this section

Class Description (UserSignalSegments)	257
To describe the class and its attributes.	
Append	257
To append a segment to the collection.	
Insert	258
To insert a segment into the collection.	
Item	259
To access a specific segment of the collection.	

[Remove](#).....260
To remove a specific segment from the collection.

Class Description (UserSignalSegments)

Syntax

```
UserSignalSegments = UserSignalSequence.Segments
```

Purpose

To administrate the segments of a global user signal.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of segments.

Methods

The class contains the following methods:

Method	Purpose
Append	To append a segment to the collection. Refer to Append on page 257.
Insert	To insert a segment into the collection. Refer to Insert on page 258.
Item	To access a specific segment of the collection. Refer to Item on page 259.
Remove	To remove a specific segment from the collection. Refer to Remove on page 260.

Related topics

References

[UserSignalSequence](#).....264

Append

Class

UserSignalSegments

Syntax

```
UserSignalSegment = UserSignalSegments.Append()
```

Purpose To append a segment to the collection.

Parameters –

Return value The method returns an object of the following type:

Type	Description
UserSignalSegment ¹⁾	The new UserSignalSegment object.

¹⁾ Refer to [UserSignalSegment](#) on page 260.

Related topics

References

[Class Description \(UserSignalSegments\)](#)..... 257

Insert

Class UserSignalSegments

Syntax `UserSignalSegment = UserSignalSegments.Insert(ID, Type)`

Purpose To insert a segment into the collection.

Parameters The method uses the following parameters:

Parameter	Type	Description
ID	Variant	The index of a specific UserSignalSegment object. The value can be specified as the UserSignalSegment object or integer (constant type).
Type	InsertTypes ¹⁾	Specifies whether the new UserSignalSegment object is inserted before or after the specified UserSignalSegment object specified by the ID parameter.

¹⁾ Refer to [InsertTypes](#) on page 368.

Return value

The method returns an object of the following type:

Type	Description
UserSignalSegment ¹⁾	The new UserSignalSegment object.

¹⁾ Refer to [UserSignalSegment](#) on page 260.

Related topics**References**

[Class Description \(UserSignalSegments\)](#)..... 257

Item

Class

UserSignalSegments

Syntax

```
UserSignalSegment = UserSignalSegments.Item(ID)
```

Purpose

To access a specific segment of the collection.

Parameters

The method uses the following parameters:

Parameter	Type	Description
ID	Variant	The index of the specific UserSignalSegment object. The value can be specified as string (name) or integer (index).

Return value

The method returns an object of the following type:

Type	Description
UserSignalSegment ¹⁾	The specific UserSignalSegment object.

¹⁾ Refer to [UserSignalSegment](#) on page 260.

Related topics**References**

[Class Description \(UserSignalSegments\)](#)..... 257

Remove

Class UserSignalSegments

Syntax `Result = UserSignalSegments.Remove(ID)`

Purpose To remove a specific segment from the collection.

Parameters The method uses the following parameters:

Parameter	Type	Description
ID	Variant	The index of the specific UserSignalSegment object. The value can be specified as UserSignalSegment object or integer (index).

Return value The method returns an object of the following type:

Type	Description
Bool	True if successful

Related topics

References

[Class Description \(UserSignalSegments\)](#)..... 257

UserSignalSegment

Purpose To access a segment of a global user signal.

Class Description (UserSignalSegment)

Syntax `UserSignalSegment = UserSignalSegments.Item(Index)`

Purpose To access a segment of a global user signal.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get/set the name of the segment.
Comment	String	To get/set the comment of the segment.
Valid	Bool	To get the valid state.
Activity	UserSignalValue ¹⁾	To get the value.
Transition	UserSignalTransition ²⁾	To get UserSignalTransition object.
Successor	UserSignalSegment ³⁾	To get the UserSignalSegment of the successor.

¹⁾ Refer to [UserSignalValue](#) on page 265.

²⁾ Refer to [UserSignalTransition](#) on page 265.

³⁾ Refer to [UserSignalSegment](#) on page 260.

Related topics**References**

[UserSignalSegments](#)..... 256

UserSignalSequences

Purpose

To administrate the sequences of a global user signal.

Where to go from here**Information in this section**

Class Description (UserSignalSequences)	261
To describe the class and its attributes.	
Add	262
To add a UserSignalSequence object to the collection.	
Item	263
To access a specific UserSignalSequence object.	
Remove	263
To remove a UserSignalSequence object from the collection.	

Class Description (UserSignalSequences)

Syntax

```
UserSignalSequences = GlobalUserSignal.Sequences
```

Purpose To administrate the sequences of a global user signal.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of sequences of the collection.

Methods The class contains the following methods:

Method	Purpose
Add	To add a UserSignalSequence object to the collection. Refer to Add on page 262.
Item	To access a specific UserSignalSequence object. Refer to Item on page 263.
Remove	To remove a UserSignalSequence object from the collection. Refer to Remove on page 263.

Related topics

References

[GlobalUserSignal](#)..... 255

Add

Class UserSignalSequences

Syntax `UserSignalSequence = UserSignalSequences.Add()`

Purpose To add a UserSignalSequence object to the collection.

Parameters —

Return value The method returns an object of the following type:

Type	Description
UserSignalSequence ¹⁾	The new UserSignalSequence object.

¹⁾ Refer to [UserSignalSequence](#) on page 264.

Related topics**References**

[Class Description \(UserSignalSequences\)](#)..... 261

Item

Class

UserSignalSequences

Syntax

```
UserSignalSequence = UserSignalSequences.Item(Variant Index)
```

Purpose

To access a specific UserSignalSequence object.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The index of the specific UserSignalSequence object. The value can be specified as integer (index).

Return value

The method returns an object of the following type:

Type	Description
UserSignalSequence ¹⁾	The specific UserSignalSequence object.

¹⁾ Refer to [UserSignalSequence](#) on page 264.

Related topics**References**

[Class Description \(UserSignalSequences\)](#)..... 261

Remove

Class

UserSignalSequences

Syntax

```
Result = UserSignalSequences.Remove(Variant Index)
```

Purpose To remove a UserSignalSequence object from the collection.

Parameters The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The index of the specific UserSignalSequence object. The value can be specified as UserSignalSequence object or integer (index).

Return value The method returns an object of the following type:

Type	Description
Bool	True if successful

Related topics

References

[Class Description \(UserSignalSequences\)](#)..... 261

UserSignalSequence

Purpose To access a sequence of a global user signal.

Class Description (UserSignalSequence)

Syntax `UserSignalSequence = UserSignalSequences.Item(variant Index)`

Purpose To access a sequence of a global user signal.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Segments	UserSignalSegments ¹⁾	To get the UserSignalSegments object.
StartCondition	UserSignalTransition ²⁾	To get the UserSignalTransition object that is used as start condition.

¹⁾ Refer to [UserSignalSegments](#) on page 256.

²⁾ Refer to [UserSignalTransition](#) on page 265.

Related topics**References**

[UserSignalSequences](#)..... 261

UserSignalTransition

Purpose

To get information on the transition of a segment.

Class Description (UserSignalTransition)

Syntax

```
UserSignalTransition = UserSignalSegment.Transition
UserSignalTransition = UserSignalSequence.StartCondition
```

Purpose

To get information on the transition of a segment.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get/set the name of the transition.
Comment	String	To get/set the comment of the transition.
Delay	Double	To get/set the time delay.
Conditions	TransitionsConditions ¹⁾	To get the transition conditions.

¹⁾ Refer to [TransitionsConditions](#) on page 278.

Related topics**References**

[UserSignalSegment](#)..... 260
[UserSignalSequence](#)..... 264

UserSignalValue

Purpose

To get the value of the global user signal.

Where to go from here

Information in this section

Class Description (UserSignalValue)	266
To describe the class and its attributes.	
Activate	266
To activate a source type for the UserSignalValue object.	

Class Description (UserSignalValue)

Syntax

```
UserSignalValue = UserSignalElement.Activity
```

Purpose

To get the value of the global user signal.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Delay	Double	To get/set a time delay.
ActiveElement	SimpleConstant ¹⁾	To get the active source type object.
AvailableElement	Strings (collection)	To get the names of all available source types.

¹⁾ Refer to [SimpleConstant](#) on page 348.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a source type for the UserSignalValue object. Refer to Activate on page 266.

Related topics

References

UserSignalSegment	260
---	-----

Activate

Class

UserSignalValue

Syntax

```
UserSignalValueType = UserSignalValue.Activate(Variant Type)
```

Purpose

To activate a source type for the UserSignalValue object.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific UserSignalValue object. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
SimpleConstant ¹⁾	The activated type for the UserSignalValue object.

¹⁾ Refer to [SimpleConstant](#) on page 348.

Related topics**References**

[Class Description \(UserSignalValue\)](#)..... 266

Classes for Configuring Conditions

Where to go from here

Information in this section

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To get a distance condition.	
ConditionDistanceRefLine	269
To get/set a distance (along road reference line) condition.	
ConditionDrivenDistance	270
To get/set a driven distance condition.	
ConditionDuration	271
To get/set a duration condition.	
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To get/set an endless condition.	
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To get/set an immediate condition.	
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To get/set a position condition.	
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To get/set a time gap condition.	
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To get/set a trigger condition.	
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You can change the parameters of the scenario.	

ConditionDistance

Purpose To get a distance condition.

Class Description (ConditionDistance)

Syntax `ConditionDistance = TransitionsConditions.Add(ConditionType)`

Purpose To get information about a distance condition.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.
Distance	Double	To get/set the distance.
Operator	ApproximationOperators ²⁾	To get the approximation operators.
Object1	RelativeVehicleSelection ³⁾	To get the reference vehicle of object 1.
Object2	RelativeVehicleSelection ³⁾	To get the reference vehicle of object 2.

¹⁾ Refer to [ConditionTypes](#) on page 367.

²⁾ Refer to [ApproximationOperators](#) on page 309.

³⁾ Refer to [RelativeVehicleSelection](#) on page 337.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionDistanceRefLine

Purpose To get/set a distance (along road reference line) condition.

Class Description (ConditionDistanceRefLine)

Syntax

```
ConditionDistanceRefLine = TransitionsConditions.Add(ConditionType)
```

Purpose

To get/set a distance (along road reference line) condition.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.
Distance	Double	To get/set the distance.
Object1	RelativeVehicleSelection ²⁾	To get the first reference vehicle.
Object1ReferencePoint	ReferencePointTypes ³⁾	To get/set the reference point of object 1.
Object2	RelativeVehicleSelection ²⁾	To get the second reference vehicle.
Object2ReferencePoint	ReferencePointTypes ³⁾	To get/set the reference point of object 2.
Operator	ApproximationOperators ⁴⁾	To get the approximation operators.

¹⁾ Refer to [ConditionTypes](#) on page 367.

²⁾ Refer to [RelativeVehicleSelection](#) on page 337.

³⁾ Refer to [ReferencePointTypes](#) on page 369.

⁴⁾ Refer to [ApproximationOperators](#) on page 309.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionDrivenDistance

Purpose

To get/set a driven distance condition.

Class Description (ConditionDrivenDistance)

Syntax

```
ConditionDistanceRefLine = TransitionsConditions.Add(ConditionType)
```

Purpose To get information about a driven distance condition.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.
Distance	Double	To get/set the distance.
ReferenceVehicle	RelativeVehicleSelection ²⁾	To get the reference vehicle.

¹⁾ Refer to [ConditionTypes](#) on page 367.

²⁾ Refer to [RelativeVehicleSelection](#) on page 337.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionDuration

Purpose To get/set a duration condition.

Class Description (ConditionDuration)

Syntax `ConditionDuration = TransitionsConditions.Add(ConditionType)`

Purpose To get information about a duration condition.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.
Duration	Double	To get/set the duration.

¹⁾ Refer to [ConditionTypes](#) on page 367.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionEndless

Purpose

To get/set an endless condition.

Class Description (ConditionEndless)

Syntax

```
ConditionDuration = TransitionsConditions.Add(ConditionType)
```

Purpose

To get information about an endless condition.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.

¹⁾ Refer to [ConditionTypes](#) on page 367.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionImmediate

Purpose

To get/set an immediate condition.

Class Description (ConditionImmediate)

Syntax

```
ConditionImmediate = TransitionsConditions.Add(ConditionType)
```

Purpose

To get information about an immediate condition.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.

¹⁾ Refer to [ConditionTypes](#) on page 367.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionLateralAcceleration

Purpose

To get a lateral acceleration condition (only for a maneuver).

Class Description (ConditionLateralAcceleration)

Syntax

```
ConditionLateralAcceleration =  
TransitionsConditions.Add(ConditionType)
```

Purpose

To get information about a lateral acceleration condition (only for a maneuver).

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Acceleration	Double	To get/set the acceleration.
Name	String	To get the condition name.

Attributes	Type	Purpose
Operator	RelativeOperators ¹⁾	To get/set the relative operator.
Type	ConditionTypes ²⁾	To get the condition type.

¹⁾ Refer to [RelativeOperators](#) on page 335

²⁾ Refer to [ConditionTypes](#) on page 367.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionPosition

Purpose To get/set a position condition.

Class Description (ConditionPosition)

Syntax

```
ConditionPosition = TransitionsConditions.Add(ConditionType)
```

Purpose To get information about a position condition.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.
Position	Double	To get/set the position.
ReferenceVehicle	RelativeVehicleSelection ²⁾	To get the reference vehicle.

¹⁾ Refer to [ConditionTypes](#) on page 367.

²⁾ Refer to [RelativeVehicleSelection](#) on page 337.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionTimegap

Purpose To get/set a time gap condition.

Class Description (ConditionTimegap)

Syntax `ConditionTimegap = TransitionsConditions.Add(ConditionType)`

Purpose To get information about a time gap condition.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.
Time	Double	To get/set the time gap.
Operator	ApproximationOperators ²⁾	To get the approximation operators.
Object1	RelativeVehicleSelection ³⁾	To get the reference vehicle of object 1.
Object2	RelativeVehicleSelection ³⁾	To get the reference vehicle of object 2.

¹⁾ Refer to [ConditionTypes](#) on page 367.

²⁾ Refer to [ApproximationOperators](#) on page 309.

³⁾ Refer to [RelativeVehicleSelection](#) on page 337.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionTrigger

Purpose To get/set a trigger condition.

Class Description (ConditionTrigger)

Syntax

```
ConditionTrigger = TransitionsConditions.Add(ConditionType)
```

Purpose

To get information about a trigger condition.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.
Index	Integer	To get/set the index of the trigger.

¹⁾ Refer to [ConditionTypes](#) on page 367.

Related topics

References

[TransitionsConditions](#)..... 278

ConditionVelocity

Purpose

To get/set a velocity condition.

Class Description (ConditionVelocity)

Syntax

```
ConditionTimegap = TransitionsConditions.Item(Index)
```

Purpose

To get information about a velocity condition.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the condition name.
Type	ConditionTypes ¹⁾	To get the condition type.
Velocity	Double	To get/set the velocity.

Attributes	Type	Purpose
Operator	RelativeOperators ²⁾	To get the relative operators.
ReferenceVehicle	RelativeVehicleSelection ³⁾	To get the reference vehicle.

¹⁾ Refer to [ConditionTypes](#) on page 367.

²⁾ Refer to [RelativeOperators](#) on page 335.

³⁾ Refer to [RelativeVehicleSelection](#) on page 337.

Related topics

References

[TransitionsConditions](#)..... 278

Continue

Purpose

The class contains information about the **Continue** lateral type.

Class Description (Continue (LateralProfile))

Syntax

No direct creation

Purpose

To get information about the **Continue** lateral type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the lateral type.
Type	LateralProfileTypes ¹⁾	To get the lateral type constant.

¹⁾ Refer to [LateralProfileTypes](#) on page 368.

Continue

Purpose

To get information about the **Continue** longitudinal type.

Class Description (Continue (LongitudinalProfile))

Syntax No direct creation

Purpose To get information about the **Continue** longitudinal type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the longitudinal type.
Type	LongitudinalProfileTypes ¹⁾	To get the type constant of the longitudinal type.

¹⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

TransitionsConditions

Purpose To get transition conditions.

Where to go from here Information in this section

Class Description (TransitionsConditions).....	278
To describe the class and its attributes.	
Add.....	279
To add a transition condition to the TransitionConditions collection.	
FindItem.....	280
To search for a condition type in the TransitionConditions collection.	
Item.....	281
To access a transition condition.	
Remove.....	282
To remove a transition condition.	

Class Description (TransitionsConditions)

Syntax

```

TransitionConditions = FellowTransition.Conditions
TransitionConditions = UserSignalTransition.Conditions
TransitionConditions = ManeuverTransition.Conditions

```

Purpose To get transition conditions.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of conditions.

Methods The class contains the following methods:

Method	Purpose
Add	To add a transition condition to the TransitionConditions collection. Refer to Add on page 279.
Item	To access a transition condition. Refer to Item on page 281.
FindItem	To search for a condition type in the TransitionConditions collection. Refer to FindItem on page 280.
Remove	To remove a transition condition. Refer to Remove on page 282.

Related topics

References

Class Description (FellowTransition)	242
Class Description (ManeuverTransition)	223
Class Description (UserSignalTransition)	265

Add

Class TransitionsConditions

Syntax `TransitionCondition = TransitionsConditions.Add(ConditionType Type)`

Purpose To add a transition condition to the TransitionConditions collection.

Parameters The method uses the following parameters:

Parameter	Type	Description
Type	ConditionTypes ¹⁾	The type of the transition condition.

¹⁾ Refer to [ConditionTypes](#) on page 367.

Return value

The method returns an object of the following type:

Type	Description
ConditionLateralAcceleration ¹⁾ , ConditionDistance, ConditionDrivenDistance, ConditionDuration, ConditionEndless, ConditionImmediate, ConditionPosition, ConditionTimegap, ConditionTrigger, ConditionVelocity, or ConditionDistanceRefLine on page 269 ²⁾	The transition condition object.

¹⁾ Only for a ManeuverTransition

²⁾ Refer to [ConditionLateralAcceleration](#) on page 273, [ConditionDistance](#) on page 269, [ConditionDrivenDistance](#) on page 270, [ConditionDuration](#) on page 271, [ConditionEndless](#) on page 272, [ConditionImmediate](#) on page 272, [ConditionPosition](#) on page 274, [ConditionTimegap](#) on page 275, [ConditionTrigger](#) on page 275, [ConditionVelocity](#) on page 276, or [ConditionDistanceRefLine](#) on page 269.

Related topics**References**

[Class Description \(TransitionsConditions\)](#)..... 278

FindItem

Class

TransitionsConditions

Syntax

```
TransitionCondition = TransitionsConditions.FindItem(ConditionType Type,
int Index = 0)
```

Purpose

To search for a condition type in the TransitionConditions collection.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	ConditionTypes ¹⁾	The type of the transition condition.
Index	Integer	The index from which the search starts.

¹⁾ Refer to [ConditionTypes](#) on page 367.

Return value

The method returns an object of the following type:

Type	Description
ConditionDistance, ConditionDrivenDistance, ConditionDuration, ConditionEndless,	The transition condition object.

Type	Description
ConditionImmediate, ConditionPosition, ConditionTimegap, ConditionTrigger, ConditionVelocity ¹⁾	

¹⁾ Refer to [ConditionDistance](#) on page 269, [ConditionDrivenDistance](#) on page 270, [ConditionDuration](#) on page 271, [ConditionEndless](#) on page 272, [ConditionImmediate](#) on page 272, [ConditionPosition](#) on page 274, [ConditionTimegap](#) on page 275, [ConditionTrigger](#) on page 275, or [ConditionVelocity](#) on page 276.

Related topics

References

[Class Description \(TransitionsConditions\)](#)..... 278

Item

Class TransitionsConditions

Syntax `TransitionCondition = TransitionsConditions.Item(Variant Index)`

Purpose To access a transition condition.

Parameters The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The identifier (integer) of the transition condition.

Return value

The method returns an object of the following type:

Type	Description
ConditionLateralAcceleration ¹⁾ , ConditionDistance, ConditionDrivenDistance, ConditionDuration, ConditionEndless, ConditionImmediate, ConditionPosition, ConditionTimegap, ConditionTrigger, ConditionVelocity, or ConditionDistanceRefLine ²⁾	The transition condition object.

¹⁾ Only for a ManeuverTransition

²⁾ Refer to [ConditionLateralAcceleration](#) on page 273, [ConditionDistance](#) on page 269, [ConditionDrivenDistance](#) on page 270, [ConditionDuration](#) on page 271, [ConditionEndless](#) on page 272, [ConditionImmediate](#) on page 272, [ConditionPosition](#) on page 274, [ConditionTimegap](#) on page 275, [ConditionTrigger](#) on page 275, [ConditionVelocity](#) on page 276, or [ConditionDistanceRefLine](#) on page 269.

Related topics**References**

[Class Description \(TransitionsConditions\)](#)..... 278

Remove

Class

TransitionsConditions

Syntax

```
ReVal = TransitionsConditions.Remove(Variant Index)
```

Purpose

To remove a transition condition.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	Variant	The identifier of the transition condition to be removed. The identifier can be its index (integer type) or the TransitionCondition object.

Return value

The method returns an object of the following type:

Type	Description
Boolean	True if successful

Related topics**References**

[Class Description \(TransitionsConditions\)](#)..... 278

Classes for Model Configuration

Where to go from here	Information in this section
	ListElements.....283 To access a list of elements. ListElement.....287 To access an element of a list. ManeuverConfiguration.....288 To modify the maneuver configuration. ModelConfiguration.....288 To modify the contents of the model configuration. ReadOnlyListElements.....290 To access a read-only list of elements. ScenarioConfiguration.....292 To modify the scenario configuration.
	Information in other sections Automation Using Python Scripts (ModelDesk Automation ) Gives information on how you can automate ModelDesk by using Python scripts.

ListElements

Purpose	To access a list of elements.
Where to go from here	Information in this section Class Description (ListElements).....284 To describe the class and its attributes. Add.....284 To add an element. Item.....285 To get a specific element of a list. Move.....286 To move an element in the list.

[Remove](#).....286
To remove an element from the list.

Class Description (ListElements)

Syntax

```
FellowUserSignals = ScenarioConfiguration.FellowUserSignals
GlobalUserSignals = ScenarioConfiguration.GlobalUserSignals
Triggers = ScenarioConfiguration.Triggers
```

Purpose

To access a list of elements.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of elements in the list.

Methods

The class contains the following methods:

Method	Purpose
Add	To add an element. Refer to Add on page 284.
Item	To get a specific element. Refer to Item on page 285.
Move	To move an element in the list. Refer to Move on page 286.
Remove	To remove an element from the list. Refer to Remove on page 286.

Related topics

Basics

[ScenarioConfiguration](#).....292

Add

Class

ListElements

Syntax

```
RetVal = ListElements.Add()
```

Purpose	To add an element.				
Parameters	–				
Return value	The method returns the following parameter: <table border="1"> <thead> <tr> <th>Type</th><th>Description</th></tr> </thead> <tbody> <tr> <td>Boolean</td><td>True if successful</td></tr> </tbody> </table>	Type	Description	Boolean	True if successful
Type	Description				
Boolean	True if successful				
Related topics	References Class Description (ListElements)..... 284				

Item

Class

ListElements

Syntax

ListElement = ListElements.Item(object Index)

Purpose

To get a specific element of a list.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	object	The element of the specific index.

Return value

The method returns the following parameter:

Type	Description
ListElement ¹⁾	The specific list element.

1)

Refer to [ListElement](#) on page 287.

Related topics

References

Class Description (ListElements)

284

Move

Class ListElements

Syntax `ListElements.Move(object Value, integer NewIndex)`

Purpose To move an element in the list.

Parameters The method uses the following parameters:

Parameter	Type	Description
Value	Object	The element that is moved.
NewIndex	Integer	The position in the list to which the element is moved.

Return value —

Related topics References

[Class Description \(ListElements\)](#)..... 284

Remove

Class ListElements

Syntax `RetVal = ListElements.Remove(ListElement Item)`

Purpose To remove an element from the list.

Parameters The method uses the following parameters:

Parameter	Type	Description
Item	ListElement ¹⁾	Element to be removed.

¹⁾ Refer to [ListElement](#) on page 287.

Return value

The method returns the following parameter:

Type	Description
Boolean	True if successful.

Related topics**References**

[Class Description \(ListElements\)](#)..... 284

ListElement

Purpose

To access an element of a list.

Class Description (ListElement)

Syntax

```
ListElement = ListElements.Item(object Index)
ListElement = ReadOnlyListElements.Item(object Index)
```

Purpose

To get information on a list element.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Description	String	To get/set a description of the element.
Index	Integer	To get the index of the element.
Name	String	To get/set the name of the element.

Methods

—

Related topics**References**

[ListElements](#)..... 283
[ReadOnlyListElements](#)..... 290

ManeuverConfiguration

Purpose To modify the maneuver configuration.

Class Description (ManeuverConfiguration)

Syntax `ManeuverConfiguration = ModelConfiguration.ManeuverConfiguration`

Purpose To modify the maneuver configuration.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Triggers	ReadOnlyListElements ¹⁾	To get the collection of triggers.
UserSignals	ReadOnlyListElements ¹⁾	To get the collection of user signals.

¹⁾ Refer to [ReadOnlyListElements](#) on page 290.

Methods —

Related topics

References

[ModelConfiguration](#)..... 288

ModelConfiguration

Purpose To modify the contents of the model configuration.

Where to go from here

Information in this section

[Class Description \(ModelConfiguration\)](#)..... 289
To describe the class and its attributes.

[Reset](#)..... 289
To reset the model configuration.

Information in other sections

[Environment Configuration Dialog](#)..... 88
To configure settings for the simulation model.

Class Description (ModelConfiguration)

Syntax

```
ModelConfiguration = Project.ModelConfiguration
```

Purpose

To modify the contents of the model configuration.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ManeuverConfiguration	ManeuverConfiguration ¹⁾	To get the maneuver configuration.
ScenarioConfiguration	ScenarioConfiguration ²⁾	To get the scenario configuration.

¹⁾ Refer to [ManeuverConfiguration](#) on page 288.

²⁾ Refer to [ScenarioConfiguration](#) on page 292.

Methods

The class contains the following methods:

Method	Purpose
Reset	To reset the model configuration. Refer to Reset on page 289.

Related topics

References

[Environment Configuration Dialog](#)..... 88
[Project \(ModelDesk Project and Experiment Management\)](#) 

Reset

Class

ModelConfiguration

Syntax

```
RetVal = ModelConfiguration.Reset()
```

Purpose

To reset the model configuration.

Parameters

–

Return value

The method returns the following parameter:

Type	Description
Boolean	True if successful.

Related topics**References**

[Class Description \(ModelConfiguration\)](#)..... 289

ReadOnlyListElements

Purpose

To access a read-only list of elements.

Where to go from here**Information in this section**

Class Description (ReadOnlyListElements)	290
To describe the class and its attributes.	
Item	291
To get a specific element of the list.	
Move	292
To move an element in the list.	

Class Description (ReadOnlyListElements)

Syntax

```
ReadOnlyListElements = ManeuverConfiguration.Triggers
ReadOnlyListElements = ManeuverConfiguration.UserSignals
```

Purpose To access a read-only list of elements.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of elements in the list.

Methods The class contains the following methods:

Method	Purpose
Item	To get a specific element. Refer to Item on page 291.
Move	To move an element in the list. Refer to Move on page 292.

Related topics

References

[ManeuverConfiguration](#)..... 288

Item

Class ReadOnlyListElements

Syntax `ListElement = ReadOnlyListElements.Item(object Index)`

Purpose To get a specific element of the list.

Parameters The method uses the following parameters:

Parameter	Type	Description
Index	object	Index of the specific element.

Return value The method returns the following parameter:

Type	Description
ListElement ¹⁾	The specific list element.

¹⁾ Refer to [ListElement](#) on page 287.

Related topics**References**[Class Description \(ReadOnlyListElements\)](#)..... 290

Move

Class

ReadOnlyListElements

Syntax`ReadOnlyListElements.Move(object Value, integer NewIndex)`**Purpose**

To move an element in the list.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Value	object	The element to be moved.
NewIndex	Integer	The new position in the list to which the element is moved.

Return value

—

Related topics**References**[Class Description \(ReadOnlyListElements\)](#)..... 290

ScenarioConfiguration

Purpose

To modify the scenario configuration.

Class Description (ScenarioConfiguration)

Syntax`ScenarioConfiguration = ModelConfiguration.ScenarioConfiguration`

Purpose To modify the scenario configuration.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
FellowCount	Integer	To get the number of fellows.
FellowUserSignals	ListElements ¹⁾	To get the list of fellow user signals.
GlobalUserSignals	ListElements ¹⁾	To get the list of global user signals.
Triggers	ListElements ¹⁾	To get the list of triggers.

¹⁾ Refer to [ListElements](#) on page 283.

Methods —

Related topics	References
	ModelConfiguration 288

Classes for Configuring Traffic Drivers

Where to go from here

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DrivingStyle.....	294
To specify the driving style of a traffic driver.	
RuleCompliantDriving.....	295
To specify the rule compliant driving of a traffic driver.	
TrafficDrivers.....	296
To manage all the traffic driver of a ModelDesk project.	
TrafficDriverTypes.....	298
To activate a traffic driver in the maneuver or a fellow definition.	
TrafficDriver.....	300
To handle a traffic driver.	
TrafficSignRecognition.....	303
To specify the traffic sign recognition of a traffic driver.	

Information in other sections

Overview of the Object Model for Configuring Traffic Drivers.....	182
This gives you a quick overview of the classes for for configuring traffic drivers.	
Automating Scenarios in Python.....	170
You can change the parameters of the scenario.	

DrivingStyle

Purpose

To specify the driving style of a traffic driver.

Class Description (DrivingStyle)

Syntax

```
DrivingStyle = TrafficDriver.DrivingStyle
```

Purpose

To specify the driving style of a traffic driver.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
DistanceAtStandstill	Double	To get/set distance of the driver's vehicle to the vehicle ahead at standstill in meters
DistanceTime	Double	To get/set the distance of the driver's vehicle to the vehicle ahead in seconds while driving. This value is used to calculate the required distance in meters.
DetectionRange	Double	To get/set the range in front of the driver's vehicle in meters within other traffic participants are detected along the road reference line.
MaximumLongitudinalAcceleration	Double	To get/set the maximum acceleration in longitudinal direction in m/s ² in everyday traffic.
ComfortableDeceleration	Double	To get/set the comfortable deceleration in m/s ² in everyday traffic.
MaximumLateralAcceleration	Double	To get/set the maximum lateral acceleration in m/s ² in curves.

Methods

—

Related topics

References

Traffic Driver Dialog	96
TrafficDriver	300

RuleCompliantDriving

Purpose

To specify the rule compliant driving of a traffic driver.

Class Description (RuleCompliantDriving)

Syntax

```
RuleCompliantDriving = TrafficDriver.RuleCompliantDriving
```

Purpose

To specify the rule compliant driving of a traffic driver.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
DistanceAtStandstill	Double	To get/set the distance between the driver's vehicle and a traffic light or priority sign at standstill in meters.
SpeedLimitOffset	Double	To get/set the offset to the specified speed limit at which the driver's vehicle can drive slower or faster than the specified speed limit in km/h.
SignCompliantDrivingStart	Double	To get/set the offset from the position of the speed limit or no passing start sign to the position where the driver's vehicle complies with the traffic sign in meters.
SignCompliantDrivingEnd	Double	To get/set the offset from the position of the speed limit or no passing end sign to the position where the driver's vehicle complies with the traffic sign in meters.
YieldSpeedFactor	Double	To get/set the deceleration factor before junctions in percent. This value is used to calculate the velocity the vehicle slows down when approaching a junction where the right-of-way status must be checked.
SpeedLimitMode	SpeedLimitModes ¹⁾	To get/set the mode for road scenery speed limit use.

¹⁾ Refer to [SpeedLimitModes](#) on page 371.

Methods

—

Related topics**References**

Traffic Driver Dialog	96
TrafficDriver	300

TrafficDrivers

Purpose

To manage all the traffic drivers of a ModelDesk project.

Where to go from here

Information in this section

Class Description (TrafficDrivers)	297
To describe the class and its attributes.	
Item	297
To get a specific traffic driver.	

Class Description (TrafficDrivers)

Syntax

```
TrafficDrivers = ActiveProject.TrafficDrivers
```

Purpose

To manage all the traffic drivers of the ModelDesk project.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of traffic drivers.
AvailableElements	String[]	To get the names of all available traffic drivers.

Methods

The class contains the following methods:

Method	Purpose
Item	To get a specific traffic driver. Refer to Item on page 297.

Related topics

References

[Project \(ModelDesk Project and Experiment Management !\[\]\(56549452e01ca28bdf2500ced9653143_img.jpg\)](#))

Item

Class

TrafficDrivers

Syntax

```
TrafficDriver = TrafficDrivers.Item(object index)
```

Purpose To get a specific traffic driver.

Parameters The method uses the following parameters:

Parameter	Type	Description
index	object	The index of the traffic driver.

Return value The method returns the following parameter:

Type	Description
TrafficDriver ¹⁾	The specific traffic driver.

¹⁾ Refer to [TrafficDriver](#) on page 300.

Related topics

References

[Class Description \(TrafficDrivers\)](#)..... 297

TrafficDriverTypes

Purpose To activate a traffic driver in the maneuver or a fellow definition.

Where to go from here

Information in this section

[Class Description \(TrafficDriverTypes\)](#)..... 298
To describe the class and its attributes.

[Activate](#)..... 299
To select the traffic driver to be activated.

Class Description (TrafficDriverTypes)

Syntax

```
TrafficDriverType = Maneuver.TrafficDriverType
```

or

```
TrafficDriverType = Fellow.TrafficDriverType
```

Purpose To activate a traffic driver for the maneuver or a fellow definition.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
AvailableElements	String[]	To get a list of all the available traffic drivers.
ActiveElement	String	To get the name of the active traffic driver.

Methods The class contains the following methods:

Method	Purpose
Activate	To select the traffic driver to be activated. Refer to Activate on page 299.

Related topics

References

Fellow	229
Maneuver	209

Activate

Class TrafficDriverType

Syntax `RetVal = TrafficDriverType.Activate(object item)`

Purpose To select the traffic driver to be activated.

Parameters The method uses the following parameters:

Parameter	Type	Description
item	object	The traffic driver.

Return value The method returns the following parameter:

Type	Description
Boolean	True if successful.

Related topics**References**[Class Description \(TrafficDriverTypes\).....](#) 298

TrafficDriver

Purpose

To handle a traffic driver.

Where to go from here**Information in this section**

[Class Description \(TrafficDriver\).....](#) 300
To describe the class and its attributes.

[DiscardChanges.....](#) 301
To discard changes.

[Save.....](#) 302
To save the properties.

[SaveAs.....](#) 302
To save the traffic driver under another file name.

Class Description (TrafficDriver)

Syntax`TrafficDriver= TrafficDriver.Item()`**Purpose**

To handle a traffic driver.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Author	String	To get/set the name of the author.
Description	String	To get/set the description.
DrivingStyle	DrivingStyle ¹⁾	To get/set the driving style.
Name	String	To get the name.

Attributes	Type	Purpose
RuleCompliantDriving	RuleCompliantDriving ²⁾	To get/set the rule compliant driving.
TrafficSignRecognition	TrafficSignRecognition ³⁾	To get/set the traffic sign recognition.

- 1) Refer to [DrivingStyle](#) on page 294.
- 2) Refer to [RuleCompliantDriving](#) on page 295.
- 3) Refer to [TrafficSignRecognition](#) on page 303.

Methods

The class contains the following methods:

Method	Purpose
DiscardChanges	To discard changes. Refer to DiscardChanges on page 301.
Save	To save the properties of the traffic driver. Refer to Save on page 302.
SaveAs	To save the traffic driver under another name. Refer to SaveAs on page 302.

Related topics

References

TrafficDrivers	296
--------------------------------------	-----

DiscardChanges

Class

TrafficDriver

Syntax

```
TrafficDriver.DiscardChanges()
```

Purpose

To discard changes.

Parameters

—

Return value

—

Related topics

References

Class Description (TrafficDriver)	300
---	-----

Save

Class TrafficDriver

Syntax `TrafficDriver.Save()`

Purpose To save the properties.

Parameters –

Return value The method returns the following parameter:

Type	Description
Boolean	True if successful.

Related topics

References

[Class Description \(TrafficDriver\)](#)..... 300

SaveAs

Class TrafficDriver

Syntax `TrafficDriver.SaveAs(FileName, OverwriteExisting)`

Purpose To save the traffic driver under another file name.

Parameters The method uses the following parameters:

Parameter	Type	Description
FileName	String	The file name.
OverwriteExisting	Boolean	Saving the traffic driver overwrites an existing file called FileName if true.

Return value	The method returns the following parameter:	
	Type	Description
	Boolean	True if successful.
Related topics	References	
	Class Description (TrafficDriver)..... 300	

TrafficSignRecognition

Purpose	To specify the traffic sign recognition of a traffic driver.
---------	--

Class Description (TrafficSignRecognition)

Syntax	TrafficSignRecognition = TrafficDriver.TrafficSignRecognition	
Purpose	To specify the traffic sign recognition of a traffic driver.	
Attributes	The class contains the following attributes:	
	Attributes	Type Purpose
	DistanceFront	Double To get/set the distance for traffic sign recognition in front of the vehicle in meters.
	DistanceRight	Double To get/set the distance for traffic sign recognition to the right of the vehicle in meters.
	DistanceLeft	Double To get/set the distance for traffic sign recognition to the right of the vehicle in meters.
	DistanceVertical	Double To get/set the distance for traffic sign recognition above the vehicle in meters.
	RelativeSignAngle	Double To get/set the maximum angle between the traffic and the vector from the sign to the vehicle main point in degrees.
Methods	—	

Related topics

References

Traffic Driver Dialog	96
TrafficDriver	300

Miscellaneous Classes for Scenarios

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To get information about the AccelerationSource type.	
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The overview gives you a quick overview of the classes for the Scenario Editor.	
Automating Scenarios in Python	170
You can change the parameters of the scenario.	

Acceleration

Purpose To get information about the **Acceleration** longitudinal type.

Class Description (Acceleration)

Syntax No direct creation

Purpose To get information about the **Acceleration** longitudinal type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
SourceType	AccelerationSource ¹⁾	To get the AccelerationSource object for defining acceleration parameters.
Name	String	To get the name of the longitudinal type.
Type	LongitudinalProfileTypes ²⁾	To get the type constant of the longitudinal type.

¹⁾ Refer to [AccelerationSource](#) on page 308.

²⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

AccelerationSource

Purpose To get information about the **AccelerationSource** type.

Where to go from here Information in this section

Class Description (AccelerationSource).....	308
To describe the class and its attributes.	
Activate.....	308
To activate a Acceleration source type.	

Class Description (AccelerationSource)

Syntax No direct creation

Purpose To get information about the **Acceleration** source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Ramp, Constant, or Table ¹⁾	To get the active Acceleration source type object.
AvailableElements	Strings (collection)	To get the names of all available Acceleration source types.

¹⁾ Refer to [Ramp](#) on page 347, [Constant](#) on page 343, or [Table](#) on page 351.

Methods The class contains the following methods:

Method	Purpose
Activate	To activate an Acceleration source type. Refer to Activate on page 308.

Activate

Class AccelerationSource

Syntax

AccelerationSourceType = AccelerationSource.Activate(Variant Type)

Purpose

To activate a Acceleration source type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific source type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
Ramp, Constant, or Table ¹⁾	A source type object according to the given parameter.

¹⁾ Refer to [Ramp](#) on page 347, [Constant](#) on page 343, or [Table](#) on page 351.

Related topics

References

[Class Description \(AccelerationSource\)](#)..... 308

ApproximationOperators

Purpose

To get the approximation operator type.

Where to go from here

Information in this section

[Class Description \(ApproximationOperators\)](#)..... 309

To describe the class and its attributes.

[Activate](#)..... 310

To activate an approximation operator.

Class Description (ApproximationOperators)

Syntax

ApproximationOperators = [Parent].ApproximationOperator

Purpose To get the approximation operator type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	ApproximationOperator ¹⁾	To get the active approximation operator object.
AvailableElements	Strings (collection)	To get the names of all available approximation operators.

¹⁾ Refer to [ApproximationOperator](#) on page 311.

Methods The class contains the following methods:

Method	Purpose
Activate	To activate an approximation operator. Refer to Activate on page 310.

Activate

Class ApproximationOperators

Syntax `ApproximationOperator = ApproximationOperators.Activate(Variant Item)`

Purpose To activate an approximation operator.

Parameters The method uses the following parameters:

Parameter	Type	Description
Item	Variant	The identifier of the approximation operator. The value can be a string (name) or an integer (constant type).

Return value The method returns an object of the following type:

Type	Description
ApproximationOperator	The activated approximation operator object. Refer to ApproximationOperator on page 311

Related topics**References**

[Class Description \(ApproximationOperators\)](#)..... 309

ApproximationOperator

Purpose

To get information of an approximation operator.

Class Description (ApproximationOperator)

Syntax

`ApproximationOperator = [Parent].ApproximationOperator`

Purpose

To get information of an approximation operator.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of an approximation operator.
Type	ApproximationType ¹⁾	To get the approximation operator type.

¹⁾ Refer to [ApproximationType](#) on page 366.

BasicRoad

Purpose

To specify a BasicRoad profile.

Class Description (BasicRoad)

Syntax

No direct creation.

Purpose

To specify a BasicRoad profile.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Straight or Circle ¹⁾	To get the active element.
AvailableElements	String[]	To get the names of all the available elements.
Name	String	To get the name of the BasicRoad profile.
Type	LateralProfileTypes ²⁾	To get the type of the lateral profile.

¹⁾ Refer to [Straight](#) on page 350 or [Circle](#) on page 343.

²⁾ Refer to [LateralProfileTypes](#) on page 368.

Methods

—

DependencyTypes

Purpose

To select the dependency type.

Where to go from here**Information in this section**

Class Description (DependencyTypes)	312
To describe the class and its attributes.	
Activate	313
To activate a dependency type.	

Class Description (DependencyTypes)

Syntax

DependencyType = [Parent].DependencyTypes

Purpose

To select the dependency type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	DependencyTypeAbsolute or DependencyTypeRelative ¹⁾	To get the active dependency type object.
AvailableElements	Strings (collection)	To get the names of all available dependency types.

¹⁾ Refer to [DependencyTypeAbsolute](#) on page 314 or [DependencyTypeRelative](#) on page 314.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a dependency type. Refer to Activate on page 338.

Activate

Class

DependencyTypes

Syntax

```
DependencyType = DependencyTypes.Activate(Variant Item)
```

Purpose

To activate a dependency type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Item	Variant	The identifier of the dependency type. You can use strings ('absolute', 'relative') or type constant (see DependencyTypeConstants on page 367).

Return value

The method returns an object of the following type:

Type	Description
DependencyTypeAbsolute or DependencyTypeRelative ¹⁾	The activated dependency type object.

¹⁾ Refer to [DependencyTypeAbsolute](#) on page 314 or [DependencyTypeRelative](#) on page 314.

Related topics**References**[Class Description \(DependencyTypes\)](#)..... 312

DependencyTypeAbsolute

Purpose To get information on an absolute dependency type.

Class Description (DependencyTypeAbsolute)

Syntax No direct creation

Purpose To get information on an absolute dependency type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the dependency type.
Type	DependencyTypeConstants ¹⁾	To get the type of the dependency type.

¹⁾ Refer to [DependencyTypeConstants](#) on page 367.

DependencyTypeRelative

Purpose To get information on a relative dependency type.

Class Description (DependencyTypeRelative)

Syntax No direct creation

Purpose To get information on a relative dependency type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
RelativeVehicle	RelativeVehicleSelection ¹⁾	To get the relative vehicle selection object.
Name	String	To get the name of the dependency type.
Type	DependencyTypeConstants ²⁾	To get the type of the dependency type.

¹⁾ Refer to [RelativeVehicleSelection](#) on page 337.

²⁾ Refer to [DependencyTypeConstants](#) on page 367.

DistanceMeter

Purpose

To get information about the **DistanceMeter** longitudinal type.

Class Description (DistanceMeter)

Syntax

No direct creation

Purpose

To get information about the **DistanceMeter** longitudinal type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the longitudinal type.
PointOnReferenceVehicle	ReferencePointTypes ¹⁾	To get/set the reference point on the referenced vehicle. Valid values are front, rear, and main.
ReferencePoint	ReferencePointTypes ¹⁾	To get/set the reference point on the ASM vehicle. Valid values are front, rear, and main.
RelativeVehicle	RelativeVehicleSelection ²⁾	To get the RelativeVehicleSelection object.
SourceType	DistanceMeterSource ³⁾	To get the DistanceMeterSource object for defining distance parameters.
Type	LongitudinalProfileTypes ⁴⁾	To get the type constant of the longitudinal type.

¹⁾ Refer to [ReferencePointTypes](#) on page 369.

²⁾ Refer to [RelativeVehicleSelection](#) on page 337.

³⁾ Refer to [DistanceMeterSource](#) on page 316.

⁴⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

DistanceMeterSource

Purpose To get information about the `DistanceMeterSource` type.

Where to go from here Information in this section

Class Description (DistanceMeterSource)	316
To describe the class and its attributes.	
Activate	317
To activate a DistanceMeter source type.	

Class Description (DistanceMeterSource)

Syntax No direct creation

Purpose To get information about the `DistanceMeter` source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Constant, Ramp, Extern, or FinalDistance ¹⁾	To get the active DistanceMeter source type object.
AvailableElements	Strings (collection)	To get the names of all available DistanceMeter source types.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Extern](#) on page 343, or [FinalDistance](#) on page 344.

Methods The class contains the following methods:

Method	Purpose
Activate	To activate a DistanceMeter source type. Refer to Activate on page 317.

Activate

Class DistanceTimeSource

Syntax `DistanceMeterSourceType = DistanceMeterSource.Activate(Variant Type)`

Purpose To activate a DistanceMeter source type.

Parameters The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific source type. The value can be specified as string (name) or integer (constant type).

Return value The method returns an object of the following type:

Type	Description
Constant, Ramp, Extern, or FinalDistance ¹⁾	A source type object according to the given parameter.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Extern](#) on page 343, or [FinalDistance](#) on page 344.

Related topics

References

[Class Description \(DistanceTimeSource\)](#)..... 318

DistanceTime

Purpose To get information about the DistanceTime longitudinal type.

Class Description (DistanceTime)

Syntax No direct creation

Purpose To get information about the **DistanceTime** longitudinal type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
SourceType	DistanceTimeSource ¹⁾	To get the DistanceTimeSource object for defining distance parameters.
RelativeVehicle	RelativeVehicleSelection ²⁾	To get the RelativeVehicleSelection object.
Name	String	To get the name of the longitudinal type.
Type	LongitudinalProfileTypes ³⁾	To get the type constant of the longitudinal type.

¹⁾ Refer to [DistanceTimeSource](#) on page 318.

²⁾ Refer to [RelativeVehicleSelection](#) on page 337.

³⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

DistanceTimeSource

Purpose To get information about the **DistanceTimeSource** type.

Where to go from here Information in this section

Class Description (DistanceTimeSource)	318
To describe the class and its attributes.	
Activate	319
To activate a DistanceTime source type.	

Class Description (DistanceTimeSource)

Syntax No direct creation

Purpose To get information about the **DistanceTime** source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Constant, Ramp, Extern, FinalDistance, or Table ¹⁾	To get the active DistanceTime source type object.
AvailableElements	Strings (collection)	To get the names of all available DistanceTime source types.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Extern](#) on page 343, [FinalDistance](#) on page 344, or [Table](#) on page 351.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a DistanceTime source type. Refer to Activate on page 319.

Activate

Class

DistanceTimeSource

Syntax

```
DistanceTimeSourceType = DistanceTimeSource.Activate(Variant Type)
```

Purpose

To activate a DistanceTime source type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific source type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
Constant, Ramp, Extern, FinalDistance, or Table ¹⁾	A source type object according to the given parameter.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Extern](#) on page 343, [FinalDistance](#) on page 344, or [Table](#) on page 351.

Related topics

References

[Class Description \(DistanceTimeSource\)](#)..... 318

FollowRoad

Purpose

To specify a FollowRoad profile.

Class Description (FollowRoad)

Syntax

No direct creation.

Purpose

To specify a FollowRoad profile.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
DriverTolerance	Double	To get/set the value of the driver tolerance.
DriverToleranceEnabled	Boolean	To get/set the use of the driver tolerance.
DriverType	DriverTypes ¹⁾	To get/set the driver type.
DriverTypeValues	DriverType ²⁾	To get the values of the driver type.
ForceToRoad	Boolean	To get/set the ForceToRoad flag.
LaneIndex	Constant, FinalLane, Table, Manual, or Extern ³⁾	To get the lane index.
LateralOffset	Constant, Table, Manual, or Extern ⁴⁾	To get the lateral offset.
Name	String	To get the name of the profile.
ReferenceType	ReferencedVehicleTypes ⁵⁾	To get/set the reference type (relative to lane = 0 or relative to object = 1)
Type	LateralProfileTypes ⁶⁾	To get the type of the lateral profile.

¹⁾ Refer to [DriverTypes](#) on page 367.

²⁾ Refer to [DriverType](#) on page 203.

- 3) Refer to [Constant](#) on page 343, [FinalLane](#) on page 345, [Table](#) on page 351, [Manual](#) on page 346, or [Extern](#) on page 343.
- 4) Refer to [Constant](#) on page 343, [Table](#) on page 351, [Manual](#) on page 346, or [Extern](#) on page 343.
- 5) Refer to [ReferencedVehicleTypes](#) on page 369.
- 6) Refer to [LateralProfileTypes](#) on page 368.

Methods

—

LateralAcceleration

Purpose

To specify a LateralAcceleration profile.

Class Description (LateralAcceleration)

Syntax

No direct creation

Purpose

To specify a LateralAcceleration profile.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the LateralAcceleration profile.
SourceType	Constant, Table, or Ramp ¹⁾	To get the source type for the acceleration pedal.
Type	LongitudinalProfileTypes ²⁾	To get the longitudinal profile type.

¹⁾ Refer to [Constant](#) on page 343, [Table](#) on page 351, or [Ramp](#) on page 347.

²⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

Methods

—

LateralDeviation

Purpose To get information about the lateral deviation type of a fellow.

Class Description (LateralDeviation)

Syntax No direct creation

Purpose To get information about the lateral deviation type of a fellow.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the lateral type.
Type	LateralProfileTypes ¹⁾	To get the lateral type constant.
DependencyType	DependencyType ²⁾	To get the DependencyType object.
SourceType	LateralDeviationSource ³⁾	To get the LateralDeviationSource object.
PointOnReferenceVehicle	ReferencePointTypes ⁴⁾	To get/set the point on the reference vehicle that is used to define the lateral distance between both objects if the dependency type is relative. Valid values are left, right, front, rear, and main.
ReferencePoint	ReferencePointTypes ⁴⁾	To get/set the point on the current fellow that is used to define the lateral distance between both objects if the dependency type is relative. Valid values are left, right, front, rear, and main.

¹⁾ Refer to [LateralProfileTypes](#) on page 368.

²⁾ Refer to [DependencyTypes](#) on page 312.

³⁾ Refer to [LateralDeviationSource](#) on page 322.

⁴⁾ Refer to [ReferencePointTypes](#) on page 369.

LateralDeviationSource

Purpose The class contains the lateral deviation types of the fellow definition.

Where to go from here

Information in this section

Class Description (LateralDeviationSource)	323
To describe the class and its attributes.	
Activate	323
To get a specific lateral deviation type.	

Class Description (LateralDeviationSource)

Syntax

No direct creation

Purpose

To get information about the lateral deviation type of a fellow.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Constant, Ramp, Extern, FinalDeviation, SmoothLaneChange, or Table ¹⁾	To get the deviation from the center line.
AvailableElements	Strings (collection)	To get the names of all available lateral deviation types.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Extern](#) on page 343, [FinalDeviation](#) on page 344, [SmoothLaneChange](#) on page 349, or [Table](#) on page 351.

Methods

The class contains the following methods:

Method	Purpose
Activate	To get a specific lateral deviation type. Refer to Activate on page 323.

Activate

Class

LateralDeviationSource

Syntax

```
LateralDeviationType = LateralDeviationSource.Activate(Variant Type)
```

Purpose To get a specific lateral deviation source.

Parameters The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific lateral deviation source. The value can be specified as string (name) or integer (constant type, see SourceTypeConstants on page 370).

Return value The method returns an object of the following type:

Type	Description
Constant, Ramp, Extern, FinalDeviation, SmoothLaneChange, or Table ¹⁾	A lateral deviation source object according to the given parameter.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Extern](#) on page 343, [FinalDeviation](#) on page 344, or [SmoothLaneChange](#) on page 349, or [Table](#) on page 351.

Related topics

References

[Class Description \(LateralDeviationSource\)](#)..... 323

LaneSelection

Purpose To get information about the lane selection of a fellow.

Class Description (LaneSelection)

Syntax No direct creation

Purpose To get information about the lane selection type of a fellow.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the lateral type.
Type	LateralProfileTypes ¹⁾	To get the lateral type constant.

Attributes	Type	Purpose
DependencyType	DependencyType ²⁾	To get the DependencyType object.
SourceType	LaneSelectionSource ³⁾	To get the LaneSelectionSource object.
ForceToRoad	Bool	To get/set the ForceToRoad state.

¹⁾ Refer to [LateralProfileTypes](#) on page 368.

²⁾ Refer to [DependencyTypes](#) on page 312.

³⁾ Refer to [LaneSelectionSource](#) on page 325.

LaneSelectionSource

Purpose The class contains the lane selection types of the fellow definition.

Where to go from here Information in this section

[Class Description \(LaneSelectionSource\)](#)..... 325

To describe the class and its attributes.

[Activate](#)..... 326

To get a specific lane selection type.

Class Description (LaneSelectionSource)

Syntax No direct creation

Purpose To get information about the lane selection type of a fellow.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Constant, LaneChange, SmoothLaneChange, Extern, FinalLane, or Table ¹⁾	To get the lane selection type.
AvailableElements	Strings (collection)	To get the names of all available lane selection types.

¹⁾ Refer to [Constant](#) on page 343, [LaneChange](#) on page 346, [SmoothLaneChange](#) on page 349, [Extern](#) on page 343, [FinalLane](#) on page 345, or [Table](#) on page 351.

Methods

The class contains the following methods:

Method	Purpose
Activate	To get a specific lane selection type. Refer to Activate on page 326.

Activate

Class

LaneSelectionSource

Syntax

```
LaneSelectionType = LaneSelectionSource.Activate(Variant Type)
```

Purpose

To get a specific lane selection source.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific lane selection source. The value can be specified as string (name) or integer (constant type, see SourceTypeConstants on page 370).

Return value

The method returns an object of the following type:

Type	Description
Constant, LaneChange, SmoothLaneChange, Extern, FinalLane, or Table ¹⁾	A lane selection deviation source object according to the given parameter.

¹⁾ Refer to [Constant](#) on page 343, [LaneChange](#) on page 346, [SmoothLaneChange](#) on page 349, [Extern](#) on page 343, [FinalLane](#) on page 345, or [Table](#) on page 351.

Related topics**References**

[Class Description \(LaneSelectionSource\)](#)..... 325

NamedItem

Purpose

To get information of a NamedItem object.

Class Description (NamedItem)

Syntax

NamedItem = NamedItemList.Item()

Purpose

To access a NamedItem object.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name.
Description	String	To get the description.

Related topics

References

Class Description (NamedItemList)..... 328

NamedItemList

Purpose

To access a NamedItemList object.

Where to go from here

Information in this section

Class Description (NamedItemList)..... 328

To describe the class and its attributes.

Add..... 328

To add a NamedItem object.

Item..... 329

To access a specific NamedItem object.

Remove..... 329

To remove a NamedItem object.

Class Description (NamedItemList)

Syntax No direct creation

Purpose To access a NamedItemList object.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	The number of NamedItem objects.

Methods The class contains the following methods:

Method	Purpose
Add	To add a NamedItem object. Refer to Add on page 328.
Item	To access a specific NamedItem object. Refer to Item on page 329.
Remove	To remove a NamedItem object. Refer to Remove on page 329.

Add

Class NamedItemList

Syntax `NamedItem = NamedItemList.Add(String Name)`

Purpose To add a NamedItem object.

Parameters The method uses the following parameters:

Parameter	Type	Description
Name	String	Name of the added NamedItem object.

Return value The method returns an object of the following type:

Type	Description
NamedItem ¹⁾	The added NamedItem object.

¹⁾ Refer to [NamedItem](#) on page 326.

Related topics**References**

[Class Description \(NamedItemList\)](#)..... 328

Item

Class

NamedItemList

Syntax

```
NamedItem = NamedItemList.Item(object Index)
```

Purpose

To access a specific NamedItem object.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Index	object	The index of the specific NamedItem object.

Return value

The method returns an object of the following type:

Type	Description
NamedItem ¹⁾	The specific NamedItem object.

¹⁾ Refer to [NamedItem](#) on page 326.

Related topics**References**

[Class Description \(NamedItemList\)](#)..... 328

Remove

Class

NamedItemList

Syntax

```
RetVal = NamedItemList.Item(object Index)
```

Purpose To remove a NamedItem object.

Parameters The method uses the following parameters:

Parameter	Type	Description
Index	object	The index of the NamedItem object to be removed.

Return value The method returns an object of the following type:

Type	Description
Boolean	True if successful.

Related topics

References

[Class Description \(NamedItemList\).....](#) 328

NotUsed

Purpose The class contains information about the **NotUsed** lateral type.

Class Description (NotUsed (LateralProfile))

Syntax No direct creation

Purpose To get information about the **NotUsed** lateral type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the lateral type.
Type	LateralProfileTypes ¹⁾	To get the lateral type constant.

¹⁾ Refer to [LateralProfileTypes](#) on page 368.

NotUsed

Purpose The class contains information about the **NotUsed** longitudinal type.

Class Description (NotUsed (LongitudinalProfile))

Syntax No direct creation

Purpose To get information about the **NotUsed** longitudinal type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the longitudinal type.
Type	LongitudinalProfileTypes ¹⁾	To get the longitudinal type constant.

¹⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

PedalStimulus

Purpose To specify a PedalStimulus profile.

Class Description (PedalStimulus)

Syntax No direct creation

Purpose To specify a PedalStimulus profile.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
AccelerationPedal	Manual, Extern, Constant, Table, or Ramp ¹⁾	To get the source type for the acceleration pedal.
BrakePedal	Manual, Extern, Constant, Table, or Ramp ¹⁾	To get the source type for the brake pedal.
Name	String	To get the name of the PedalStimulus profile.
Type	LongitudinalProfileTypes ²⁾	To get the longitudinal profile type.

¹⁾ Refer to [Manual](#) on page 346, [Extern](#) on page 343, [Constant](#) on page 343, [Table](#) on page 351, or [Ramp](#) on page 347.

²⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

Methods

—

Position

Purpose

To get information about the **Position** longitudinal type.

Class Description (Position)

Syntax

No direct creation

Purpose

To get information about the **Position** longitudinal type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
SourceType	PositionSource ¹⁾	To get the PositionSource object for defining distance parameters.
Name	String	To get the name of the longitudinal type.
Type	LongitudinalProfileTypes ²⁾	To get the type constant of the longitudinal type.

¹⁾ Refer to [PositionSource](#) on page 333.

²⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

PositionSource

Purpose To get information about the **PositionSource** type.

Where to go from here **Information in this section**

Class Description (PositionSource).....	333
To describe the class and its attributes.	
Activate.....	333
To activate a position source type.	

Class Description (PositionSource)

Syntax No direct creation

Purpose To get information about the **Position** source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	YawAngleConstant, Ramp, Extern, or Table ¹⁾	To get the active position source type object.
AvailableElements	Strings (collection)	To get the names of all available position source types.

¹⁾ Refer to [YawAngleConstant](#) on page 351, [Ramp](#) on page 347, [Extern](#) on page 343, or [Table](#) on page 351.

Methods The class contains the following methods:

Method	Purpose
Activate	To activate a position source type. Refer to Activate on page 333.

Activate

Class PositionSource

Syntax

```
PositionSourceType = PositionSource.Activate(Variant Type)
```

Purpose

To activate a position source type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific source type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
YawAngleConstant, Ramp, Extern, or Table ¹⁾	A source type object according to the given parameter.

¹⁾ Refer to [YawAngleConstant](#) on page 351, [Ramp](#) on page 347, [Extern](#) on page 343, or [Table](#) on page 351.

Related topics**References**

[Class Description \(PositionSource\)](#)..... 333

RelativeOperators

Purpose

To get information on a relative operator.

Class Description (RelativeOperator)

Syntax

```
RelativeOperator = [Parent].RelativeOperator
```

Purpose

To get information on a relative operator.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the relative operator ('<=' or '>=').
Type	RelativeOperatorType ¹⁾	To get the type constant of relative operator.

¹⁾ Refer to [RelativeOperatorType](#) on page 370.

RelativeOperators

Purpose

To select the relative operator type.

Where to go from here**Information in this section**

Class Description (RelativeOperators).....	335
To describe the class and its attributes.	
Activate.....	336
To activate a relative operator.	

Class Description (RelativeOperators)

Syntax

```
RelativeOperators = [Parent].RelativeOperators
```

Purpose

To select the relative operator type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	RelativeOperators ¹⁾	To get the active relative operator object.
AvailableElements	Strings (collection)	To get the names of all available relative operators.

¹⁾ Refer to [RelativeOperators](#) on page 334.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a relative operator. Refer to Activate on page 336.

Activate

Class

RelativeOperators

Syntax

```
RelativeOperator = RelativeOperators.Activate(Variant Item)
```

Purpose

To activate a relative operator.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Item	Variant	The identifier of the relative operator. Valid values are integer (constant type) and string(name).

Return value

The method returns an object of the following type:

Type	Description
RelativeOperators ¹⁾	The activated relative operator object.

¹⁾ Refer to [RelativeOperators](#) on page 334.

Related topics**References**

[Class Description \(RelativeOperators\)](#)..... 335

RelativeObject

Purpose

To get information of a relative object.

Class Description (RelativeObject)

Syntax

```
RelativeObject = [Parent].RelativeObject
```

Purpose

To get information on a relative object.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of a relative object.
Index	Integer	To get the index of a relative object.
Type	RelativeObjectType ¹⁾	To get the relative object type.
Valid	Boolean	True if valid

¹⁾ Refer to [RelativeObjectType](#) on page 370.

RelativeVehicleSelection

Purpose

To select the RelativeObject.

Where to go from here

Information in this section

Class Description (RelativeVehicleSelection)	337
To describe the class and its attributes.	
Activate	338
To activate a relative vehicle.	

Class Description (RelativeVehicleSelection)

Syntax

```
RelativeVehicleSelection = [Parent].RelativeVehicleSelection
```

Purpose

To select the RelativeObject.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	RelativeObject ¹⁾	To get the active relative object (vehicle or position marker).
AvailableElements	Strings (collection)	To get the names of all available relative objects.

¹⁾ Refer to [RelativeObject](#) on page 336.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a relative object. Refer to Activate on page 338.

Activate

Class

RelativeVehicleSelection

Syntax

```
RelativeVehicle = RelativeVehicleSelection.Activate(Variant Item)
```

Purpose

To activate a relative object.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Item	Variant	The identifier of the relative vehicle. Only strings are valid. The initial substring of the complete fellow names are allowed, for example, 'M1', 'F1', or 'F3'.

Return value

The method returns an object of the following type:

Type	Description
RelativeObject ¹⁾	The activated relative vehicle object.

¹⁾ Refer to [RelativeObject](#) on page 336.

Related topics**References**

[Class Description \(RelativeVehicleSelection\)](#)..... 337

Route

Purpose To get information of a route.

Class Description (Route)

Syntax Route = [Parent].Route

Purpose To get information on a route.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the route.
Description	String	To get the description of the route.
Valid	Boolean	To get the valid state of the route.
ReversePossible	Boolean	To get the information whether the route can be used in reverse direction.
Index	Long	To get the index of the route.

RouteSelection

Purpose To select a route.

Where to go from here Information in this section

Class Description (RouteSelection)	340
To describe the class and its attributes.	
Activate	340
To activate a route.	

Class Description (RouteSelection)

Syntax

```
RouteSelection = ManeuverSequence.RouteSelection
RouteSelection = FellowSequence.RouteSelection
```

Purpose

To select the route.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Route ¹⁾	To get the active route object.
AvailableElements	Strings (collection)	To get the names of all available routes.
Direction	DrivingDirectionConstants ²⁾	To get/set the direction of travel.
UseExternal	Boolean	To get/set the flag to indicate that external route definitions are used.

¹⁾ Refer to [Route](#) on page 339.

²⁾ Refer to [DrivingDirectionConstants](#) on page 368.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a route. Refer to Activate on page 340.

Related topics

References

FellowSequence	238
ManeuverSequence	218

Activate

Class

RouteSelection

Syntax

```
Route = RouteSelection.Activate(Variant Item)
```

Purpose

To activate a route.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Item	Variant	The identifier of the route. Only strings are valid. The initial substring of the complete route names are allowed, for example, 'R1', or 'R2'.

Return value

The method returns an object of the following type:

Type	Description
Route ¹⁾	The activated route object.

¹⁾ Refer to [Route](#) on page 339.

Related topics

References

Class Description (RouteSelection)	340
--	-----

SourceTypes

Purpose

To specify the source type.

Where to go from here

Information in this section

Circle	343
To define a Circle source type.	
Constant	343
To define a Constant source type.	
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To define an Extern source type.	
FinalDeviation	344
To define a FinalDeviation source type.	
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To define a FinalDistance source type.	
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To define a FinalLane source type.	
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To define a FinalVelocity source type.	

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To define a Fixed source type.	
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To define a LaneChange source type.	
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To define a Manual source type.	
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To define a SimpleConstant source type.	
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To define a Sine source type.	
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To define a SineWithDwell source type.	
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To define a SmoothLaneChange source type.	
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To define a Step source type.	
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Straight.....	350
To define a Straight source type.	
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To define a Table source type.	
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To define an UntilStop source type.	
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To define a YawAngleConstant source type.	

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The source properties are used in several properties to specify the kind of source.	

Circle

Syntax `Circle = [Parent].Circle`

Purpose To define a Circle source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Radius	Double	To get/set the radius in meters.
Direction	DirectionTypes	To get/set the direction.
ControlMode	ControlModeTypes ¹⁾	To get/set the control mode.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

¹⁾ Refer to [ControlModeTypes](#) on page 367.

Constant

Syntax `Constant = [Parent].Constant`

Purpose To define a Constant source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Constant	Double	To get/set the constant value.
HoldCurrentValue	Boolean	To get/set the hold current value flag.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

Extern

Syntax `Extern = [Parent].Extern`

Purpose To define an Extern source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

FinalDeviation

Syntax

`FinalDeviation = [Parent].FinalDeviation`

Purpose

To define a FinalDeviation source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Deviation	Double	To get/set the deviation value.
LateralVelocity	Double	To get/set the lateral velocity value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

FinalDistance

Syntax

`FinalDistance = [Parent].FinalDistance`

Purpose

To define a FinalDistance source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Distance	Double	To get/set the distance value.
Duration	Double	To get/set the duration value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

FinalLane

Syntax `FinalLane = [Parent].FinalLane`

Purpose To define a FinalLane source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
FinalLane	Double	To get/set the final lane value.
LateralVelocity	Double	To get/set the lateral velocity value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

FinalVelocity

Syntax `FinalVelocity = [Parent].FinalVelocity`

Purpose To define a FinalVelocity source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Velocity	Double	To get/set the velocity value.
NegativeAcceleration	Double	To get/set the negative acceleration value.
PositiveAcceleration	Double	To get/set the positive acceleration value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

Fixed

Syntax `Fixed = [Parent].Fixed`

Purpose To define a Fixed source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

LaneChange

Syntax

`LaneChange = [Parent].LaneChange`

Purpose

To define a LaneChange source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
StartLane	Double	To get/set the start lane value.
EndLane	Double	To get/set the end lane value.
Duration	Double	To get/set the duration value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

Manual

Syntax

`Manual = [Parent].Manual`

Purpose

To define a Manual source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

Pulse

Syntax `Pulse = [Parent].Pulse`

Purpose To define a Pulse source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
SteeringWheelAngle	Double	To get/set the angle of the steering wheel.
Direction	DirectionTypes ¹⁾	To get/set the direction.
InterpretationType	InterpretationTypes ²⁾	To get/set the interpretation type.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.
X1	Double	To get/set the X 1 value.
X2	Double	To get/set the X 2 value.
X3	Double	To get/set the X 3 value.
X4	Double	To get/set the X 4 value.

¹⁾ Refer to [DirectionTypes](#) on page 367.

²⁾ Refer to [InterpretationTypes](#) on page 368.

Ramp

Syntax `Ramp = [Parent].Ramp`

Purpose To define a Ramp source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Start	Double	To get/set the start value.
End	Double	To get/set the end value.
Duration	Double	To get/set the duration value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

SimpleConstant

Syntax `SimpleConstant = [Parent].SimpleConstant`

Purpose To define a SimpleConstant source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Constant	Double	To get/set the constant value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

Sine

Syntax `Sine = [Parent].Sine`

Purpose To define a Sine source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Amplitude	Double	To get/set the amplitude value.
StartFrequency	Double	To get/set the start frequency value.
EndFrequency	Double	To get/set the end frequency value.
Duration	Double	To get/set the duration value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

SineWithDwell

Syntax `SineWithDwell = [Parent].SineWithDwell`

Purpose To define a SineWithDwell source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
SteeringWheelAngle	Double	To get/set the angle of the steering wheel.
Direction	DirectionTypes ¹⁾	To get/set the direction.
InterpretationType	InterpretationTypes ²⁾	To get/set the interpretation type.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.
X1	Double	To get/set the X 1 value.
X2	Double	To get/set the X 2 value.

¹⁾ Refer to [DirectionTypes](#) on page 367.

²⁾ Refer to [InterpretationTypes](#) on page 368.

SmoothLaneChange

Syntax

`SmoothLaneChange = [Parent].SmoothLaneChange`

Purpose

To define a SmoothLaneChange source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
StartDeviation	Double	To get/set the start deviation value.
EndDeviation	Double	To get/set the end deviation value.
Duration	Double	To get/set the duration value.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

Step

Syntax

`Step = [Parent].Step`

Purpose

To define a Step source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
SteeringWheelAngle	Double	To get/set the angle of the steering wheel.
Direction	DirectionTypes ¹⁾	To get/set the direction.
InterpretationType	InterpretationTypes ²⁾	To get/set the interpretation type.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.
X1	Double	To get/set the X 1 value.
X2	Double	To get/set the X 2 value.

¹⁾ Refer to [DirectionTypes](#) on page 367.

²⁾ Refer to [InterpretationTypes](#) on page 368.

StopWithin

Syntax

StopWithin = [Parent].StopWithin

Purpose

To define a StopWithin source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
HoldCurrentValue	Boolean	To get/set the hold current value flag.
ApproachVelocity	Double	To get/set the approach velocity value.
StopDistance	Double	To get/set the stop distance value.
Deceleration	Double	To get/set the deceleration value.
Name	String	To get the source type name.
SourceType	Integer	To get the source type constant.

Straight

Syntax

Straight = [Parent].Straight

Purpose

To define a Straight source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

Table

Syntax

`Table = [Parent].Table`

Purpose

To define a Table source type.

Attributes

Refer to [Class Description \(Table\)](#) on page 358.

Methods

Refer to [Class Description \(Table\)](#) on page 358.

UntilStop

Syntax

`UntilStop = [Parent].UntilStop`

Purpose

To define an UntilStop source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

YawAngleConstant

Syntax

`YawAngleConstant = [Parent].YawAngleConstant`

Purpose To define a YawAngleConstant source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Constant	Double	To get/set the constant value.
YawAngle	Double	To get/set the yaw angle value
HoldCurrentValue	Boolean	To get/set the hold current value flag.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

Standstill

Purpose To specify a Standstill profile.

Class Description (Standstill)

Syntax No direct creation

Purpose To specify a Standstill profile.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Name	String	To get the name of the Standstill profile.
Type	LongitudinalProfileTypes ¹⁾	To get the longitudinal profile type.

¹⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

Methods —

StimulusAngle

Purpose To specify a StimulusAngle profile.

Class Description (StimulusAngle)

Syntax No direct creation.

Purpose To specify a StimulusAngle profile.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Constant, Ramp, Table, Manual, Extern, Fixed, Step, Pulse, Sine, or SineWithDwell ¹⁾	To get the active element.
AvailableElements	String[]	To get a list of all the available elements.
Name	String	To get the name of the StimulusAngle profile.
Type	LateralProfileTypes ²⁾	To get the type of the lateral profile.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Table](#) on page 351, [Manual](#) on page 346, [Extern](#) on page 343, [Fixed](#) on page 345, [Step](#) on page 349, [Pulse](#) on page 347, [Sine](#) on page 348, or [SineWithDwell](#) on page 348.

²⁾ Refer to [LateralProfileTypes](#) on page 368.

Methods —

StimulusTorque

Purpose To specify a StimulusTorque profile.

Class Description (StimulusTorque)

Syntax No direct creation.

Purpose To specify a StimulusTorque profile.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Constant, Ramp, Table, Manual, or Extern ¹⁾	To get the active element.
AvailableElements	String[]	To get a list of all the available elements.
Name	String	To get the name of the StimulusAngle profile.
Type	LateralProfileTypes ²⁾	To get the type of the lateral profile.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Table](#) on page 351, [Manual](#) on page 346, or [Extern](#) on page 343.

²⁾ Refer to [LateralProfileTypes](#) on page 368.

Methods —

Stop

Purpose To specify a Stop profile.

Class Description (Stop)

Syntax No direct creation

Purpose To specify a Stop profile.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	StopWithin or UntilStop ¹⁾	To get the active element.
AvailableElements	String[]	To get all available elements
Name	String	To get the name of the Stop profile.
Type	LongitudinalProfileTypes ²⁾	To get the longitudinal profile type.

¹⁾ Refer to [StopWithin](#) on page 350 or [UntilStop](#) on page 351.

²⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

Methods

—

StopWithin

Purpose

To get information about the **StopWithin** longitudinal type.

Class Description (StopWithin)

Syntax

No direct creation

Purpose

To get information about the **StopWithin** longitudinal type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
SourceType	StopWithinSource ¹⁾	To get the StopWithinSource object for defining distance parameters.
Name	String	To get the name of the longitudinal type.
Type	LongitudinalProfileTypes ²⁾	To get the type constant of the longitudinal type.

¹⁾ Refer to [StopWithinSource](#) on page 356.

²⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

StopWithinSource

Purpose To get information about the `StopWithinSource` type.

Where to go from here Information in this section

Class Description (StopWithinSource)	356
To describe the class and its attributes.	
Activate	356
To activate a <code>StopWithinSource</code> type.	

Class Description (StopWithinSource)

Syntax No direct creation

Purpose To get information about the `StopWithinSource` type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	StopWithin ¹⁾	To get the active <code>StopWithinSource</code> type object.
AvailableElements	Strings (collection)	To get the names of all available <code>StopWithinSource</code> types.

¹⁾ Refer to [StopWithin](#) on page 350.

Methods The class contains the following methods:

Method	Purpose
Activate	To activate a <code>StopWithinSource</code> type. Refer to Activate on page 356.

Activate

Class `StopWithinSource`

Syntax

StopWithinSourceType = StopWithinSource.Activate(Variant Type)

Purpose

To activate a StopWithinSource type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific source type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
StopWithin ¹⁾	A source type object according to the given parameter.

¹⁾ Refer to [StopWithin](#) on page 350.

Related topics

References

[Class Description \(StopWithinSource\)](#)..... 356

Table

Purpose

To define a Table source type.

Where to go from here

Information in this section

[Class Description \(Table\)](#)..... 358

To describe the class and its attributes.

[Export](#)..... 358

To write the values of a table to a file.

[Import](#)..... 359

To read values of a table from a file.

[Insert](#)..... 360

To insert an entry in a table.

[Item](#)..... 360

To get a specific entry of a table.

[Remove.....](#) 361
To remove an entry of a table.

Class Description (Table)

Syntax `Table = [Parent].Table`

Purpose To define a Table source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
Count	Integer	To get the number of entries.
InterpretationType	InterpretationTypes ¹⁾	To get/set the interpretation type.
Name	String	To get the source type name.
Type	Integer	To get the source type constant.

¹⁾ Refer to [InterpretationTypes](#) on page 368.

Methods The class contains the following methods:

Method	Purpose
Export	To write the values of a table to a file. Refer to Export on page 358.
Import	To read values of a table from a file. Refer to Import on page 359.
Insert	To insert an entry in a table. Refer to Insert on page 360.
Item	To get a specific entry of a table. Refer to Item on page 360.
Remove	To remove an entry of a table. Refer to Remove on page 361.

Export

Class Table

Syntax `RetVal = Table.Export(String File)`

Purpose To write the values of a table to a file.

Parameters

The method uses the following parameters:

Parameter	Type	Description
File	String	The file name to which the values are written.

Return value

The method returns an object of the following type:

Type	Description
Boolean	True if successful.

Related topics**References**

[Class Description \(Table\)](#)..... 358

Import

Class

Table

Syntax

```
RetVal = Table.Import(String File)
```

Purpose

To read values of a table from a file.

Parameters

The method uses the following parameters:

Parameter	Type	Description
File	String	The name of the file which values are read.

Return value

The method returns an object of the following type:

Type	Description
Boolean	True if successful.

Related topics**References**

[Class Description \(Table\)](#)..... 358

Insert

Class Table

Syntax `TableValue = Table.Insert(double x, double y)`

Purpose To insert an entry in a table.

Parameters The method uses the following parameters:

Parameter	Type	Description
x	Double	The x value.
y	Double	The y value.

Return value The method returns an object of the following type:

Type	Description
TableValue ¹⁾	The inserted value of the table.

¹⁾ Refer to [TableValue](#) on page 362.

Related topics

References

[Class Description \(Table\)](#)..... 358

Item

Class Table

Syntax `TableValue = Table.Item(object Variable)`

Purpose To get a specific entry of a table.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Variable	object	The entry to be accessed.

Return value

The method returns an object of the following type:

Type	Description
TableValue ¹⁾	A specific entry of a table.

¹⁾ Refer to [TableValue](#) on page 362.

Related topics**References**

[Class Description \(Table\)](#)..... 358

Remove

Class

Table

Syntax

```
RetVal = Table.Remove(object Item)
```

Purpose

To remove an entry of a table.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Item	object	Item to be removed from the table.

Return value

The method returns an object of the following type:

Type	Description
Boolean	True if successful.

Related topics**References**

[Class Description \(Table\)](#)..... 358

TableValue

Purpose To access an entry of a Table source type.

Class Description (TableValue)

Syntax

```
TableValue = Table.Item()
TableValue = Table.Insert()
```

Purpose To access an entry of a Table source type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
X	Double	To get the X value.
Y	Double	To get the Y value.

Methods —

Related topics

References

[Class Description \(Table\)](#)..... 358

Velocity

Purpose To get information about the **Velocity** longitudinal type.

Class Description (Velocity)

Syntax No direct creation

Purpose To get information about the **Velocity** longitudinal type.

Attributes The class contains the following attributes:

Attributes	Type	Purpose
SourceType	VelocitySource ¹⁾	To get the VelocitySource object for defining distance parameters.
DependencyType	DependencyTypes ²⁾	To get the DependencyTypes object.
Name	String	To get the name of the longitudinal type.
Type	LongitudinalProfileTypes ³⁾	To get the type constant of the longitudinal type.

¹⁾ Refer to [VelocitySource](#) on page 363.

²⁾ Refer to [DependencyTypes](#) on page 312.

³⁾ Refer to [LongitudinalProfileTypes](#) on page 369.

VelocitySource

Purpose To get information about the **VelocitySource** type.

Where to go from here Information in this section

Class Description (VelocitySource)	363
To describe the class and its attributes.	
Activate	364
To activate a velocity source type.	

Class Description (VelocitySource)

Syntax No direct creation

Purpose To get information about the **Velocity** source type.

Attributes

The class contains the following attributes:

Attributes	Type	Purpose
ActiveElement	Constant, Ramp, Extern, FinalVelocity, Sine, or Table ¹⁾	To get the active velocity source type object.
AvailableElements	Strings (collection)	To get the names of all available velocity source types.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Extern](#) on page 343, [FinalVelocity](#) on page 345, [Sine](#) on page 348, or [Table](#) on page 351.

Methods

The class contains the following methods:

Method	Purpose
Activate	To activate a velocity source type. Refer to Activate on page 364.

Activate

Class

VelocitySource

Syntax

```
VelocitySourceType = VelocitySource.Activate(Variant Type)
```

Purpose

To activate a velocity source type.

Parameters

The method uses the following parameters:

Parameter	Type	Description
Type	Variant	The type of the specific source type. The value can be specified as string (name) or integer (constant type).

Return value

The method returns an object of the following type:

Type	Description
Constant, Ramp, Extern, FinalVelocity, Sine, or Table ¹⁾	A source type object according to the given parameter.

¹⁾ Refer to [Constant](#) on page 343, [Ramp](#) on page 347, [Extern](#) on page 343, [FinalVelocity](#) on page 345, [Sine](#) on page 348, or [Table](#) on page 351.

Related topics

References

Class Description (VelocitySource)	363
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Enumerations for Scenarios

Where to go from here

Information in this section

[Enumerations for Scenarios..... 366](#)
You can use predefined constants in the tool automation to improve the readability of the code.

Information in other sections

[Overview of the Classes for Creating Scenarios..... 177](#)
The overview gives you a quick overview of the classes for the Scenario Editor.

[Automating Scenarios in Python..... 170](#)
You can change the parameters of the scenario.

Enumerations for Scenarios

Introduction

You can use predefined constants in the tool automation to improve the readability of the code.

Constants

The following constants exist to automate the scenario creation.

ApproximationType The following constants are used to specify an approximation type:

Value	Description
Approach = 0	Distance becomes smaller than specified value.
Depart = 1	Distance becomes greater than specified value.

ClutchAndGearTypes The following constants are used to specify the setting of clutch and gear:

Value	Description
OpenClutch = 0	Open clutch.
Driver = 1	Clutch and gear are set by the driver model.
Stimulus = 2	Clutch and gear are stimulated.
ReferenceGear = 3	Reference gear is specified.

ConditionTypes The following constants are used to specify a condition:

Value	Description
Duration = 0	Condition is Duration type.
Immediate = 1	Condition is Immediate type.
Endless = 2	Condition is Endless type.
Trigger = 3	Condition is Trigger type.
Position = 4	Condition is Position type.
DrivenDistance = 5	Condition is DrivenDistance type.
Distance = 6	Condition is Distance type.
Timegap = 7	Condition is Timegap type.
Velocity = 8	Condition is Velocity type.
LateralAcceleration = 9	Condition is LateralAcceleration type.
DistanceRefLine = 10	Condition is Distance (along road reference line) type.

ControlModeTypes The following constants are used to specify the control mode:

Value	Description
Position = 0	The control mode is position.
YawAngle = 1	The control mode is yaw angle.

DependencyTypeConstants The following constants are used to specify a dependency:

Value	Description
Absolute = 0	Absolute
Relative = 1	Relative

DirectionTypes The following constants are used to specify a direction:

Value	Description
Left = 0	Left direction.
Right = 1	Right direction.

DownloadSelectionTypes The following constants are used to select the elements for downloading:

Value	Description
All = 0	Maneuver and traffic is downloaded.
Maneuver = 1	The maneuver is downloaded.
Traffic = 2	Traffic content is downloaded.

DriverTypes The following constants are used to specify the driver type:

Value	Description
Slow = 0	The slow driver is used.
Normal = 1	The normal driver is used.
Fast = 2	The fast driver is used.
Custom = 3	Custom settings are used for the driver.
AdaptToRoad = 4	The driver adapts to the road.

DrivingDirectionConstants The following constants are used to specify a direction of travel:

Value	Description
Direct = 0	Direct direction
Oncoming = 1	Oncoming direction

GlobalPropertiesLaneIndexInterpretationTypes The following constants are used to specify the lane index interpretation type of the global properties:

Value	Description
RelativeToPreferredLane = 0	Relative to the preferred lane
RelativeToCenterLane = 1	Relative to the center lane

HeightModeTypes The following constants are used to specify the height mode:

Value	Description
Offset = 0	Height is specified as an offset to the tire radius.
Relative = 1	Height is specified relatively to the road surface.

InsertTypes The following constants are used to specify an insert type:

Value	Description
Before = 0	A new element is inserted before a specified one.
After = 1	A new element is inserted after a specified one.

InterpretationTypes The following constants are used to specify how values are interpreted:

Value	Description
Time = 0	Values are interpreted as time in seconds.
Distance = 1	Values are interpreted as the driven distance in meters.
DistanceRoute = 2	Values are interpreted as distance on the route in meters.

LateralControllerTypes The following constants are used to specify the lateral controller type:

Value	Description
Controller1 = 1	The controller 1 is used. This controller is recommended for maneuvers with road.
Controller2 = 2	The controller 2 is used. This controller is recommended for maneuvers without road.

LateralProfileTypes The following constants are used to specify the lateral profile:

Value	Description
NotUsed = 0	Lateral profile is not used.
Continue = 2	Lateral profile is continue.
Deviation = 3	Lateral profile is deviation.
LaneSelection = 4	Lateral profile is lane selection.
StimulusAngle = 5	Lateral profile is stimulated by an angle.

Value	Description
StimulusTorque = 6	Lateral profile is stimulated by a torque.
BasicRoad = 7	Lateral profile is specified by a basic road.
FollowRoad = 8	Lateral profile is specified by the road.

LineLengthCalculationType The following constants are used to specify how the line length:

Value	Description
ReferenceLine = 0	The actual length of lines and gaps differs from the specified values depending on the distance to the reference line. The specified values are the lengths on the reference line.
Absolute = 1	The actual length of lines and gaps is equal to the specified values.

LongitudinalProfileTypes The following constants are used to specify the longitudinal profile:

Value	Description
NotUsed = 0	Not used
Continue = 2	Continue the last valid definition
Acceleration = 3	Acceleration profile with absolute values
DistanceMeter = 4	Distance of a fellow vehicle to another fellow vehicle in meters
DistanceTime = 5	Distance of a vehicle to another fellow vehicle in seconds
Position = 6	Position of the fellow vehicle from the road start position
Velocity = 7	Velocity of the fellow vehicle
StopWithin = 8	Stop within a distance
PedalStimulus = 9	Stimulus of a pedal
Standstill = 10	Standstill
LateralAcceleration = 11	Lateral acceleration
Stop = 12	Stop

ReferencedVehicleTypes The following constants are used to specify the reference type of a follow road maneuver:

Value	Description
RelativeToLane = 0	Relative to a lane
RelativeToObject = 1	Relative to an object

ReferencePointTypes The following constants are used to specify a reference point of an object.

Value	Description
Main = 0	Main point (front axle center)
Rear = 1	Rear side center
Front = 2	Front side center
Left = 3	Left side center
Right = 4	Right side center
Nearest = 5	Nearest side center

RelativeObjectType The following constants are used to specify a relative object type:

Value	Description
Vehicle = 0	Object is a vehicle.
Marker = 1	Object is a position marker.

RelativeOperatorType The following constants are used to specify a relative operator:

Value	Description
GreaterThanOrEqual = 0	Relation is greater than or equal.
LessThanOrEqual = 1	Relation is less than or equal.

ReplacementTypes The following constants are used to specify the replacement type:

Value	Description
Maneuver = 1	Replaces the maneuver.

SelectorLeverTypes The following constants are used to specify a selector lever type:

Value	Description
Driver = 1	Selector lever is specified by the driver model.
Stimulus = 2	Selector lever is stimulated.

SourceTypeConstants The following constants are used to specify the source type:

Value	Description
Constant = 0	The source type is Constant.
Ramp = 1	The source type is Ramp.
FinalDistance = 2	The source type is FinalDistance.
FinalVelocity = 3	The source type is FinalVelocity.
FinalDeviation = 4	The source type is FinalDeviation.
Sine = 5	The source type is Sine.
SmoothLaneChange = 6	The source type is SmoothLaneChange.
External = 7	The source type is External.
FinalLane = 8	The source type is FinalLane.
LaneChange = 9	The source type is LaneChange.
StopWithin = 10	The source type is StopWithin.
SimpleConstant = 11	The source type is SimpleConstant.
YawAngleConstant = 12	The source type is YawAngleConstant.
Table = 13	The source type is Table.
UntilStop = 14	The source type is UntilStop.
Fixed = 15	The source type is Fixed.
Step = 16	The source type is Step.
Pulse = 17	The source type is Pulse.
SineWithDwell = 18	The source type is SineWithDwell.

Value	Description
Circle = 19	The source type is Circle.
Straight = 20	The source type is Straight.
Manual = 21	The source type is Manual.

SpeedLimitModes The following constants are used to specify the speed limit mode for a traffic driver:

Value	Description
IgnoreScenery = 0	The speed limit specified in the scenery of the road is ignored.
UseScenery = 1	The speed limit specified in the scenery of the road is evaluated.

StopTypes The following constants are used to specify the stop type:

Value	Description
StopWithin = 8	The ego-vehicle stops within a specified value.
BrakeUntilStop = 9	The ego-vehicle brakes until it stops.

Related topics

HowTos

[How to Use Enumerations Predefined in ModelDesk \(ModelDesk Automation !\[\]\(a6f0dfe010077374846fc436b284615e_img.jpg\)\)](#)

References

[Overview of the Classes for Creating Scenarios..... 177](#)

Limitations

Limitations of the Scenario Editor

Limitations

Tool automation The tool automation interface of the Scenario Editor has been changed in ModelDesk 4.1. It is not possible to use older scripts with the new Scenario Editor.

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